

Targeting Without Transfers*

Filip Tokarski
Stanford GSB

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Abstract

I study the welfare-maximizing allocation of heterogeneous goods when monetary transfers are prohibited. Agents have private values, and the designer chooses a mechanism subject to incentive compatibility and aggregate supply constraints. I characterize the optimal mechanism for two kinds of goods, and show that it either offers one pure option per good or adds a bundle that delivers a larger total quantity. Including the bundle is optimal when narrow preference margins between goods are sufficiently predictive of greater need, allowing the designer to target high-value agents through their willingness to accept mixing. I then consider the case with N types of goods and characterize when the optimal mechanism takes the form of a simple menu, where each option offers some amount of one kind of good and none of the others. When this is the case, it can be implemented as a competitive equilibrium with equal incomes or a choice-based lottery.

1 Introduction

When designing mechanisms without transfers, it is often natural to evaluate them using criteria that avoid interpersonal utility comparisons. This approach is especially appealing when the policymaker has explicitly non-welfarist goals (such as fairness) or when participants' valuations for the allocated goods are plausibly similar. Indeed, the literature on mechanisms without money has largely focused on notions based on Pareto efficiency and ordinal welfare rankings.¹ Nevertheless, criteria agnostic to cardinal values are less fitting for settings like social programs, where policymakers view applicants as differing sharply in terms of need and aim to target those for whom receiving the goods has the greatest social value. For instance, affordable housing programs in many European countries serve a broad population, including

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¹See, among others, Hylland and Zeckhauser (1979); Abdulkadiroğlu and Sönmez (1998); Bogomolnaia and Moulin (2001); Abdulkadiroğlu and Sönmez (2013).

families facing eviction as well as middle-class households with stable employment (Whitehead and Scanlon, 2007). In the U.S. context, Cook et al. (2023) find that affordable housing recipients differ substantially in various measures of need, and that this heterogeneity persists even after conditioning on observables.

Motivated by such settings, I consider a mechanism design problem without transfers where the designer allocates heterogeneous goods to maximize cardinal welfare. Importantly, agents' valuations are their private information; this prevents the designer from simply giving the available supply to those who need it most. I make two main points. The first is that, under reasonably permissive conditions on value distributions, the optimal mechanism gives every agent some amount of one good and none of the others. In my model, this allocation rule can be implemented in several natural ways, including as a competitive equilibrium with equal incomes—a mechanism that has often been proposed in the literature because of its desirable fairness and efficiency properties (Varian, 1973; Hylland and Zeckhauser, 1979; Budish, 2011; Azevedo and Budish, 2019). Second, I show that the designer can sometimes improve upon this simple mechanism by letting agents choose large, mixed bundles alongside smaller pure options. Intuitively, this is the case when agents with higher values for the goods tend to be *less picky*: they value the available goods similarly, and so care less about which particular good they receive. These agents are then willing to give up choice in exchange for a larger total allocation, while more selective agents, whose values are lower on average, choose the smaller pure options.

Thus, the optimal mechanism depends on the statistical relationship between agents' *absolute* level of need and the strength of their *relative* preferences across goods. When higher-value agents tend to be more picky, mechanisms offering only pure options are likely to be optimal. In the reverse case, the designer can use mixed bundles to reward agents with narrower preference margins. Both correlation patterns are plausible in different settings. For instance, Cook et al. (2023) find that lower-income households are less selective when applying for affordable housing: they are more willing to trade off assignment to a preferred unit for a higher probability of receiving an offer *somewhere*. In other settings, however, the correlation may go in the opposite direction. For example, consider school choice environments with specialized curricula, such as dual-language immersion programs. Families who place disproportionate weight on admission to such programs may do so because of a child's idiosyncratic needs, aptitudes, or interests. In such cases, an intense *relative* preference for a particular option may instead signal a higher *absolute* value for receiving it.

I prove two main results. First, I fully characterize the welfare-maximizing mechanism in the case of two kinds of goods: the mechanism always offers two pure options, and sometimes also includes a single mixed bundle that gives agents some of each good and delivers a larger total quantity. In the symmetric case, I show that introducing the mixed bundle is welfare-improving precisely when weak relative preferences are sufficiently predictive of higher total value, so that the informational gain from targeting outweighs the allocative inefficiency from mixing. I then turn to the general N -good case and characterize when the mechanism offering

only pure options is optimal. The condition is stated in terms of monotone transports on an appropriately constructed signed measure, which captures the designer’s marginal value of increasing different types’ allocations. I then derive a simpler sufficient condition in the special case of symmetric goods: there, the pure option mechanism is optimal if agents whose cardinal values for their favorite goods are higher tend to be more selective, in a precise stochastic sense. Intuitively, this is because any distortion away from the pure option mechanism—which is the unique incentive-compatible Pareto-efficient allocation—must reallocate resources toward less selective types. The condition ensures, however, that the higher-value types are precisely the *more* selective ones, so such distortions move resources in the wrong direction.

I also discuss the implications of my results for market design in settings such as allocating donations to food banks and affordable housing. In these settings, existing mechanisms can be improved when the agents the designer wants to prioritize are also those willing to accept less tailored allocations. For food banks, the results suggest a natural modification of the CEEI mechanism currently used to allocate donations: the system could offer discounts for bundles of food, allowing food banks that receive little support from their local donors to choose larger allocations. In the case of affordable housing, the model speaks to the design of waitlists and repeated lotteries. Indeed, I show in Appendix A that the baseline model can be interpreted as a reduced-form description of such mechanisms, once allocations are redefined appropriately. There, one could let applicants choose between a priority waitlist where they must accept any suitable unit offered to them, and longer waitlists that let them select specific units or locations. Finally, I discuss how my framework could be extended to incorporate observable characteristics, such as income or household size.

My paper contributes to the literature on allocating heterogeneous goods without transfers, and connects most directly to work on pseudomarkets and competitive equilibrium with equal incomes (CEEI). Hylland and Zeckhauser (1979) introduce CEEI in an assignment setting by giving agents equal budgets of artificial currency and letting them purchase probability shares of different goods; the resulting lotteries are *ex ante* Pareto-efficient. Subsequent work has developed pseudomarket mechanisms for richer assignment environments, including approximate CEEI for combinatorial assignment (Budish, 2011) and pseudo-markets with priorities and other constraints (He et al., 2018). The literature also studies mechanisms based only on agents’ rankings over objects. Random serial dictatorship lets agents choose sequentially in random order (Abdulkadiroğlu and Sönmez, 1998), while probabilistic serial assigns agents fractional shares by letting them continuously claim their most-preferred available objects, yielding ordinally efficient outcomes (Bogomolnaia and Moulin, 2001). Other papers analyze the large-market properties of these mechanisms: Azevedo and Budish (2019) formalize a notion of strategy-proofness in the large for assignment and related mechanisms, while Che and Kojima (2010) show that random serial dictatorship and probabilistic serial become asymptotically equivalent.

While the study of allocating heterogeneous goods without transfers has focused mainly on criteria that avoid interpersonal utility comparisons, a smaller body of work allows for *cardi-*

nal objectives and studies mechanisms that maximize them (Miralles, 2012; Chakravarty and Kaplan, 2013; Ashlagi and Shi, 2016; Dogan and Uyanik, 2020; Akyol, 2025). My paper is closest to Miralles (2012), who studies welfare-maximizing mechanisms with cardinal utilities in a symmetric, two-good setting with finitely many agents. He shows that although the welfare optimum may differ from CEEI in finite markets, CEEI becomes optimal in a large-market limit. This, however, is because the problem is solved under a condition that turns out to exclude precisely the kind of relationship between relative preference intensity and cardinal need that makes screening with bundles useful.

A related literature studies eliciting preference intensities—information about how strongly agents prefer some options over others. In school choice, Abdulkadiroğlu et al. (2011) observe that the Boston mechanism can elicit the *extent* to which families prefer certain schools—a property that deferred acceptance does not have. In a paper closely related to mine, Ortoleva et al. (2021) consider optimal mechanisms in a setting without transfers where agents have a common ranking over goods but differ in their sensitivity to quality. My paper, by contrast, does not impose such structure and considers heterogeneously differentiated goods. This leads to different and complementary results. Indeed, the authors show that the first-best allocations may offer lotteries between qualities, and that second-best allocations always involve lotteries and may involve free disposal; neither of these results holds in my setting. Similarly to my work, they show that CEEI allocations, despite being Pareto-efficient, do not always maximize weighted welfare.

Finally, my work builds on methods developed in the multidimensional screening literature. To obtain a characterization of the optimality of the pure option mechanism, I invoke ideas used in the study of the multi-product monopoly problem (Armstrong, 1996; Rochet and Choné, 1998; Manelli and Vincent, 2006; Kleiner and Manelli, 2019). In particular, my optimality conditions rely on stochastic dominance and transport arguments related to those in Daskalakis et al. (2013, 2017).

The rest of the paper is structured as follows. Section 2 presents the model and Section 3 illustrates the core intuitions through simple two-good examples. Section 4 then formally separates absolute and relative values, and reformulates the designer’s problem in terms of relative-value profiles. Section 5 specializes the model to two goods and fully characterizes the welfare-maximizing mechanism. Section 6 defines the pure option mechanism in the general N -good case, shows how it can be implemented through a CEEI, a representative endowment economy, or a choice-based lottery, and characterizes when it is optimal. Section 7 outlines the proof of the characterization. Finally, Section 8 discusses implications for market design.

2 Model

The designer has N different kinds of goods indexed by $i \in \{1, \dots, N\}$ with $N \geq 2$. She possesses a fixed mass of each, with the supplies given by $s = (s_1, s_2, \dots, s_N) > 0$. There is a unit mass of

agents, each of whom has a profile of values $v = (v_1, v_2, \dots, v_N)$ for the goods; the values are private information and are distributed according to F with support on a bounded set $\mathcal{V} \subset \mathbb{R}_+^N$. This distribution puts no mass on $\mathbf{0}$ and assigns positive mass to agents with $v_i > 0$ for every good i . The designer chooses an allocation rule for the goods, $y = (y_1, y_2, \dots, y_N) : \mathcal{V} \rightarrow \mathbb{R}_+^N$, to maximize utilitarian welfare:

$$\int_{\mathcal{V}} v \cdot y(v) dF(v). \quad (\text{O})$$

She faces incentive compatibility and supply constraints:

$$v \cdot y(v) \geq v \cdot y(v') \quad \text{for all } v, v', \quad (\text{IC})$$

$$\int y(v) dF(v) \leq s. \quad (\text{S})$$

An allocation rule $y : \mathcal{V} \rightarrow \mathbb{R}_+^N$ is *feasible* if it satisfies (IC) and (S).

2.1 Discussion of the model

I briefly discuss the interpretation of the model. First, it assumes that the designer's preferences align with those of agents: if an agent chooses one allocation over another, the designer also considers giving her that allocation more socially valuable.

Second, the model allows for multiple interpretations of agents' values v ; for instance, one can identify v_i with an agent's (latent) willingness to pay for a unit of good i . While these values are not directly elicitable without money, they remain meaningful for the designer's welfare objective. Alternatively, and more generally, one can view them as the designer's subjective conviction about the social value of giving goods to different agents. She may, for example, place higher weights on individuals with certain characteristics (need, vulnerability, family size, etc.), and believe that these characteristics are correlated with the pattern of preferences agents reveal over the available goods.

Finally, note that at baseline we interpret $y_i(v)$ as the *amount* of good i given to type v . However, some settings of interest feature unit demand, so applying the model to them requires interpreting allocations as *probabilities* of receiving a certain good. This in turn requires imposing an additional constraint that $\sum_i y_i(v) \leq 1$ for all v . While this restriction changes the problem in general, the following result shows that imposing it is without loss when supply is sufficiently scarce relative to the population.

Proposition 1. *Consider the model augmented with the probability constraint*

$$\sum y_i(v) \leq 1 \quad \text{for all } v \in \mathcal{V}. \quad (\text{P})$$

There exists $\bar{\eta} > 0$ such that for every $\eta \in (0, \bar{\eta}]$ and every allocation rule $y : \mathcal{V} \rightarrow \mathbb{R}_+^N$ that is feasible with supplies ηs , constraint (P) is slack for all $v \in \mathcal{V}$.

Intuitively, this is because when overall supply is sufficiently small, the designer cannot afford to offer any option that delivers some good with certainty. If she did, the mass of agents requesting such an option would be so large that supply constraints would be violated. This kind of extreme mismatch between demand and supply is plausible in settings like public-housing lotteries, where units are exceptionally scarce relative to the number of applicants.²

Proposition 1 also implies that the set of feasible allocation rules is uniformly bounded for any supply s . A standard compactness argument then shows that an optimal mechanism exists.

3 Examples

To preview the paper’s core intuitions, I begin with illustrative examples featuring just two goods; I verify them in Appendix C.

Example 1. Fix any supplies $s_1, s_2 > 0$ and let values be distributed uniformly on $[0, 1]^2$. Then the optimal mechanism offers agents two options:

$$\{q_1 \text{ of good 1}\}, \quad \{q_2 \text{ of good 2}\}.$$

The quantities q_1, q_2 are chosen so that the supply constraint holds with equality when all agents pick their preferred option.

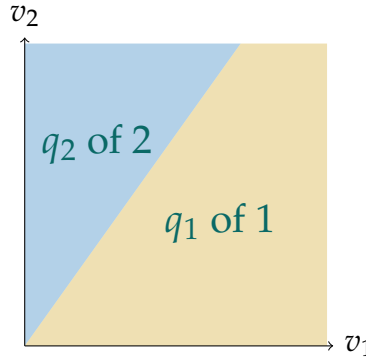


Figure 1: Optimal allocation in Example 1.

Under this mechanism, agents for whom $q_1 v_1 > q_2 v_2$ select the former option, while those for whom $q_1 v_1 < q_2 v_2$ select the latter. As shown in Figure 1, these two sets of types are separated by a ray from the origin defined by

$$\frac{v_1}{v_2} = \frac{q_2}{q_1}. \quad (1)$$

Let us note two things about this allocation. First, it is Pareto-efficient among all allocation rules satisfying the supply constraint (S). Second, it depends only on the *ratio* of agents’ values

²Under this unit-demand interpretation, $y_i(v)$ is the probability that a type- v agent receives good i . Given appropriate technical conditions, the exact law of large numbers allows such interim probabilities to be implemented by independent individual lotteries that satisfy exact aggregate feasibility; see Sun (2006).

for goods 1 and 2, but not on how large v_1 and v_2 are in absolute terms. Indeed, while the designer would like to allocate the goods only to agents whose v_1 and v_2 are high, she cannot do so in an incentive-compatible way. She is therefore restricted to conditioning the allocation on how much agents like one good *relative to the other*. This highlights a useful distinction: an agent's *absolute values*, (v_1, v_2) , capture the overall intensity of need for the goods, while her *relative values*, which I define as $(\frac{v_1}{v_1+v_2}, \frac{v_2}{v_1+v_2})$, capture the strength of her preference between the goods. Since all agents with the same relative values always rank all offered options the same way, no incentive-compatible mechanism can meaningfully elicit absolute values among agents with the same profile of relative values.

The simple mechanism in Example 1 is not always optimal, however. For other value distributions, the designer may want to offer a richer menu.

Example 2. Let $s_1 = s_2$. Fix a small $\varepsilon > 0$. An ε -share of agents have values distributed uniformly on $[0, 1]^2$. A $(1 - \varepsilon)/2$ -share of agents have values distributed uniformly on

$$[1 - \varepsilon, 1]^2.$$

Finally, a $(1 - \varepsilon)/4$ -share have values distributed uniformly on

$$[0, 1] \times [0, \varepsilon] \quad \text{and} \quad [0, \varepsilon] \times [0, 1].$$

Then, for all sufficiently small $\varepsilon > 0$, the optimal mechanism offers three options:

$$\{q_L \text{ of good 1}\}, \quad \{q_L \text{ of good 2}\}, \quad \{q_B \text{ of good 1 and } q_B \text{ of good 2}\},$$

where $q_B > q_L/2$. Thus the bundle delivers a larger total quantity than either pure option.

Under this mechanism, each agent can pick between a low amount of her favorite good and a bundle of the two goods that offers a larger total amount. Agents with strong relative preferences between the two goods pick the pure allocations while those with narrow preference margins choose the bundle.

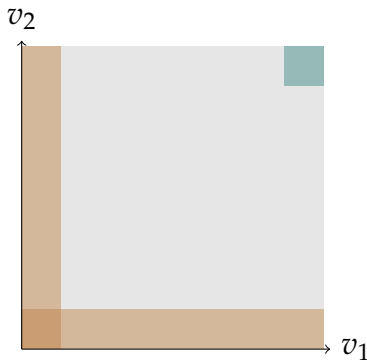


Figure 2a: Value distribution in Example 2.

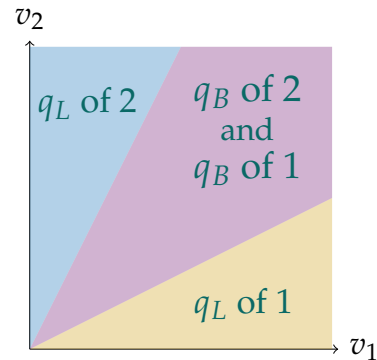


Figure 2b: Optimal allocation in Example 2.

To see why this menu is optimal, note that the value distribution, illustrated in Figure 2a, concentrates its mass on two kinds of agents. The first are *common-need* agents, who are close to indifferent about which good they get but have high values for both. The second kind are *specialized-need* agents, who tend to value one good substantially more than the other. However, their highest values tend to be lower than those of the common-need agents. Here too, all agents with the same relative values $(\frac{v_1}{v_1+v_2}, \frac{v_2}{v_1+v_2})$ receive the same allocation. However, agents whose relative values are close together choose the bundle and thus receive higher total allocations. Since these agents also tend to have higher *absolute* values (v_1, v_2) , offering a bundle gives the designer an incentive-compatible way of directing more goods to agents in greater need. More generally, doing so can help the designer if relative and absolute values are statistically related. In such cases, she can sometimes proxy for high absolute values by offering more attractive options to agents with certain relative preferences.

Note, however, that the optimal allocation in Example 2 is not Pareto-efficient among all allocation rules satisfying the supply constraint (S). Indeed, agents who receive the bundle could profitably trade among themselves so that types above and below the 45-degree line in Figure 2b each receive only the good they prefer.³ This illustrates the designer’s tradeoff: bundling reduces allocative efficiency, but helps screen agents by the strength of their relative preferences, allowing scarce resources to be targeted toward those with greater need.

4 Absolute and relative values

Motivated by the preceding examples, I now formally separate absolute and relative values. Since the designer cannot elicit absolute values of agents who share the same profile of relative values, we can, without loss, identify types with the latter. Let Γ be the $(N - 1)$ -simplex of relative-value profiles:

$$\Gamma := \{\theta \in \mathbb{R}_+^N : \sum \theta_i = 1\}.$$

Define V as the random variable describing the value vector v of an agent drawn from F and let Θ be the following Γ -valued random variable⁴

$$\Theta := \frac{V}{\sum_j V_j}.$$

The renormalization thus maps all types that are identical up to scaling to the same renormalized type $\theta \in \Gamma$. Let G denote the distribution of Θ , which is the push-forward of F under the map $v \mapsto v / \sum_j v_j$. Throughout, we impose the following assumption on G :

³While the optimal allocation is not Pareto-efficient subject to (S) alone, it is Pareto-efficient within the class of allocation rules satisfying both (S) and (IC).

⁴Recall $V = \mathbf{0}$ occurs with probability zero. We therefore exclude this null type without loss: it values every allocation at zero, and assigning it zero allocation makes all IC constraints involving it trivial. Thus Θ is defined only for $\sum_j V_j > 0$, avoiding division by zero.

Assumption 1. G admits a density g that is strictly positive a.e. on Γ . Moreover, g and its first weak derivatives along Γ are square-integrable.

While the designer cannot elicit absolute values, they are still important for her objective. We therefore let $\lambda : \Gamma \rightarrow \mathbb{R}_+$ be the conditional expectation of total value given the renormalized type:

$$\lambda(\Theta) = \mathbb{E} \left[\sum_j V_j \middle| \Theta \right].$$

Intuitively, $\lambda(\theta)$ is the expected total value associated with agents whose types v got mapped to θ .⁵ Since $\mathcal{V}$ is bounded, we can fix a version of this conditional expectation that is bounded on all of Γ . We then rewrite the designer's problem as follows:

Problem 1. Choose an allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ to maximize weighted expected utility:

$$\int_{\Gamma} \lambda(\theta) U(\theta) dG(\theta), \quad (\text{O}')$$

where $U(\theta) = x(\theta) \cdot \theta$, subject to:

$$\theta \cdot x(\theta) \geq \theta \cdot x(\theta') \quad \text{for all } \theta, \theta' \in \Gamma, \quad (\text{IC}')$$

$$\int_{\Gamma} x(\theta) dG(\theta) \leq s. \quad (\text{S}')$$

Indeed, Problem 1 is equivalent to the designer's original problem in the following sense:

Lemma 1. For any feasible allocation rule $y : \mathcal{V} \rightarrow \mathbb{R}_+^N$, there exists $x : \Gamma \rightarrow \mathbb{R}_+^N$ satisfying

$$x(\theta) = \mathbb{E} [y(V) \mid \Theta = \theta] \quad \text{a.e.} \quad (2)$$

that is feasible in Problem 1. Moreover, welfare from y equals the renormalized welfare from x :

$$\int_{\mathcal{V}} v \cdot y(v) dF(v) = \int_{\Gamma} \lambda(\theta) \theta \cdot x(\theta) dG(\theta). \quad (3)$$

Conversely, for any feasible x in Problem 1, the allocation rule given by $y(\mathbf{0}) = 0$ and $y(v) := x(v / \sum_i v_i)$ for $v \neq \mathbf{0}$ is feasible for the original problem, and the two allocation rules satisfy (3).

This renormalization has a clear economic interpretation. The type θ contains the minimal information needed to describe behavior and is the object that can be empirically identified from choices. By contrast, $\lambda(\theta)$ captures the expected *scale* of values conditional on θ , and therefore affects the problem only through the designer's objective. Economically, λ encodes the designer's prior about how need (i.e., absolute value) varies across preference profiles, and is relevant only for the normative ranking of feasible allocations.

⁵Related renormalizations appear in Weitzman (1977); Dworczak et al. (2021). There, values are normalized by agents' marginal utility of money, so the analogue of λ is the expected marginal utility of money among agents with a given willingness to pay for a good.

5 Two kinds of goods

I now characterize the optimal mechanism when there are only two types of goods. In this case, the reparametrization from Section 4 effectively makes types one-dimensional, removing the difficulties of multidimensional screening. Specifically, each renormalized type $\theta = (\theta_1, \theta_2) \in \Gamma$ is pinned down by the single coordinate $t := \theta_2 \in [0, 1]$, and so the designer's problem reduces to choosing a one-dimensional allocation rule. Indeed, in this case the optimal mechanism takes a simple form:

Theorem 1. *There exists an optimal mechanism with one of the following two forms. It either offers two pure options,*

$$\{q_1 \text{ of good 1}\}, \quad \{q_2 \text{ of good 2}\},$$

or it offers two pure options and one bundle,

$$\{q_1 \text{ of good 1}\}, \quad \{q_2 \text{ of good 2}\}, \quad \{q_1^m \text{ of good 1 and } q_2^m \text{ of good 2}\}.$$

While the proof is in the appendix, I explain its core logic as well as the reason for the simple structure of the optimal mechanism. I show that every allocation rule satisfying (IC) can be written as a positive combination of binary threshold rules of the form

$$x^z(1-t, t) = \begin{cases} (z, 0), & t < z, \\ (0, 1-z), & t \geq z. \end{cases} \quad (4)$$

This rule is induced by a two-option menu offering either z units of good 1 or $1-z$ units of good 2; z is exactly the cutoff at which agents switch from preferring the pure good-1 option to the pure good-2 option, and the cutoff type $\theta = (1-z, z)$ is indifferent between the two.

This representation turns the designer's problem into a linear program over positive measures on such threshold rules. Each threshold rule uses up a certain amount of good 1 and a certain amount of good 2, so the supply constraints impose two linear inequalities on the measure. Since this linear program has only two resource constraints, a standard extreme-point argument implies that an optimal solution places weight on at most two thresholds, say $a \leq b$. These two thresholds partition the type interval into three regions,

$$t < a, \quad a \leq t < b, \quad t \geq b,$$

and therefore generate at most three allocation levels: a pure good-1 option, a bundle, and a pure good-2 option.

Thus, the optimal mechanism can take one of two forms. It either offers two pure options, one for each good, or it offers two smaller pure options together with a larger mixed bundle, as in Example 2. The next corollary gives an explicit recipe for the optimal menu. To state it, denote

the vector of supplies used by the threshold rule x^z by

$$\varphi(z) := (z \mathbb{P}(\Theta_2 \leq z), (1 - z) \mathbb{P}(\Theta_2 \geq z))$$

and let

$$w(z) := \mathbb{E}[\lambda(\Theta) \max\{z\Theta_1, (1 - z)\Theta_2\}]$$

denote the welfare it generates. The optimal menu can then be constructed as follows:

Corollary 1. *Let*

$$\rho^* \in \arg \min_{\rho > 0} (\rho s_1 + s_2) \max_{z \in [0,1]} \frac{w(z)}{\rho \varphi_1(z) + \varphi_2(z)},$$

and define

$$\mathcal{P} := \left\{ (a, b) \in (0, 1)^2 : a \leq b, \quad a, b \in \arg \max_{z \in (0,1)} \frac{w(z)}{\rho^* \varphi_1(z) + \varphi_2(z)}, \quad \frac{\varphi_2(a)}{\varphi_1(a)} \geq \frac{s_2}{s_1} \geq \frac{\varphi_2(b)}{\varphi_1(b)} \right\}.$$

Then $\mathcal{P}$ is nonempty. For every $(a, b) \in \mathcal{P}$, there exist $m_a, m_b \geq 0$ satisfying

$$m_a \varphi(a) + m_b \varphi(b) = s.$$

For any such m_a, m_b , the menu with quantities

$$q_1 = a m_a + b m_b, \quad q_2 = (1 - a) m_a + (1 - b) m_b, \quad q_1^m = b m_b, \quad q_2^m = (1 - a) m_a \quad (5)$$

is optimal. If $a = b$, or if one of the weights is zero, the mechanism reduces to two pure options.

To understand the construction, consider the shadow costs associated with the two goods' supplies. Let us normalize the shadow cost of good 2 to unity and use $\rho > 0$ to denote the shadow cost of good 1. Under these costs, threshold rule z has total resource cost $\rho \varphi_1(z) + \varphi_2(z)$, and therefore generates welfare per unit of cost equal to

$$\frac{w(z)}{\rho \varphi_1(z) + \varphi_2(z)}.$$

Note that for any fixed ρ , the best such ratio gives an upper bound on what any feasible combination of threshold rules can achieve. Indeed, the total cost of the available supply is $\rho s_1 + s_2$, and each unit of cost can generate at most

$$\max_{z \in [0,1]} \frac{w(z)}{\rho \varphi_1(z) + \varphi_2(z)}$$

units of welfare, and thus every feasible mechanism gives welfare of at most

$$(\rho s_1 + s_2) \max_{z \in [0,1]} \frac{w(z)}{\rho \varphi_1(z) + \varphi_2(z)}.$$

The value ρ^* is chosen to make this upper bound as small as possible. At the optimum, the mechanism can therefore put positive weight only on threshold rules that maximize the welfare-cost ratio under ρ^* . Now, since there are two goods, at most two such threshold rules are needed to exhaust supply. The condition defining $\mathcal{P}$ says exactly that the selected thresholds $a \leq b$ straddle the aggregate supply ratio s_2/s_1 : one threshold uses relatively more good 2, while the other uses relatively more good 1. Hence some positive combination of their supply vectors can equal s . Once the two threshold rules are combined with weights m_a and m_b , types below a receive only good 1, types above b receive only good 2, and types between a and b receive the mixed bundle. This yields the menu quantities in (5).

Remark 1. *The optimal menu construction simplifies when supplies are equal and g, λ are exchangeable. Then $\rho^* = 1$ and the optimal thresholds a and b can be chosen so that $a = 1 - b$.*

Let us now discuss conditions under which introducing the bundle is optimal. The intuition is starkest in the symmetric case. To that end, consider the following specialization of Theorem 1:

Corollary 2. *Suppose g and λ are exchangeable and $s_1 = s_2$. Then the symmetric two-option mechanism is optimal if and only if, for every $z \in [1/2, 1]$,*

$$\mathbb{E} \left[\lambda(\Theta) (z - \max_i \Theta_i) \mid \max_i \Theta_i \leq z \right] \leq (2z - 1) \mathbb{E} \left[\lambda(\Theta) \max_i \Theta_i \right]. \quad (6)$$

In particular, this holds if $t \lambda(1 - t, t)$ is non-decreasing in t for $t \in [1/2, 1]$.

To understand the result, consider first the mechanism with two symmetric options and order all agents according to their highest value: $\max_i \Theta_i$. Note that agents for whom this quantity is closer to $1/2$, i.e. those who have weaker relative preferences over which good they get, are more willing to accept mixtures of goods in exchange for receiving a larger total allocation. Now, start with a symmetric pure option mechanism and consider a perturbation under which all agents with $\max_i \Theta_i \leq \tau$ get some of their less-preferred good alongside their favorite one, while the allocations of all agents' preferred goods are reduced. To maintain incentive compatibility, these changes must be calibrated so that the types with $\max_i \Theta_i = \tau$ remain indifferent between the pure options and bundles. Also, the reduction in all types' favorite-good allocation is chosen so that the perturbation does not violate the supply constraint. The difference between the left- and right-hand sides of (6) then captures the welfare effects of such a perturbation. If it is welfare-improving for some τ , the pure option mechanism cannot be optimal. Since Theorem 1 lets us restrict attention to mechanisms with at most one bundle in the symmetric case, the absence of such a profitable perturbation is also sufficient for optimality.

It is then intuitive that introducing such a bundle would not be beneficial under the monotonicity condition in Corollary 2. Echoing the intuition from Example 2 and Section 6.1, offering the bundle serves to direct rewards toward agents with less extreme relative preferences. If such agents tend to have lower values for their favorite good, as measured by $t\lambda(1-t, t)$, doing so is counterproductive. Importantly, however, the opposite monotonicity of $t\lambda(1-t, t)$ is *not* sufficient to conclude that the designer should introduce the bundle. This is because mixing goods is an intrinsically distortionary screening device: to direct rents toward less extreme types, the mechanism must give them some of the good they value less, and must finance this by reducing other agents' allocations of their preferred good to satisfy the supply constraint. Thus, even if agents with less extreme relative preferences tend to have higher expected total values, this correlation must be strong enough to compensate for the resulting inefficiency.

6 The pure option mechanism

As shown in Example 1, the optimal mechanism sometimes takes a particularly simple form: it offers one pure option for each good, and each agent chooses her favorite. In this section, I characterize when such a mechanism is indeed optimal in the general case.

Definition 1. An allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ is a **pure option mechanism** with a vector of **pure options** $q = (q_1, \dots, q_N) \in \mathbb{R}_{++}^N$ if for every $\theta \in \Gamma$,

$$x(\theta) \in \arg \max_{i \in \{1, \dots, N\}} \theta \cdot q_i e_i, \quad \text{and} \quad \int_{\Gamma} x(\theta) dG(\theta) = s.$$

That is, agents face a menu with N options,

$$\{q_1 \text{ of good 1}\}, \quad \{q_2 \text{ of good 2}\}, \quad \dots, \quad \{q_N \text{ of good } N\},$$

where the quantities q_i are chosen so that all the supply constraints bind when all agents choose their favorite one. In the appendix, I show that such a market-clearing menu is unique:

Fact 1. The pure option mechanism exists and is unique up to tie-breaking among a null set of types.

The pure option mechanism admits several equivalent implementations.

Definition 2. We define three implementations.

- (i) A **competitive equilibrium with equal incomes** (CEEI) is a vector of prices $p \in \mathbb{R}_+^N$ and an allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ such that the supply constraints (S') bind for all goods and all types choose utility-maximizing allocations subject to their unit budget constraint:

$$\text{for all } \theta \in \Gamma, \quad x(\theta) \in \arg \max_{z \in \mathbb{R}_+^N} \{\theta \cdot z : z \cdot p \leq 1\}.$$

Intuitively, a CEEI gives every agent one unit of artificial currency, posts per-unit market-clearing prices p , and lets everyone buy their favorite bundle z .

- (ii) A **Walrasian equilibrium of the representative endowment economy** is a price vector $p \in \mathbb{R}_+^N$ and an allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ such that markets clear and every type chooses a utility-maximizing bundle subject to the budget generated by her endowment:

$$x(\theta) \in \arg \max_{z \in \mathbb{R}_+^N} \{ \theta \cdot z : p \cdot z \leq p \cdot s \}.$$

- (iii) A **choice-based lottery** is a game where each type $\theta \in \Gamma$ chooses a good. The supply of each good is then allocated uniformly at random among the agents who selected it. Thus, each agent who chooses i receives

$$x(\theta) = \frac{s_i}{m_i} e_i,$$

with $s_i/m_i = +\infty$ if $m_i = 0$, where m_i is the mass of agents choosing good i .

Fact 2. The pure option mechanism can be implemented as a CEEI, as a Walrasian equilibrium of the representative endowment economy, and, under the unit-demand interpretation, as a pure-strategy Nash equilibrium of the choice-based lottery. Conversely, each of these implementations induces the pure option mechanism, up to tie-breaking among a null set of types.

Intuitively, the CEEI and representative-endowment implementations give all agents identical budgets to spend on goods: in the former, this budget is assigned directly, while in the latter it is the market value of their endowment s . Since utilities are linear and total allocations are not capped, almost every type spends her whole budget on the good with the highest “bang per buck,” θ_i/p_i , and so the induced allocation is pure. The allocations are also pure in any pure equilibrium of the choice-based lottery. Thus, in all three implementations, almost every type receives a pure allocation. Moreover, no type prefers another type’s allocation to her own: in the first two implementations this follows because all agents face the same budget set, while in the lottery implementation it follows from equilibrium. Since the induced pure allocations also clear the market, Fact 1 implies that they must coincide with the pure option mechanism.

I now describe the allocation induced by the pure option mechanism, that is, I characterize which types choose which pure option. To do so, I first introduce a partial order comparing how strongly different types favor a particular good relative to the other goods.

Definition 3. Take $\theta, \theta' \in \Gamma$ with $\theta_i, \theta'_i > 0$. We say θ is closer to vertex e_i than θ' , denoted by $\theta \succ_i \theta'$, if for all $k \neq i$:

$$\frac{\theta_k}{\theta_i} \leq \frac{\theta'_k}{\theta'_i}.$$

Intuitively, $\theta \succ_i \theta'$ means that θ values good i relatively more than does θ' , compared to every other good (Figure 3).

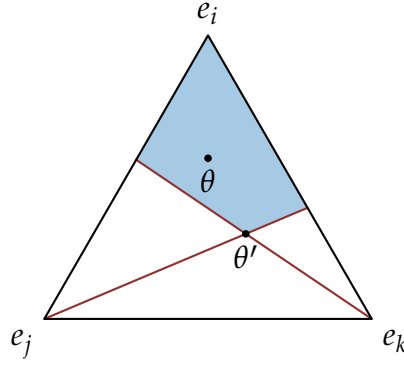


Figure 3: Types in the shaded area are closer to e_i than θ' , i.e. $\theta \succ_i \theta'$.

Fact 3. Let $q = (q_1, \dots, q_N)$ be the unique vector of pure options in the pure option mechanism, and let $\theta^0 \in \Gamma^\circ$ denote the type who is indifferent among all of them:

$$\theta^0 := \left(\frac{1/q_1}{\sum_{k=1}^N 1/q_k}, \frac{1/q_2}{\sum_{k=1}^N 1/q_k}, \dots, \frac{1/q_N}{\sum_{k=1}^N 1/q_k} \right).$$

Define

$$\Gamma_i := \{\theta : \theta \succ_i \theta^0\}.$$

Then every type $\theta \in \Gamma_i^\circ$ receives $x(\theta) = q_i e_i$.

Proof. Let $q = (q_1, \dots, q_N)$ be the unique vector of pure options. By definition of θ^0 , we have $\theta_i^0 q_i = \theta_k^0 q_k$ for all i, k . Then $\theta_k^0 / \theta_i^0 = q_i / q_k$ for all i, k . Now, fix $\theta \in \Gamma_i^\circ$. Since $\theta \succ_i \theta^0$, for every $k \neq i$,

$$\frac{\theta_k}{\theta_i} < \frac{\theta_k^0}{\theta_i^0} = \frac{q_i}{q_k}.$$

Multiplying through by $\theta_i q_k$ gives $\theta_k q_k < \theta_i q_i$ for all $k \neq i$, so option i uniquely maximizes $\theta_j q_j$ among all pure options, so $x(\theta) = q_i e_i$ in the pure option mechanism. $\square$

That is, the mechanism's allocation can be described as follows. Let q be the unique market-clearing vector of pure-option quantities. These quantities determine a unique type θ^0 that is indifferent among all options. Up to tie-breaking on a null set, option i is then chosen exactly by the types who like good i more than θ^0 , in the sense that they lie in the direction of vertex e_i from θ^0 according to the partial order defined above. Denote this region by Γ_i . Since every type lies in the direction of some vertex from θ^0 , the regions $\{\Gamma_i\}_{i=1}^N$ then form a partition of Γ , up to the null set of types who are indifferent between two or more pure options (Figure 4).

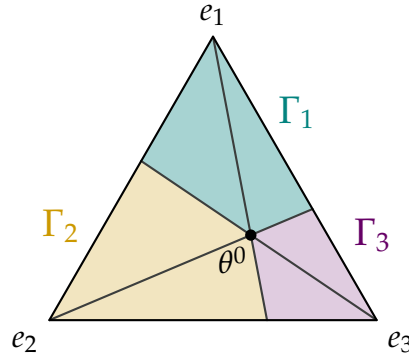


Figure 4: Each region Γ_i contains types who receive q_i of good i under the pure option mechanism.

6.1 When is the pure option mechanism optimal?

I now characterize when the pure option mechanism is welfare-maximizing; I focus on presenting the result and its high-level intuitions; the technical details underlying the proof are discussed in Section 7. To state the result, I first introduce several objects.

Shadow costs of supply. These values capture the marginal welfare gain of relaxing each supply constraint at the pure option allocation. To construct them, define

$$M_i := \int_{\Gamma_i} g(\theta) d\theta, \quad A_i := \int_{\Gamma_i} \theta_i g(\theta) \lambda(\theta) d\theta,$$

so that M_i is the mass of agents choosing option i and A_i is the designer's total value of giving each of them a unit of good i . For $i \neq j$, define:

$$T_{ij} := \int_{\Gamma_i \cap \Gamma_j} g(\theta) \theta_i d\sigma(\theta) / \sqrt{q_i^2 + q_j^2 - \frac{1}{N}(q_i - q_j)^2},$$

where $d\sigma$ denotes $(N-2)$ -dimensional Hausdorff measure on $\Gamma_i \cap \Gamma_j$. Note that for all i and $j \neq i$ we have $M_i, A_i, T_{ij} > 0$.⁶ We use these objects to construct the following matrix $J \in \mathbb{R}^{N \times N}$:

$$J_{ii} = M_i + q_i \sum_{j \neq i} T_{ij}, \quad J_{ij} = -q_j T_{ij} \quad (i \neq j).$$

Definition 4. The vector of **shadow costs** $c = (c_1, c_2, \dots, c_N)$ is given by:

$$c = J^{-1}A, \quad \text{where} \quad A := \begin{pmatrix} A_1 \\ \vdots \\ A_N \end{pmatrix},$$

⁶For A_i and M_i , this follows as $\lambda, g > 0$ and each Γ_i has positive measure. For T_{ij} , this is because the surface has a positive $(N-2)$ -dimensional Hausdorff measure and because $\theta_i > 0$ on its interior.

Fact 4. *The shadow costs $c = J^{-1}A$ are well-defined and strictly positive.*

I discuss the intuition for this shadow cost construction in Section 7. Note, however, that c is pinned down entirely by primitives—the distribution G , the weight function λ , and the market-clearing pure options q —and does not depend on any endogenous objects.

Chosen-good component. For every type θ , we define the chosen-good component $\theta_*(\theta)$ as the coordinate of θ corresponding to this type's preferred option in the pure option mechanism.⁷

$$\theta_*(\theta) := \theta_i \quad \text{for } \theta \in \Gamma_i^\circ$$

Rent measure. Define the following signed measure on the type space Γ :

$$\mu(A) = \int_{A \cap \Gamma} \theta_* \left[(\lambda - \sum_j c_j) g + \operatorname{div} \left((c - \theta \sum_j c_j) g \right) \right] d\theta - \int_{A \cap \partial \Gamma} \theta_* (c - \theta \sum_j c_j) g \cdot \nu d\sigma, \quad (7)$$

where ν is the outward unit conormal to $\partial \Gamma$ in the hyperplane containing Γ ; the divergence is also taken within this hyperplane.

Intuitively, μ captures the marginal welfare gain from increasing the size of a type's preferred option by a unit. It incorporates several effects: the direct utility to the recipient weighted by λ , the shadow cost of the additional supply required, and the incentive effects arising from tightening local incentive-compatibility constraints. Thus, μ places negative mass on types whose allocations the designer would want to *decrease* after accounting for all these effects, and positive mass on those whose allocations she would want to *increase*. In this sense, the rent measure plays a role analogous to virtual values in one-dimensional screening.

Monotone transport plan. The optimality conditions will be stated in terms of the shape of this rent measure μ . To formalize them, we use the following concept:

Definition 5. Let ρ, τ be measures on some $\Omega \subset \mathbb{R}^N$ with $\rho(\Omega) = \tau(\Omega)$, and let $\geq$ be a partial order on Ω . A $\geq$ -**monotone transport plan** from ρ to τ is a finite nonnegative Borel measure π on $\Omega \times \Omega$, supported on $\{(x, y) : x \leq y\}$, with first marginal ρ and second marginal τ ; that is,

$$\pi(A \times \Omega) = \rho(A), \quad \pi(\Omega \times A) = \tau(A) \quad \text{for all Borel } A \subseteq \Omega.$$

Intuitively, a transport plan specifies how one measure can be moved onto another, while allowing mass to move only upward according to the partial order $\geq$.

We can now state the main result of this section:

⁷On indifference boundaries, fix an arbitrary measurable tie-breaking rule.

Theorem 2. *The pure option mechanism is optimal if and only if there exists a finite signed Borel measure $\tilde{\mu}$ on Γ such that the following two conditions hold.*

1. **Preprocessing.** *For every continuous, convex $U : \Gamma \rightarrow \mathbb{R}_+$,*

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta) \leq \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\tilde{\mu}(\theta).$$

2. **Regionwise transport.** *For each i , let $\tilde{\mu}_i$ be the restriction of $\tilde{\mu}$ to Γ_i . Let $\tilde{\mu}_i^+$ and $\tilde{\mu}_i^-$ denote its positive and negative parts. Then there exists a $>_i$ -monotone transport plan from $\tilde{\mu}_i^-$ to $\tilde{\mu}_i^+$.*

Thus, the pure option mechanism is optimal if and only if, after a certain transformation, the positive and negative parts of the rent measure are suitably positioned relative to each other. This transformation of μ into an auxiliary measure $\tilde{\mu}$ is described by the preprocessing step which corresponds to ironing in one-dimensional mechanism design.⁸ Once the measure has been preprocessed into $\tilde{\mu}$, the second condition requires that, on each region Γ_i , the negative part of $\tilde{\mu}_i$ can be transported onto its positive part by moving mass only in the direction toward vertex e_i , in the sense of the partial order $>_i$.⁹

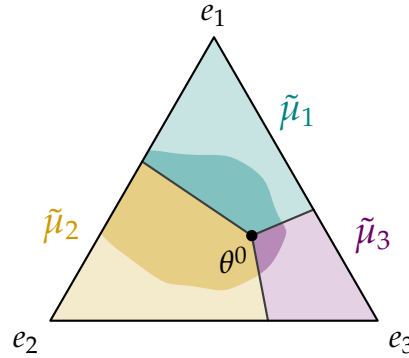


Figure 5: An example where the regionwise transport condition is satisfied. Within each region Γ_i , the negative part of $\tilde{\mu}_i$ (darker shading) can be transported onto the positive part (lighter shading) by moving mass toward vertex e_i .

Figure 5 illustrates the transport condition for a three-good example. Within each region Γ_i , the support of the positive part of $\tilde{\mu}_i$ (lighter shading) lies closer to the corresponding vertex e_i than the support of the negative part (darker shading). In other words, the types the designer would like to reward more under the pure option mechanism are precisely the more selective ones—those with stronger relative preferences for their chosen good. This foreshadows the

⁸See Rochet and Choné (1998) and Daskalakis et al. (2017) for related discussions.

⁹The transport condition is related to the stochastic-dominance certificates used by Daskalakis et al. (2013, 2017) in the study of multi-product monopoly pricing. I discuss this connection in detail in Section 7.

sufficient conditions in Section 6.2: when higher total values $\lambda(\theta)$ are associated with more extreme relative preferences, the transport condition is easier to satisfy, and the pure option mechanism is more likely to be optimal.

6.2 Sufficient conditions for optimality

We now derive a sufficient condition from Theorem 2 for the case of equal supplies and exchangeable distributions. To state it, we first introduce a stochastic monotonicity notion.

Definition 6. Let X be an $\mathcal{X}$ -valued random variable and Y be a real-valued random variable. Let $\geq$ be a partial order on $\mathcal{X}$. Fix an event E with $\mathbb{P}(E) > 0$. For any t with $\mathbb{P}(Y \geq t, E) > 0$, let $\mathcal{L}(X | Y \geq t, E)$ denote the conditional law of X given $\{Y \geq t\} \cap E$.

Then X is $\geq$ -stochastically decreasing in Y conditional on E if for all such t, t' for which $t > t'$:

$$\mathcal{L}(X | Y \geq t', E) \quad \geq\text{-stochastically dominates} \quad \mathcal{L}(X | Y \geq t, E).$$

That is, conditional on E , higher values of Y are associated with conditional distributions of X that are lower in the sense of first-order stochastic dominance with respect to $\geq$.

Proposition 2. Assume $s_1 = \dots = s_N$ and let g and λ be exchangeable. Then the pure option mechanism is optimal if, for every i , the random vector

$$\left(\frac{\Theta_1}{\Theta_i}, \dots, \frac{\Theta_N}{\Theta_i} \right)$$

is $\geq$ -stochastically decreasing in $\lambda(\Theta)\Theta_i$ conditional on $\Theta_i \geq \Theta_j$ for all $j \neq i$.

Intuitively, the condition says that the pure option mechanism is optimal when agents with higher values for their favorite good tend to be *more selective*: conditional on good i being their favorite, higher realizations of the weighted value $\lambda(\Theta)\Theta_i$ are associated with smaller ratios $(\Theta_j/\Theta_i)_{j \neq i}$ in the sense of stochastic dominance. This echoes the intuition from Example 2. There, distorting the pure option menu by introducing mixtures was beneficial precisely because *less selective* agents had higher cardinal values. Under the condition in Proposition 2, the opposite is true, and such distortions are counterproductive.

Remark 2. The stochastic monotonicity condition in Proposition 2 is related to those in Haghpahanah and Hartline (2021) and Yang (2025), who study a monopolist with access to what is essentially a wasteful screening instrument. Under a similar condition, this instrument is less costly to high-value agents, and thus can only serve to prevent misreporting from low- to high-value types. At the optimum, however, incentive constraints bind in the opposite direction: high-value types want to imitate low-value types, and thus the instrument is counterproductive and should not be used. While the proof technique used to show my result is different, the condition arises for a similar economic reason: bundling, as opposed

to letting agents choose which good they want, acts as a wasteful screening instrument. The designer is concerned about low-value types pretending to have high values. Under the right stochastic monotonicity condition, however, high-value agents tend to be pickier, so bundling is less attractive to them than to the agents who try to imitate them. Consequently, this instrument screens “in the wrong direction”.

Proposition 2 also yields the following corollary, which provides a simple, closed-form condition that can be verified directly from the marginal distribution of values:

Corollary 3. Assume $s_1 = \dots = s_N$, and suppose that the unnormalized values $V_1, \dots, V_N$ are i.i.d. with common distribution F_M . Suppose F_M is supported on $[0, \bar{v}]$ and admits a density f_M that is positive on $(0, \bar{v})$. If

$$x \frac{f_M(x)}{F_M(x)} \text{ is non-increasing on } (0, \bar{v}), \quad (8)$$

then the pure option mechanism is optimal.

The condition is satisfied for several standard distribution families:

Example 3. Assume $s_1 = \dots = s_N$, and suppose $V_1, \dots, V_N$ are i.i.d. on $[0, \bar{v}]$. Then the pure option mechanism is optimal if the common distribution F_M is any of the following:

1. the uniform distribution, $F_M(x) = \frac{x}{\bar{v}}$;
2. a power distribution, $F_M(x) = \left(\frac{x}{\bar{v}}\right)^\alpha$ with $\alpha \geq 2$;
3. a truncated exponential distribution, $F_M(x) = \frac{1 - \exp(-\beta x)}{1 - \exp(-\beta \bar{v})}$ with $\beta > 0$;
4. a truncated Weibull distribution, $F_M(x) = \frac{1 - \exp(-\beta x^\alpha)}{1 - \exp(-\beta \bar{v}^\alpha)}$ with $\beta > 0$ and $\alpha \in \{1\} \cup [2, \infty)$.

We end this section with an example in which the pure option mechanism is *not* optimal.

Example 4. Suppose $s_1 = \dots = s_N$. Fix $p \in (0, 1)$ and suppose that a fraction p of agents have unnormalized values v distributed uniformly on $[1 - \varepsilon, 1]^N$. Also, for each $j \in \{1, \dots, N\}$, a mass $(1 - p)/N$ of agents have values v distributed uniformly on

$$[0, \varepsilon]^{j-1} \times [0, 1] \times [0, \varepsilon]^{N-j}.$$

Then, for all sufficiently small $\varepsilon > 0$, the pure option mechanism is not optimal.

This is an N -good analogue of Example 2. A fraction p of agents have *common need*: they value all goods highly and nearly equally. The remaining agents have *specialized need*: such agents are associated with one good j which they tend to strongly prefer. Since common-need agents tend to have higher values, directing rents to them by introducing a bundle is welfare-improving.

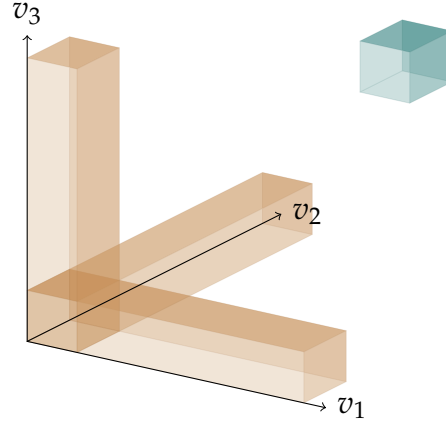


Figure 6: The distribution in Example 4 when $N = 3$. The brown regions are the specialized-need groups, and the green cube is the common-need group.

6.3 Why is the pure option mechanism often optimal?

The preceding results show that the pure option mechanism is optimal for a non-trivial set of primitives. In the symmetric i.i.d. case, for example, the reverse-hazard-rate condition in Corollary 3 is relatively permissive. While it is harder to state equally simple sufficient conditions in the general asymmetric case, the next result shows that the optimality of the pure option mechanism is not a knife-edge implication of symmetry. It shows that if optimality conditions hold with an appropriately-defined *strict slack*, then they continue to hold after small, possibly asymmetric, perturbations of the primitives. To formulate this notion of slack, we introduce the following definition:

Definition 7. Fix primitives $p = (\lambda, g, s)$, and let $\Gamma_i(p)$ and μ^p be the associated pure-option regions and rent measure. A set $C \subseteq \Gamma_i(p)$ is an $>_i$ -upper set if

$$\theta \in C, \quad \theta <_i \theta', \quad \theta' \in \Gamma_i(p) \quad \implies \quad \theta' \in C.$$

We say that p has **upper-set slack** if there exists $\eta > 0$ such that, for every i and every closed $>_i$ -upper set $C \subseteq \Gamma_i(p)$,

$$\mu^p(C) \geq \eta \min\{\alpha(C), \alpha(\Gamma_i(p) \setminus C)\}. \quad (\text{BUS})$$

Here $\alpha := m + \sigma$, where m is $(N - 1)$ -dimensional Lebesgue measure on Γ , and σ is $(N - 2)$ -dimensional Hausdorff measure on $\partial\Gamma$.

This upper-set formulation arises from Strassen's theorem, which links the existence of monotone transports to positivity on all upper sets. We then get the following result:

Proposition 3. Let $p^* := (\lambda^*, g^*, s^*)$ be baseline primitives. Suppose $g^* \in C^2(\Gamma)$, $\lambda^* \in C^1(\Gamma)$, and the associated rent measure μ^* satisfies upper-set slack.

Then there exists $\varepsilon > 0$ such that, for every primitive vector $p = (\lambda, g, s)$ satisfying

$$\lambda = e^\ell \lambda^*, \quad g = \frac{e^h g^*}{\int_\Gamma e^h g^* dm}, \quad \|\ell\|_{C^1(\Gamma)} + \|h\|_{C^2(\Gamma)} + \|s - s^*\| < \varepsilon,$$

the pure option mechanism is optimal.

I use the exponential parametrization to ensure that λ and g remain positive after the perturbation. Note also that, as the primitives are perturbed, the market-clearing pure options q_i and the regions Γ_i generally change. The result then says that the corresponding pure option mechanism, with its perturbed quantities and regions, is optimal.

It is also worth noting that the upper-set slack condition is not excessively demanding. For example, it holds in the uniform symmetric environment:

Example 5. Assume $s_1 = \dots = s_N$, and suppose unnormalized values V are distributed uniformly on $[0, 1]^N$. Then the associated primitives (λ, g, s) have upper-set slack.

These results raise a question: why is such a simple mechanism optimal in a rich class of cases? To see this, we first observe that the pure option mechanism is essentially the only one that is both Pareto-efficient (given the available supply) and incentive compatible.

Proposition 4. Suppose the allocation rule x is Pareto-efficient subject to the supply constraint (S'), that is, there does not exist an allocation rule $\tilde{x}$ satisfying (S') such that

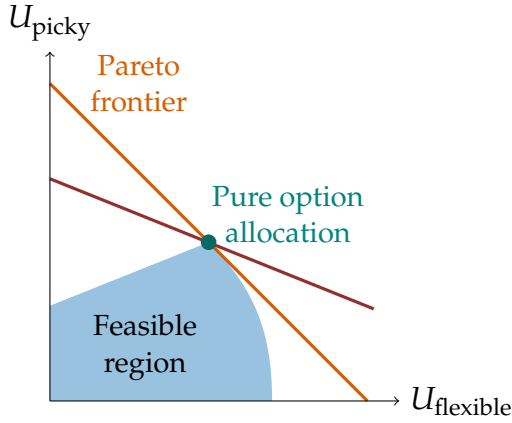
$$\theta \cdot \tilde{x}(\theta) \geq \theta \cdot x(\theta) \quad \text{for all } \theta,$$

with a strict inequality for a positive mass of types. Then, if x satisfies (IC'), it coincides with the allocation of the pure option mechanism almost everywhere. That is, if q is the unique market-clearing vector of pure options, then

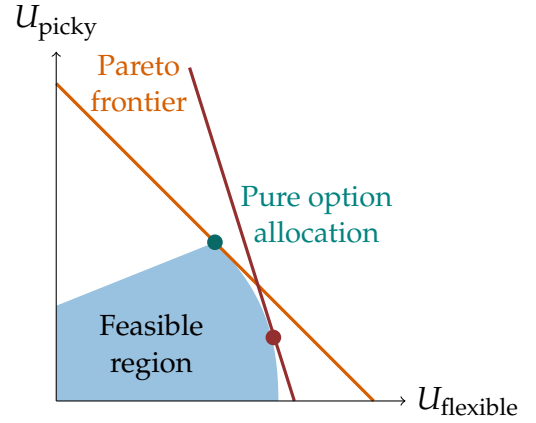
$$x(\theta) = q_i e_i \quad \text{for almost every } \theta \in \Gamma_i.$$

Intuitively, any such mechanism coincides with the pure option mechanism up to how zero-mass indifferences are resolved. To understand why this is the case, note first that the supply constraint must bind at any Pareto-efficient allocation: if some supply were left over, distributing it uniformly across everyone would strictly raise welfare while preserving incentives. Moreover, note that Pareto efficiency is inconsistent with assigning bundles to a positive mass of types. This is because the recipients of bundles could then profitably trade among themselves so that each good in the mixture would go to those who value it relatively more. Such trades would be supply-preserving and Pareto-improving. Therefore, any Pareto-efficient allocation must be pure for almost all types. However, Fact 1 shows that there is only one pure allocation that exhausts the whole supply and satisfies (IC').

Consequently, any welfare improvement over the pure option mechanism must come from Pareto-inefficient distortions. These produce mixed allocations which are relatively more attractive to agents whose values for the bundled goods are close together.¹⁰ Hence, any departure from Pareto efficiency necessarily rewards agents who are *less picky*, at least with respect to the goods being mixed. This clarifies why the pure option mechanism is optimal for one special class of distributions: those under which being less picky is always a signal of *lower* values. When this is the case, any such distortion shifts rents toward lower-value types, so the designer does better by adhering to the Pareto-efficient outcome. Thus, broadly speaking, the pure option mechanism is often optimal because it is associated with an entire “direction” of the designer’s Pareto weights λ (Figure 7).



Pareto weights skew towards picky agents.



Pareto weights skew towards flexible agents.

Figure 7: A heuristic illustration showing that moving away from the pure option mechanism can reward flexible agents, but not picky ones. If the designer’s Pareto weights are tilted towards picky agents, she always prefers the Pareto-efficient allocation. If they are tilted towards flexible ones, she might want to distort it.

The next result formalizes this directional intuition. Suppose the pure option mechanism is already optimal for primitives (s, g, λ) . Now, hold s and g fixed and change the designer’s Pareto weights from λ to $\tilde{\lambda}$. If, within each pure-option region, the new weights place relatively more value on more selective types, then the pure option mechanism remains optimal.

Corollary 4. *Suppose the pure option mechanism is optimal for (s, g, λ) . Consider any $\tilde{\lambda} : \Gamma \rightarrow \mathbb{R}_+$. For each i , define finite measures v_i and $\tilde{v}_i$ on Γ_i by*

$$dv_i(\theta) = \theta_i \lambda(\theta) g(\theta) d\theta, \quad d\tilde{v}_i(\theta) = \theta_i \tilde{\lambda}(\theta) g(\theta) d\theta.$$

Suppose that, for every i , there exists a $>_i$ -monotone transport plan from v_i to $\tilde{v}_i$. Then the pure option mechanism is also optimal for $(s, g, \tilde{\lambda})$.

¹⁰The only other possible distortion is to discard some of the supply.

The above results can be interpreted as saying that, in this environment, the designer has very limited instruments for reallocating rents across agents. This distinguishes my model from environments with transfers: in the optimal taxation model of Mirrlees (1971), for example, a Pareto-efficient allocation is optimal only under knife-edge social preferences. This is because the designer almost always has some way of distorting this allocation in a way that directs rents her preferred way: if she wants to help low-productivity agents, she taxes labor and redistributes; if she wants to help high-productivity agents, she taxes lump-sum and subsidizes labor. Here, the designer has no such freedom. Since transfers are unavailable, she can only create indirect redistribution devices, such as offering large, mixed bundles to target flexible types. However, there is no such device the designer can construct that would redistribute rents in the opposite direction.

To conclude this section, one should note that while the discussion above has focused on targeting agents based on the strength of their relative preferences, the designer could also screen them based on *which specific goods* they like. Nevertheless, any potentially productive distortion of the pure option mechanism will still produce mixed bundles, so the forces discussed above will remain relevant. Consequently, if the association between the strength of relative preference and cardinal values is strong, the pure option mechanism is likely to remain optimal even when strong preferences for some goods correlate with high cardinal values.

7 Proof of Theorem 2

I now present the key steps in the proof of Theorem 2 along with additional technical intuitions; supporting facts and lemmas are shown in the appendix.

Characterizing incentive compatibility. We first characterize incentive-compatible allocation rules in the renormalized problem using the partial order introduced in Definition 3.

Proposition 5. *A function $U : \Gamma \rightarrow \mathbb{R}_+$ is the indirect utility function of some allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ satisfying (IC') if and only if U is convex and satisfies the following condition:*

$$\text{for every } i \text{ and every } \theta, \theta' \text{ in } \Gamma \text{ such that } \theta \succ_i \theta', \quad \frac{U(\theta')}{\theta'_i} \geq \frac{U(\theta)}{\theta_i}. \quad (\text{R})$$

It is not surprising that incentive-compatible allocation rules must produce convex indirect utility functions U , as they are maxima of affine functions of θ :

$$U(\theta) = \max_{\theta' \in \Gamma} \theta \cdot x(\theta').$$

Condition (R) additionally restricts how fast indirect utility $U(\theta)$ can grow as θ moves towards the vertex e_i . To understand why (R) is necessary for incentive compatibility, fix any good i and

two types such that $\theta \succ_i \theta'$. Note that normalizing $U(\theta)$ by θ_i gives:

$$\frac{U(\theta)}{\theta_i} = \sum_{k \neq i} \frac{\theta_k}{\theta_i} x_k(\theta) + x_i(\theta).$$

We can then equivalently think of type- θ agents as maximizing their scaled utilities $U(\theta)/\theta_i$. Recall also that by the definition of the $\succ_i$ -order, all the ratios θ'_k/θ'_i are higher for θ' than for θ . This implies that type θ' can always guarantee a higher scaled indirect utility than type θ :

$$\frac{U(\theta')}{\theta'_i} = \sum_{k \neq i} \frac{\theta'_k}{\theta'_i} x_k(\theta') + x_i(\theta') \geq \sum_{k \neq i} \frac{\theta_k}{\theta_i} x_k(\theta) + x_i(\theta) = \frac{U(\theta)}{\theta_i}.$$

Indeed, since $\theta'_k/\theta'_i \geq \theta_k/\theta_i$ for all $k \neq i$, type θ' could guarantee $U(\theta')/\theta'_i$ above $U(\theta)/\theta_i$ by simply reporting θ and taking this type's allocation. As it turns out, convexity of $U(\theta)$ and (R) are also sufficient for incentive compatibility.¹¹

Expressing the problem in utility space. It will be useful to reformulate the designer's problem in terms of indirect utility functions because, as shown in Proposition 5, incentive compatibility can be characterized in terms of their shape. Indeed, writing the objective as an integral over weighted indirect utilities, rather than weighted allocations, is standard in the multidimensional screening literature.¹² The reformulation relies on the following result, which follows from Proposition 5 and Lemma 8, established in its proof:

Corollary 5. *Every indirect utility function U implemented by some (IC') allocation rule is Lipschitz continuous. Moreover, at every $\theta \in \Gamma^\circ$ where $\nabla_H U(\theta)$ exists, define:*

$$x_U(\theta) := \nabla_H U(\theta) - \mathbf{1}(\nabla_H U(\theta) \cdot \theta - U(\theta)), \quad (9)$$

where $\nabla_H U$ denotes the gradient of U intrinsic to the hyperplane containing the simplex Γ . Then $x_U(\theta) \in \mathbb{R}_+^N$, x_U is uniformly bounded, and $U(\theta) = \theta \cdot x_U(\theta)$ almost everywhere.

Thus, every allocation rule $x : \Gamma \rightarrow \mathbb{R}_+^N$ that implements U is equal to x_U almost everywhere.¹³

¹¹While working with (IC') in the simplex representation Γ is more analytically convenient in my setting, one can also characterize it in terms of an indirect utility $\tilde{U} : \mathbb{R}_+^N \rightarrow \mathbb{R}$ defined on unnormalized values v . In a quasilinear model with transfers, $\tilde{U}$ satisfies incentive compatibility if and only if the corresponding indirect utility $\tilde{U}$ is convex and nondecreasing in each coordinate (Rochet, 1987). Without transfers, incentive compatibility additionally forces $\tilde{U}$ to be positively homogeneous of degree one: for all $v \in \mathbb{R}_+^N$ and $k > 0$, $\tilde{U}(kv) = k \tilde{U}(v)$ (Lahr and Niemeier, 2024).

¹²See, for instance, Armstrong (1996); Rochet and Choné (1998); Manelli and Vincent (2006); Daskalakis et al. (2013, 2017).

¹³Indeed, if x implements U , then for every $\theta' \in \Gamma$, $U(\theta') \geq U(\theta) + x(\theta) \cdot (\theta' - \theta)$. At every point where $\nabla_H U(\theta)$ exists, this implies that the projection of $x(\theta)$ onto TH equals $\nabla_H U(\theta)$. Together with $U(\theta) = \theta \cdot x(\theta)$, this uniquely determines $x(\theta)$, and the formula is exactly (9).

Consequently, Problem 1 can be written as

$$\sup_{U \in \mathcal{U}} \int_{\Gamma} \lambda(\theta) U(\theta) g(\theta) d\theta \quad \text{subject to} \quad \int_{\Gamma} x_U(\theta) g(\theta) d\theta \leq s, \quad (10)$$

where

$$\mathcal{U} := \{U \in C(\Gamma) : U \text{ is convex and satisfies (R)}\}. \quad (11)$$

Next, I incorporate the supply constraint in (10) into the objective using the shadow costs from Definition 4, and explain the economic intuition behind their construction.

Shadow costs: construction and intuition. Consider an exercise in which the designer can allocate any amount of the N goods, but must pay per-unit costs $c = (c_1, \dots, c_N)$ for them. Starting from the pure option mechanism with quantities $q = (q_1, \dots, q_N)$, ask: what cost vector would make the designer indifferent about marginally perturbing these quantities? To pin down those costs, choose any good k and consider increasing q_k by ϵ , holding the other quantities fixed. To first order, this perturbation has two effects, illustrated in Figure 8. First, agents in Γ_k who chose option k before continue to do so, but now receive a higher quantity. This raises their utility and incurs an additional cost of $c_k \epsilon$ per agent. Second, the perturbation induces some agents who previously chose q_j , $j \neq k$, to switch to q_k . For each such agent, the designer pays $c_k q_k$ instead of $c_j q_j$. The direct welfare effects of these switchers are not first-order: both their mass and their welfare change are of order ϵ . As ϵ becomes small, the sum of these effects, divided by ϵ , converges to:

$$A_i - c_i M_i + \sum_{j \neq i} T_{ij} (c_j q_j - c_i q_i).$$

Thus, the system $Jc = A$ defining the shadow costs captures precisely the first-order conditions ensuring that such perturbations are not beneficial.

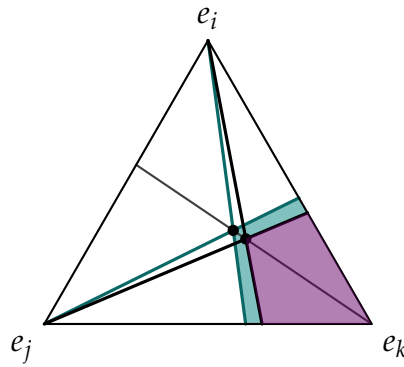


Figure 8: First-order effects of increasing the pure option q_k . Agents in the violet region receive higher quantities of k ; agents in the green region switch from other goods to k .

I now write U_{pure} for the indirect utility from the pure option mechanism. The following result

shows that we can verify the optimality of U_{pure} using the Lagrangian relaxation of (10), with the multipliers on the supply constraints set to $c = J^{-1}A$.

Proposition 6. U_{pure} solves (10) if and only if U_{pure} solves

$$\max_{U \in \mathcal{U}} \left\{ \int_{\Gamma} \lambda(\theta) U(\theta) g(\theta) d\theta - c \cdot \left(\int_{\Gamma} x_U(\theta) g(\theta) d\theta - s \right) \right\}, \quad c = J^{-1}A. \quad (12)$$

Measure representation of the objective. We now rewrite the objective in (12) using a signed measure. Moreover, we show that the measure nets to zero on each option-specific region Γ_i . Since the pure option utility is constant on each Γ_i , this zero-mass property implies that the pure option mechanism has value zero in this representation.

Lemma 2. Let $c = J^{-1}A$. Up to the constant term $c \cdot s$, the objective in (12) can be written as

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta),$$

where the signed measure μ is given by (7). Moreover, μ satisfies the following balance conditions:

$$\text{for all } i, \quad \mu(\Gamma_i) = 0. \quad (13)$$

Consequently:

$$\int_{\Gamma} \frac{U_{\text{pure}}(\theta)}{\theta_*(\theta)} d\mu(\theta) = \sum_{i=1}^N q_i \mu(\Gamma_i) = 0. \quad (14)$$

Notice that, on each region Γ_i , we have $\theta_*(\theta) = \theta_i$. Thus $U(\theta)/\theta_*(\theta)$ is the amount of good i that would give type θ the same utility as $U(\theta)$. In particular, under the pure option mechanism,

$$\frac{U_{\text{pure}}(\theta)}{\theta_*(\theta)} = q_i \quad \text{for } \theta \in \Gamma_i.$$

Thus, at the pure option mechanism, the rent measure multiplies the chosen pure allocation. For a general feasible mechanism, however, $U(\theta)/\theta_i$ is not the amount of good i assigned to type θ , but the amount of good i that would give type θ the same indirect utility. In this sense, the rent measure does not multiply the allocation directly; instead, it evaluates indirect utility expressed in units of the good chosen by that region under the pure option mechanism.¹⁴

¹⁴One could also integrate the objective by parts to obtain a representation involving the allocation rule $x(\theta)$. However, because x is a vector field, such a representation is not unique: it depends on a choice of vector-valued flows which, intuitively, correspond to sets of paths in the type space Γ along which one integrates by parts. Then, when optimizing over x to maximize such an expression, one implicitly accounts only for the effects of perturbing x that propagate through local IC constraints along these paths. In general, this can lose important information about effects propagating through other local IC constraints.

Representing the designer's objective in terms of $U(\theta)$ avoids this issue: since U is a scalar potential, the objective can be rewritten in terms of $U(\theta)$ without having to select paths along which indirect utility is integrated. As a result, this representation encodes information about effects propagating through *all* local IC constraints.

Consequently, the positive part of the preprocessed measure, $\tilde{\mu}^+$, places weight on types whose utilities the designer would like to raise, after accounting for how such changes propagate through the local IC constraints. Conversely, $\tilde{\mu}^-$ places weight on types whose utilities the designer would like to reduce. The designer cannot, however, choose U freely. Proposition 5 shows that feasible allocation rules must generate indirect utilities satisfying certain shape restrictions. In particular, (R) limits how quickly $U(\theta)$ can increase as θ moves toward the vertices of Γ . The pure option indirect utility U_{pure} is exactly the extremal profile that makes these constraints bind on each region Γ_i . The transport condition in Theorem 2 formalizes this idea. Intuitively, the pure option mechanism is optimal if, after possibly “ironing” the rent measure, each positive part $\tilde{\mu}_i^+$ lies closer to the vertex e_i than the negative part $\tilde{\mu}_i^-$. When this holds, the designer wants to raise utility most for the types closer to e_i , and the best feasible way to do so is to make $U(\theta)$ increase as rapidly as (R) permits as one moves toward that vertex. This is precisely what the pure option utility does.

Throughout the remainder of the proof, we formalize this intuition. We have already established that the pure option mechanism is optimal if and only if U_{pure} solves

$$\max_{U \in K} \int_{\Gamma} \frac{U}{\theta_*} d\mu \quad \text{subject to (R), where } K := \{U \in C(\Gamma) : U \geq 0 \text{ and } U \text{ is convex}\}.$$

To establish the transport-based optimality conditions, I formulate the problem as an infinite-dimensional linear program and use a duality approach related to those previously applied by Kleiner and Manelli (2019) and Kleiner (2022) to multi-product monopoly pricing and delegation, respectively. The argument shows that U_{pure} is optimal if and only if there exist dual multipliers for the constraints (R).

The designer’s problem as a linear program. Note K is a convex cone and for each i define

$$R_i := \{(\theta, \theta') \in \Gamma \times \Gamma : \theta_k \theta'_i \geq \theta'_k \theta_i \text{ for every } k \neq i\}.$$

This is the closed, cross-multiplied version of the order $\theta <_i \theta'$. Since R_i is a closed subset of the compact set $\Gamma \times \Gamma$, it is compact. Now, for $U \in C(\Gamma)$, define

$$(B_i U)(\theta, \theta') := \theta_i U(\theta') - \theta'_i U(\theta), \quad (\theta, \theta') \in R_i.$$

The constraint $B_i U \leq 0$ on R_i is exactly the cross-multiplied version of (R); unlike the ratio form, it remains well-defined on the boundary of the simplex. This lets us write our problem as:

$$\sup_{U \in K} \left\{ \int_{\Gamma} \frac{U}{\theta_*} d\mu : B_i U \leq 0 \text{ on } R_i \text{ for every } i \right\}. \quad (15)$$

Then, the optimality of U_{pure} in problem (15) is equivalent to the existence of a dual certificate:

Definition 8. A *dual certificate* for U_{pure} is a tuple of finite nonnegative Borel measures

$$(\gamma_1, \dots, \gamma_N), \quad \gamma_i \in \mathcal{M}_+(R_i),$$

such that, for every $U \in K$,

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{i=1}^N \int_{R_i} (B_i U)(\theta, \theta') d\gamma_i(\theta, \theta'). \quad (16)$$

Proposition 7. The pure option mechanism is optimal if and only if a dual certificate for U_{pure} exists.

We will soon relate these dual certificates to the transports in Theorem 2. Note, however, that while these transports happen only within some region Γ_i , the dual certificates from Proposition 7 may place weight on pairs of types drawn from different regions Γ_i and Γ_j . The next lemma shows that such cross-region constraints are redundant: any dual certificate can be replaced by one that operates entirely within each region.

Lemma 3. Suppose a dual certificate for U_{pure} exists. Then there also exists another dual certificate $(\tilde{\gamma}_1, \dots, \tilde{\gamma}_N)$ such that, for every j ,

$$\tilde{\gamma}_j \in \mathcal{M}_+(R_j), \quad \text{supp}(\tilde{\gamma}_j) \subseteq R_j \cap (\Gamma_j \times \Gamma_j).$$

That is, if a dual certificate exists at all, one can always find another that enforces the constraint (R) for good i only among pairs of types that both lie in Γ_i .

In the final step of the proof, I rewrite the certificate condition in the form of transports on a (possibly preprocessed) rent measure.

Deriving preprocessing and transport conditions. We first show that the conditions of Theorem 2 are sufficient. Suppose that there exists a finite signed measure $\tilde{\mu}$ satisfying both of them. Let $U \in K$ be feasible, so $B_i U \leq 0$ on R_i for every i . Fix i and let π_i be a $>_i$ -monotone transport plan from $\tilde{\mu}_i^-$ to $\tilde{\mu}_i^+$. If $\theta <_i \theta'$, then feasibility gives

$$\frac{U(\theta')}{\theta'_i} \leq \frac{U(\theta)}{\theta_i}.$$

Since π_i is supported on such pairs,

$$\int_{\Gamma_i} \frac{U(\theta)}{\theta_i} d\tilde{\mu}_i^+(\theta) = \int_{\Gamma_i \times \Gamma_i} \frac{U(\theta')}{\theta'_i} d\pi_i(\theta, \theta') \leq \int_{\Gamma_i \times \Gamma_i} \frac{U(\theta)}{\theta_i} d\pi_i(\theta, \theta') = \int_{\Gamma_i} \frac{U(\theta)}{\theta_i} d\tilde{\mu}_i^-(\theta).$$

Therefore

$$\int_{\Gamma_i} \frac{U(\theta)}{\theta_i} d\tilde{\mu}_i(\theta) \leq 0.$$

By summing over i , gluing the regionwise measures and using the fact that $\theta_*(\theta) = \theta_i$ on Γ_i :

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\tilde{\mu}(\theta) \leq 0, \quad \text{so preprocessing gives} \quad \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta) \leq 0.$$

Thus every feasible U has value at most zero. However, we know from (14) that U_{pure} attains the value zero, and so the pure option mechanism is optimal.

Conversely, suppose the pure option mechanism is optimal. By Proposition 7, there exists a dual certificate. By Lemma 3, the certificate can be chosen regionwise: for every i , there exists

$$\gamma_i \in \mathcal{M}_+(R_i), \quad \text{supp}(\gamma_i) \subseteq R_i \cap (\Gamma_i \times \Gamma_i),$$

such that, for every $U \in K$,

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta) \leq \sum_{i=1}^N \int_{R_i} B_i U(\theta, \theta') d\gamma_i(\theta, \theta').$$

For each i , let π_i be the measure on $\Gamma_i \times \Gamma_i$ defined by $d\pi_i(\theta, \theta') := \theta_i \theta'_i d\gamma_i(\theta, \theta')$. Since γ_i is supported on $R_i \cap (\Gamma_i \times \Gamma_i)$, the measure π_i is supported on pairs (θ, θ') satisfying $\theta <_i \theta'$. Let ρ_i and τ_i be the first and second marginals of π_i and define $\tilde{\mu}_i := \tau_i - \rho_i$. Then, for every $U \in K$,

$$\int_{\Gamma_i} \frac{U(\theta)}{\theta_i} d\tilde{\mu}_i(\theta) = \int_{\Gamma_i \times \Gamma_i} \left(\frac{U(\theta')}{\theta'_i} - \frac{U(\theta)}{\theta_i} \right) d\pi_i(\theta, \theta').$$

By the definition of π_i , this equals

$$\int_{R_i} \left(\frac{U(\theta')}{\theta'_i} - \frac{U(\theta)}{\theta_i} \right) \theta_i \theta'_i d\gamma_i(\theta, \theta') = \int_{R_i} B_i U(\theta, \theta') d\gamma_i(\theta, \theta').$$

Hence, gluing the regionwise measures and using the fact that $\theta_*(\theta) = \theta_i$ on Γ_i , we get:

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta) \leq \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\tilde{\mu}(\theta) \quad \text{for all } U \in K.$$

In particular, the preprocessing condition holds for every continuous, convex $U : \Gamma \rightarrow \mathbb{R}_+$.

It remains to verify the regionwise transport condition. For each i , π_i is a $>_i$ -monotone transport plan from ρ_i to τ_i . Let $\alpha_i := \rho_i \wedge \tau_i$ be the common part of ρ_i and τ_i . Since $\tilde{\mu}_i = \tau_i - \rho_i$, we have $\tilde{\mu}_i^+ = \tau_i - \alpha_i$ and $\tilde{\mu}_i^- = \rho_i - \alpha_i$. Cancelling common mass from the two marginals preserves the existence of a $>_i$ -monotone transport plan. Therefore there exists a $>_i$ -monotone transport plan from $\tilde{\mu}_i^-$ to $\tilde{\mu}_i^+$, as required.

Remark 3 (Relation to Daskalakis et al. (2013, 2017)). *The condition in Theorem 2 resembles the stochastic-dominance certificates developed in Daskalakis et al. (2013, 2017) for the problem of a multi-*

good monopolist. In particular, Daskalakis et al. (2013) provide a dominance condition for the optimality of grand bundling that is phrased in terms of a signed measure similar to mine. Our approaches are closely related: I rewrite the objective as an integral against a signed measure and certify optimality of an “extremal” indirect-utility profile through a stochastic-dominance comparison. However, several features of my environment require a different construction. First, my types live on a simplex and the planner maximizes weighted welfare rather than revenue. Second, feasibility is governed by aggregate supply constraints rather than per-agent quantity caps, so the relevant signed measure must incorporate the shadow costs of supply. Most importantly, the constraints that make the candidate solution extremal are different. In Daskalakis et al. (2013), extremality is driven by unit caps on allocations. Here, it is due to the condition (R) which bounds how fast $U(\theta)$ can grow as θ approaches a vertex. This is why the objective representation in Lemma 2 involves the transformed term $U(\theta)/\theta_*(\theta)$, rather than $U(\theta)$ alone. Relatedly, my preprocessing step requires dominance with respect to an order that differs from the one in Daskalakis et al. (2017).

8 Implications for market design

The main lesson of the paper is that in settings without transfers, the welfare-maximizing mechanism depends on how agents’ absolute and relative values covary. When high-value agents tend to be more selective, the optimal mechanism should let agents choose one kind of good and only give them that one. When high-value agents tend to be less selective, the designer can sometimes do better by offering larger bundles to agents who are willing to give up choice.

This force appears in many market-design settings. Consider, for instance, the problem of allocating food donations across food banks, studied by Prendergast (2017, 2022) and Altmann (2023). In 2005, Feeding America introduced a system under which food banks bid for donations using an artificial currency. While this currency is distributed according to a formula based on the population each food bank serves, this measure of need is imperfect because food banks also differ in the generosity of local donors operating outside the system. My results suggest that if food banks in areas with fewer local donations also have significantly weaker preferences over which food items they receive, the pseudomarket mechanism can be improved by pricing broad, pre-packaged assortments at a discount. If this relationship is weaker or reversed, however, then the pseudomarket is likely to be optimal.

A related consideration appears in affordable housing. Housing authorities often use dynamic mechanisms, such as waitlists or repeated lotteries, that allow applicants to express preferences over units. For example, in a choice-based lottery, applicants periodically list the developments they are willing to accept, and vacant units are allocated by lottery among those who listed them.¹⁵ Similarly, waitlist systems often let applicants reject offers, though sometimes at the cost of losing priority or leaving the list. As I show in Appendix A, the baseline model can

¹⁵For example, the Amsterdam housing lottery allows applicants to enter two draws per week. See <https://www.wooninfo.nl/nieuws/2013/04/nieuw-eeen-woning-via-loting/>.

be interpreted as a reduced-form description of these dynamic mechanisms, after appropriately defining $y_i(v)$ as the expected allocation of good i over an applicant's time in the system. Thus, the main lessons from my model also apply to these dynamic settings. While the literature on housing allocation has noted the trade-off between choice and targeting, it has focused on extreme mechanisms giving agents no choice, or letting them choose a specific development (Arnosti and Shi, 2020; Waldinger, 2021). My results suggest that an intermediate design, which in my baseline model corresponds to including a large bundle, may be optimal. In such a mechanism, applicants could either choose a specific development or give up choice in exchange for priority. This could be implemented in several ways. In a waitlist setting, applicants could choose between waiting for a particular development and joining a priority queue in which they cannot reject units offered to them. Alternatively, in a repeated lottery, they could choose between applying to a unit-specific lottery and entering a pooled lottery that assigns them randomly across several units. The evidence in Cook et al. (2023)—that lower-income households are less selective when applying for affordable housing—suggests that including such mixed options may be welfare-improving.

Finally, it is worth noting that my baseline model abstracts from observable characteristics that are important in many applications. For instance, social programs often condition priorities or eligibility on variables such as income or household size. While I do not model these explicitly, my framework could be extended to allow for such observable labels. Then, conditional on each observable label, the designer would face a screening problem of the kind studied here; across labels, these problems would be linked by the common supply constraints (see e.g. Akbarpour et al. (2024)). Although I do not pursue this extension here, a natural conjecture is that, under certain distributional conditions, the pure-option logic applies within each observable group and the resulting mechanism can be implemented as a competitive equilibrium with *unequal* incomes, where budgets depend on the observable label.

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A Waitlist interpretation

I now show that the baseline model from Section 2 can be interpreted as a reduced-form description of a steady-state waitlist setting.

Waitlist model. Consider a stationary waitlist environment in discrete time. There are N different kinds of goods indexed by $i \in \{1, \dots, N\}$ with $N \geq 2$. The designer possesses a fixed mass of each, with the supplies given by $s = (s_1, s_2, \dots, s_N) > 0$. In each period, a unit mass of new agents arrives; each of them has a profile of values $v = (v_1, v_2, \dots, v_N)$ for the goods. The values are private information and, in each arriving cohort, are distributed according to F with support on a bounded set $\mathcal{V} \subset \mathbb{R}_+^N$. This distribution puts no mass on $\mathbf{0}$ and assigns positive mass to agents with $v_i > 0$ for every good i . Agents remain in the system from one period to the next with probability $\delta \in [0, 1)$. The departure process is exogenous and independent of the agent's type, report and allocation history.

Agents may be assigned objects in every period after they arrive in the system. Let $\mathcal{A} := \{\emptyset, 1, \dots, N\}$ denote the set of feasible one-period assignments for an individual agent, where $\emptyset$ denotes receiving no good. A lifetime assignment path is an element $a = (a_0, a_1, a_2, \dots) \in \mathcal{A}^{\mathbb{N}_0}$, where a_t is the assignment made to the agent in the t -th period after entry, conditional on the agent still being in the system.

The utility of type v from a realized lifetime assignment path $a \in \mathcal{A}^{\mathbb{N}_0}$ is then $\sum_{t=0}^{\infty} \delta^t v_{a_t}$, with the convention $v_{\emptyset} := 0$. If the assignment path is random, with distribution $Z \in \Delta(\mathcal{A}^{\mathbb{N}_0})$, the agent's expected utility from it is

$$\mathbb{E}_{a \sim Z} \left[\sum_{t=0}^{\infty} \delta^t v_{a_t} \right].$$

Mechanisms. We restrict attention to stationary mechanisms that admit a steady state. That is, the same mechanism is applied to every entering cohort, and the induced cross-sectional distribution of active agents, assignment histories, and occupied goods is constant over time. Using stationarity and the Revelation Principle, we can assume that the designer selects a stationary allocation rule $z : \mathcal{V} \rightarrow \Delta(\mathcal{A}^{\mathbb{N}_0})$. She chooses it to maximize steady-state welfare

$$\max_z \int \mathbb{E}_{a \sim z(v)} \left[\sum_{t=0}^{\infty} \delta^t v_{a_t} \right] dF(v) \quad (\text{O''})$$

subject to incentive compatibility and supply constraints:

$$\mathbb{E}_{a \sim z(v)} \left[\sum_{t=0}^{\infty} \delta^t v_{a_t} \right] \geq \mathbb{E}_{a \sim z(v')} \left[\sum_{t=0}^{\infty} \delta^t v_{a_t} \right] \quad \text{for all } v, v', \quad (\text{IC''})$$

$$\sum_{t=0}^{\infty} \delta^t \int \mathbb{E}_{a \sim z(v)} [\mathbf{1}\{a_t = i\}] dF(v) \leq s_i \quad \text{for every } i. \quad (\text{S''})$$

The supply constraint says that, in the steady-state period, the total mass of agents assigned good i cannot exceed the stock s_i . To see why it takes this form, note that in any calendar period there is a unit mass of newly arrived agents, a mass δ of agents who entered one period earlier, a mass δ^2 of agents who entered two periods earlier, and so on. The sum therefore represents

the steady-state use of good i .¹⁶

Reduction to the static model. We now show that this dynamic problem reduces to the static model in Section 2. Given any stationary direct allocation rule $z : \mathcal{V} \rightarrow \Delta(\mathcal{A}^{\mathbb{N}_0})$, define the reduced-form allocation $y : \mathcal{V} \rightarrow \mathbb{R}_+^N$ by

$$y_i(v) := \mathbb{E}_{a \sim z(v)} \left[\sum_{t=0}^{\infty} \delta^t \mathbf{1}\{a_t = i\} \right].$$

Thus $y_i(v)$ is the expected survival-weighted number of periods in which an agent reporting v receives good i . By the convention $v_{\emptyset} = 0$, we have

$$\mathbb{E}_{a \sim z(v')} \left[\sum_{t=0}^{\infty} \delta^t v_{a_t} \right] = \sum_{i=1}^N v_i y_i(v') = v \cdot y(v').$$

Under this notation, (O''), (IC'') and (S'') reduce to (O), (IC), and (S). This also implies the following bound on the reduced-form allocation:

$$\sum_{i=1}^N y_i(v) \leq \frac{1}{1-\delta} \quad \text{for every } v,$$

because an agent can receive at most one good in any period. Thus the waitlist model maps directly into the baseline model with this additional individual feasibility constraint. This constraint, however, is analogous to the probability constraint discussed in Subsection 2.1 and therefore, by Proposition 1, imposing it is without loss when supply s is sufficiently scarce.

Remark 4. Under this reduction, distinct waitlist and lottery mechanisms can yield equivalent reduced-form allocations, in the spirit of Arnosti and Shi (2020). Consider, for example, a good-specific waitlist, in which each agent joins the waitlist for one good and receives it once she reaches the front, or a repeated choice-based lottery, in which each waiting agent chooses one good-specific lottery to enter in each period and receives that good if she wins. When supply is sufficiently scarce, both mechanisms implement reduced-form allocations that correspond to the pure-option mechanism in the baseline model.

B Omitted proofs

B.1 Technical preliminaries

B.1.1 Strassen's theorem.

Definition 9. Let $\geq$ be a partial order on Ω . A set $C \subseteq \Omega$ is an $\geq$ -upper set if $\theta \in C$, $\theta \leq \theta'$ implies $\theta' \in C$. A function $\eta : \Omega \rightarrow \mathbb{R}$ is $\geq$ -increasing if $\theta \leq \theta'$ implies $\eta(\theta') \geq \eta(\theta)$.

¹⁶As in the static unit-demand interpretation, the constraint is stated at the reduced-form aggregate level. Implementing the associated interim probabilities by individual randomizations while satisfying exact aggregate feasibility requires the usual exact-law-of-large-numbers argument; see footnote 2.

The following is a special case of Strassen's theorem stated in Fritz (2018):

Theorem 3 (Strassen (1965); Kellerer (1984); Edwards (1978)). *Let ρ, τ be measures on some $\Omega \subset \mathbb{R}^N$ with $\rho(\Omega) = \tau(\Omega)$ and let $\geq$ be a partial order on Ω such that the set $\{(x, y) \in \Omega \times \Omega : x \geq y\}$ is closed in $\Omega \times \Omega$. Then $\tau \geq$ -stochastically dominates ρ if and only if any of the following conditions holds:*

1. $\rho(C) \leq \tau(C)$ for every closed $\geq$ -upper set $C \subseteq \Omega$.
2. For every bounded, lower semicontinuous, $\geq$ -increasing $\eta : \Omega \rightarrow \mathbb{R}$,

$$\int_{\Omega} \eta d\rho \leq \int_{\Omega} \eta d\tau.$$

3. There exists a $\geq$ -monotone transport plan from ρ to τ .

B.1.2 Differential geometry facts. Let H denote the $(N - 1)$ -dimensional hyperplane containing the simplex Γ :

$$H := \{\theta \in \mathbb{R}^N : \sum \theta_i = 1\}.$$

Note that for every $\theta \in H$, the tangent space to H at any θ is:

$$TH := \{v \in \mathbb{R}^N : \sum v_i = 0\}.$$

Let us also define the intrinsic gradient for this surface:

Definition 10. Let $\eta : H \rightarrow \mathbb{R}$ and fix $\theta \in H$. The *intrinsic gradient* $\nabla_H \eta(\theta) \in TH$ is the unique vector such that:

$$D_v \eta(\theta) = \nabla_H \eta(\theta) \cdot v \quad \text{for all } v \in TH.$$

I use the following divergence theorem on the affine hyperplane H , which follows from Green's formula in $\mathbb{R}^{N-1}$ (see e.g. Rodrigues (1987)).

Theorem 4. Let $\Omega \subset H$ be a bounded, open set such that $\partial\Omega$ is Lipschitz. Let $\eta : \overline{\Omega} \rightarrow \mathbb{R}$ be Lipschitz. Fix a tangent vector field $X : \Omega \rightarrow \mathbb{R}^N$, $X(\theta) \in TH$, such that $X \in H^1(\overline{\Omega}; TH)$. Then:

$$\int_{\Omega} \nabla_H \eta(\theta) \cdot X(\theta) dV_H(\theta) + \int_{\Omega} \eta(\theta) \operatorname{div} X(\theta) dV_H(\theta) = \int_{\partial\Omega} \eta(\theta) X(\theta) \cdot \nu(\theta) dS_{\partial\Omega}(\theta), \quad (17)$$

where dV_H denotes the $(N - 1)$ -dimensional surface measure on H , $dS_{\partial\Omega}$ denotes the $(N - 2)$ -dimensional surface measure on $\partial\Omega$, and ν is the outward unit conormal along $\partial\Omega$. Finally, $\operatorname{div} X(\theta)$ is the divergence taken in the $(N - 1)$ -dimensional subsurface H .

B.2 Proof of Proposition 1

I show that, for $\eta > 0$ sufficiently small, every incentive-compatible allocation rule y that is feasible with supplies η s satisfies $y_i(v) < 1/N$ for all i and all v . Let

$$M := \sup\{v_j : v \in \mathcal{V}, j = 1, \dots, N\} < \infty.$$

Since each good is valued positively by a positive mass of agents, for every i there exists $\delta_i > 0$ such that

$$Z_i := F(\{v \in \mathcal{V} : v_i \geq \delta_i\}) > 0.$$

Also, define

$$\bar{\eta} := \min_i \frac{\delta_i Z_i}{4MN \sum_j s_j}.$$

Now, for $k \geq 0$, define

$$m(k) := F(\{v \in \mathcal{V} : \sum_j y_j(v) \geq k\}).$$

Since y is feasible for supplies ηs , it satisfies the supply constraint:

$$\int_{\mathcal{V}} \sum_j y_j(v) dF(v) \leq \sum_j \int_{\mathcal{V}} y_j(v) dF(v) \leq \eta \sum_j s_j.$$

Hence, by Markov's inequality, for $k > 0$, $m(k) \leq \frac{1}{k} \eta \sum_j s_j$, so $m(k) \rightarrow 0$ as $k \rightarrow \infty$. In particular, if we fix any i and set $\tilde{k}_i := 2\eta \sum_j s_j / Z_i$, we get $m(\tilde{k}_i) \leq Z_i/2$. Therefore the set $S := \{v \in \mathcal{V} : v_i \geq \delta_i, \sum_j y_j(v) \leq \tilde{k}_i\}$ has mass at least $Z_i - m(\tilde{k}_i) \geq Z_i/2 > 0$. For every $v \in S$ we then have:

$$v \cdot y(v) \leq M \sum_j y_j(v) \leq M \tilde{k}_i = M \cdot \frac{2\eta \sum_j s_j}{Z_i} \leq \frac{\delta_i}{2N}.$$

Now suppose toward a contradiction that there exists some $v' \in \mathcal{V}$ with $y_i(v') \geq 1/N$. Then every $v \in S$ would strictly profitably deviate, because for such v we have $v \cdot y(v) \leq \delta_i/(2N)$, while

$$v \cdot y(v') \geq v_i y_i(v') \geq \delta_i \cdot \frac{1}{N} = \frac{\delta_i}{N}.$$

Since i was arbitrary, it follows that $y_i(v) < 1/N$ for all i and all v . For all $v \in \mathcal{V}$ we then have $\sum y_i(v) < 1$, so (P) is slack everywhere.

B.3 Proof of Lemma 1

Consider any feasible allocation rule $y : \mathcal{V} \rightarrow \mathbb{R}_+^N$ in the original problem. By the boundedness argument above, y is uniformly bounded. Let $\tilde{x}$ be a measurable version of $\mathbb{E}[y(V) \mid \Theta = \theta]$. Define

$$K := \overline{\text{co}}\{y(v) : v \in \mathcal{V}\}, \quad \text{and, for each } \theta \in \Gamma, \quad \tilde{U}(\theta) := \max_{z \in K} \theta \cdot z.$$

Since K is compact, this is well-defined. Also, since the maximum of a linear function over K is the same as the supremum over $\{y(v) : v \in \mathcal{V}\}$, we have

$$\tilde{U}(\theta) = \sup_{v' \in \mathcal{V}} \theta \cdot y(v').$$

Moreover, if $v \neq \mathbf{0}$, incentive compatibility gives

$$\frac{v}{\sum_j v_j} \cdot y(v) \geq \frac{v}{\sum_j v_j} \cdot y(v') \quad \text{for all } v' \in \mathcal{V}.$$

Thus $\Theta \cdot y(V) = \tilde{U}(\Theta)$ a.s. Note also that since $\tilde{U}$ is the support function of a compact set, it is continuous.

We now modify $\bar{x}$ on a null set so that incentive compatibility holds pointwise. Let $D \subseteq \Gamma$ be a countable dense set. Since D is countable and $q \cdot y(V) \leq \tilde{U}(q)$ a.s. for each $q \in D$, we have, after passing to a common full-measure set, $q \cdot \bar{x}(\theta) \leq \tilde{U}(q)$ for a.e. θ and every $q \in D$. Also, since $\Theta \cdot y(V) = \tilde{U}(\Theta)$ a.s., we get $\theta \cdot \bar{x}(\theta) = \tilde{U}(\theta)$ for a.e. θ . Finally, since $y(V) \in K$ a.s. and K is closed and convex, we also have $\bar{x}(\theta) \in K$ for a.e. θ . Therefore, there is a Borel set $B \subseteq \Gamma$ with $G(B) = 1$ such that, for every $\theta \in B$,

$$\bar{x}(\theta) \in K, \quad \theta \cdot \bar{x}(\theta) = \tilde{U}(\theta), \quad q \cdot \bar{x}(\theta) \leq \tilde{U}(q) \quad \text{for every } q \in D.$$

Choose a measurable selection $z^*(\theta) \in \arg \max_{z \in K} \theta \cdot z$, and define $x : \Gamma \rightarrow \mathbb{R}_+^N$ by

$$x(\theta) := \begin{cases} \bar{x}(\theta), & \theta \in B, \\ z^*(\theta), & \theta \notin B. \end{cases}$$

Then $x(\theta) = \mathbb{E}[y(V) \mid \Theta = \theta]$ a.e. We now check pointwise incentive compatibility. First, $\theta \cdot x(\theta) = \tilde{U}(\theta)$ for every $\theta \in \Gamma$. Indeed, this holds by construction if $\theta \notin B$, and by the definition of B if $\theta \in B$. Next fix $\theta, \theta' \in \Gamma$. If $\theta' \notin B$, then $x(\theta') \in K$, so $\theta \cdot x(\theta') \leq \tilde{U}(\theta)$.

If instead $\theta' \in B$, choose $q_n \in D$ with $q_n \rightarrow \theta$. Then $q_n \cdot x(\theta') = q_n \cdot \bar{x}(\theta') \leq \tilde{U}(q_n)$ for every n . Passing to the limit and using continuity of $\tilde{U}$, we get $\theta \cdot x(\theta') \leq \tilde{U}(\theta) = \theta \cdot x(\theta)$.

We now verify the supply constraint. From the tower property:

$$\int_{\Gamma} x(\theta) dG(\theta) = \int_{\Gamma} \bar{x}(\theta) dG(\theta) = \mathbb{E}[\mathbb{E}[y(V) \mid \Theta]] = \mathbb{E}[y(V)] = \int_{\mathcal{V}} y(v) dF(v) \leq s.$$

Hence x is feasible in Problem 1.

Finally, since $V = (\sum_i V_i)\Theta$ and $\Theta \cdot y(V) = \tilde{U}(\Theta)$ a.s.,

$$\begin{aligned} \int_{\mathcal{V}} v \cdot y(v) dF(v) &= \mathbb{E}[V \cdot y(V)] \\ &= \mathbb{E}\left[\left(\sum_i V_i\right) \tilde{U}(\Theta)\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\sum_i V_i \mid \Theta\right] \tilde{U}(\Theta)\right] \\ &= \int_{\Gamma} \lambda(\theta) \tilde{U}(\theta) dG(\theta) = \int_{\Gamma} \lambda(\theta) \theta \cdot x(\theta) dG(\theta). \end{aligned}$$

Conversely, let x be feasible in Problem 1. Let $y(\mathbf{0}) = 0$ and $y(v) := x(v/\sum_i v_i)$ for $v \neq \mathbf{0}$. The supply constraint follows immediately from the definition of G . Incentive compatibility follows from (IC'): for $v \neq \mathbf{0}$ and $v' \neq \mathbf{0}$,

$$v \cdot y(v) = \left(\sum_i v_i \right) \frac{v}{\sum_i v_i} \cdot x \left(\frac{v}{\sum_i v_i} \right) \geq \left(\sum_i v_i \right) \frac{v}{\sum_i v_i} \cdot x \left(\frac{v'}{\sum_i v'_i} \right) = v \cdot y(v').$$

The cases involving $\mathbf{0}$ are trivial. Since F puts no mass on $\mathbf{0}$, the welfare equality follows from the definition of λ .

B.4 Proof of Theorem 1

For each $z \in [0, 1]$, let x^z denote the threshold allocation rule defined in (4). I first show that every incentive compatible allocation rule is a positive combination of threshold rules. Let $x : \Gamma \rightarrow \mathbb{R}_+^2$ be incentive compatible and write

$$U(1-t, t) := (1-t)x_1(1-t, t) + tx_2(1-t, t), \quad \Delta(t) := x_2(1-t, t) - x_1(1-t, t).$$

By the standard one-dimensional IC characterization, after changing x on a G -null set if necessary, Δ is nondecreasing and the envelope formula holds:

$$U(1-t, t) = U(1, 0) + \int_0^t \Delta(s) ds.$$

Choose a right-continuous version of Δ , and let μ be the finite positive measure on $[0, 1]$ satisfying $\mu((0, t]) = \Delta(t) - \Delta(0)$. Then

$$U(1-t, t) = U(1, 0) + t\Delta(0) + \int_0^t \mu((0, s]) ds.$$

By Tonelli's theorem,

$$\int_0^t \mu((0, s]) ds = \int_{(0, t]} (t-z) \mu(dz).$$

Hence

$$U(1-t, t) = U(1, 0) + t\Delta(0) + \int_{(0, t]} (t-z) \mu(dz).$$

Since $x_1(1-t, t) = U(1-t, t) - t\Delta(t)$ and $x_2(1-t, t) = U(1-t, t) + (1-t)\Delta(t)$ we obtain

$$x_1(1-t, t) = U(1, 0) - \int_{(0, t]} z \mu(dz) \quad \text{and} \quad x_2(1-t, t) = U(1, 0) + \Delta(0) + \int_{(0, t]} (1-z) \mu(dz).$$

Using $U(1, 0) = x_1(1, 0)$, $U(0, 1) = x_2(0, 1)$ and $\Delta(0) = x_2(1, 0) - x_1(1, 0)$, this becomes

$$x_1(1-t, t) = x_1(0, 1) + \int_{(t, 1]} z \mu(dz) \quad \text{and} \quad x_2(1-t, t) = x_2(1, 0) + \int_{[0, t]} (1-z) \mu(dz),$$

up to values at cutoff types. Since $x \geq 0$, we have $x_1(0, 1), x_2(1, 0) \geq 0$. Thus, defining

$$\nu := \mu + x_2(1, 0)\delta_0 + x_1(0, 1)\delta_1,$$

we obtain, almost surely,

$$x_1(1-t, t) = \int_{(t, 1]} z \nu(dz) \quad \text{and} \quad x_2(1-t, t) = \int_{[0, t]} (1-z) \nu(dz). \quad (18)$$

Equivalently,

$$x(1-t, t) = \int_{[0, 1]} x^z(1-t, t) \nu(dz) \quad (19)$$

up to values at cutoff types, including the endpoints $t = 0, 1$.

Conversely, suppose x is given by (18) for some finite positive measure ν . Then $x_1, x_2 \geq 0$, and

$$\Delta(t) := x_2(1-t, t) - x_1(1-t, t) = \nu([0, t]) - \int_{[0, 1]} z \nu(dz),$$

which is nondecreasing in t . Moreover,

$$U(1-t, t) = (1-t)x_1(1-t, t) + tx_2(1-t, t) = \int_{[0, 1]} \max\{z(1-t), (1-z)t\} \nu(dz),$$

and therefore

$$U(1-t, t) = U(1, 0) + \int_0^t \Delta(s) ds.$$

Hence the one-dimensional IC characterization implies that x satisfies (IC').

Thus the designer's problem can be rewritten as a linear program over positive measures ν . For each $z \in [0, 1]$, define

$$w(z) := \mathbb{E}[\lambda(\Theta) \max\{z\Theta_1, (1-z)\Theta_2\}], \quad \varphi(z) := (z \mathbb{P}(\Theta_2 \leq z), (1-z) \mathbb{P}(\Theta_2 \geq z)).$$

Applying Fubini's theorem to (19), we obtain

$$\int_{\Gamma} \lambda(\theta) U(\theta) dG(\theta) = \int_{[0, 1]} w(z) \nu(dz) \quad \text{and} \quad \int_{\Gamma} x(\theta) dG(\theta) = \int_{[0, 1]} \varphi(z) \nu(dz).$$

Therefore, the designer's problem is

$$W^* = \sup_{\nu \geq 0} \left\{ \int_{[0, 1]} w(z) \nu(dz) : \int_{[0, 1]} \varphi_1(z) \nu(dz) \leq s_1, \quad \int_{[0, 1]} \varphi_2(z) \nu(dz) \leq s_2 \right\}. \quad (20)$$

The functions w, φ_1, φ_2 are continuous. Moreover, $\varphi_1(z) + \varphi_2(z) > 0$ for every $z \in [0, 1]$, and hence is bounded away from zero on $[0, 1]$. Thus the feasible set of (20) has uniformly bounded total mass and is weak-* compact. Since the objective is weak-* continuous, an optimal measure

exists. The dual of (20) is

$$\inf_{c_1, c_2 \geq 0} \{c_1 s_1 + c_2 s_2 : c_1 \varphi_1(z) + c_2 \varphi_2(z) \geq w(z) \text{ for all } z \in [0, 1]\}.$$

By conic duality, strong duality holds (see e.g. Shapiro et al. (2001)). Let ν^* be a primal optimizer and let (c_1^*, c_2^*) be a dual optimizer. Both resource constraints bind at ν^* . Indeed, if the first constraint were slack, adding a small mass at $z = 1$ would use only good 1 and would strictly increase welfare, since $w(1) = \mathbb{E}[\lambda(\Theta)\Theta_1] > 0$. Similarly, if the second constraint were slack, adding a small mass at $z = 0$ would use only good 2 and would strictly increase welfare, since $w(0) = \mathbb{E}[\lambda(\Theta)\Theta_2] > 0$. Therefore

$$\int_{[0,1]} \varphi(z) \nu^*(dz) = s.$$

By strong duality,

$$\int_{[0,1]} w(z) \nu^*(dz) = c_1^* s_1 + c_2^* s_2 = \int_{[0,1]} (c_1^* \varphi_1(z) + c_2^* \varphi_2(z)) \nu^*(dz).$$

Since $c_1^* \varphi_1(z) + c_2^* \varphi_2(z) \geq w(z)$ for all z , it follows that ν^* is supported on the contact set

$$E := \{z \in [0, 1] : c_1^* \varphi_1(z) + c_2^* \varphi_2(z) = w(z)\}, \quad \text{and hence} \quad s \in \text{cone}\{\varphi(z) : z \in E\} \subset \mathbb{R}^2.$$

By Carathéodory's theorem for cones in $\mathbb{R}^2$, there exist $a, b \in E$, with $a \leq b$, and $m_a, m_b \geq 0$ such that $m_a \varphi(a) + m_b \varphi(b) = s$. Because $a, b \in E$, the measure $\hat{\nu} := m_a \delta_a + m_b \delta_b$ is feasible and satisfies

$$\int_{[0,1]} w(z) \hat{\nu}(dz) = \int_{[0,1]} (c_1^* \varphi_1(z) + c_2^* \varphi_2(z)) \hat{\nu}(dz) = c_1^* s_1 + c_2^* s_2 = W^*.$$

Thus there is an optimal measure supported on at most two points.

The induced allocation rule is therefore $x(1-t, t) = m_a x^a(1-t, t) + m_b x^b(1-t, t)$, so

$$x(1-t, t) = \begin{cases} (am_a + bm_b, 0), & t < a, \\ (bm_b, (1-a)m_a), & a \leq t < b, \\ (0, (1-a)m_a + (1-b)m_b), & t \geq b. \end{cases}$$

Thus the mechanism offers at most three options: a pure good-1 option, a bundle, and a pure good-2 option. If $a = b$, or if one of the weights is zero, the bundle is redundant, and the mechanism reduces to two pure options.

B.5 Proof of Corollary 1

By the proof of Theorem 1, the designer's problem can be written as:

$$W^* = \sup_{\nu \geq 0} \left\{ \int_{[0,1]} w(z) \nu(dz) : \int_{[0,1]} \varphi_1(z) \nu(dz) \leq s_1, \quad \int_{[0,1]} \varphi_2(z) \nu(dz) \leq s_2 \right\}$$

where $w(z) := \mathbb{E}[\lambda(\Theta) \max\{z\Theta_1, (1-z)\Theta_2\}]$. The dual of this problem is:

$$\inf_{c_1, c_2 \geq 0} \left\{ c_1 s_1 + c_2 s_2 : c_1 \varphi_1(z) + c_2 \varphi_2(z) \geq w(z) \quad \text{for all } z \in [0, 1] \right\}.$$

By the argument in the proof of Theorem 1, primal and dual optimizers exist and strong duality holds. Let $(\bar{c}_1, \bar{c}_2)$ be a dual optimizer. Evaluating the dual constraint at $z = 1$ and $z = 0$ gives

$$\bar{c}_1 \geq \mathbb{E}[\lambda(\Theta)\Theta_1] > 0, \quad \bar{c}_2 \geq \mathbb{E}[\lambda(\Theta)\Theta_2] > 0.$$

Define $\bar{\rho} := \bar{c}_1/\bar{c}_2$. Then the dual constraint is equivalent to

$$\bar{c}_2 \geq \frac{w(z)}{\bar{\rho} \varphi_1(z) + \varphi_2(z)} =: H_{\bar{\rho}}(z) \quad \text{for all } z \in [0, 1], \quad (21)$$

and hence $\bar{c}_2 \geq \max_{z \in [0, 1]} H_{\bar{\rho}}(z)$. Conversely, for any $\rho > 0$, setting $c_2 := \max_{z \in [0, 1]} H_{\rho}(z)$ and $c_1 := \rho c_2$ produces a dual-feasible pair. Therefore

$$W^* \leq (\rho s_1 + s_2) \max_{z \in [0, 1]} H_{\rho}(z) \quad \text{for all } \rho > 0.$$

Applying this inequality at $\rho = \bar{\rho}$, and using strong duality, $W^* = \bar{c}_1 s_1 + \bar{c}_2 s_2 = \bar{c}_2(\bar{\rho} s_1 + s_2)$, we obtain $\bar{c}_2 \leq \max_{z \in [0, 1]} H_{\bar{\rho}}(z)$. Together with (21), this gives

$$\bar{c}_2 = \max_{z \in [0, 1]} H_{\bar{\rho}}(z), \quad \text{and hence} \quad \bar{\rho} \in \arg \min_{\rho > 0} (\rho s_1 + s_2) \max_{z \in [0, 1]} H_{\rho}(z).$$

Now, let ρ^* be any minimizer in the statement, and define $c_2^* := \max_{z \in [0, 1]} H_{\rho^*}(z)$ and $c_1^* := \rho^* c_2^*$. The preceding argument shows that (c_1^*, c_2^*) is dual feasible. Since ρ^* minimizes $(\rho s_1 + s_2) \max_{z \in [0, 1]} H_{\rho}(z)$, we have

$$c_1^* s_1 + c_2^* s_2 = (\rho^* s_1 + s_2) \max_{z \in [0, 1]} H_{\rho^*}(z) = W^*.$$

Thus (c_1^*, c_2^*) is a dual optimizer associated with the chosen ρ^* .

We next note that every maximizer of H_{ρ} is interior. Since G has a strictly positive density on Γ , $\mathbb{P}(\Theta_2 \leq z) > 0$ for every $z > 0$ and $\mathbb{P}(\Theta_2 \geq z) > 0$ for every $z < 1$. Moreover, since the zero type was removed in the renormalization, $\lambda > 0$ G -almost surely.

Fix $\rho > 0$. If $0 < z < 1/(1 + \rho)$, then

$$\rho \varphi_1(z) + \varphi_2(z) = \rho z \mathbb{P}(\Theta_2 \leq z) + (1 - z) \mathbb{P}(\Theta_2 \geq z) < 1 - z.$$

Moreover,

$$w(z) - (1 - z)w(0) = \mathbb{E}[\lambda(\Theta)(z - \Theta_2)\mathbf{1}_{\{\Theta_2 \leq z\}}] > 0.$$

Hence

$$H_{\rho}(z) = \frac{w(z)}{\rho \varphi_1(z) + \varphi_2(z)} > \frac{(1 - z)w(0)}{1 - z} = H_{\rho}(0),$$

so 0 is not a maximizer. Similarly, if $1/(1+\rho) < z < 1$, then

$$\rho\varphi_1(z) + \varphi_2(z) = \rho z\mathbb{P}(\Theta_2 \leq z) + (1-z)\mathbb{P}(\Theta_2 \geq z) < \rho z,$$

and

$$w(z) - zw(1) = \mathbb{E}\left[\lambda(\Theta)(\Theta_2 - z)\mathbf{1}_{\{\Theta_2 \geq z\}}\right] > 0.$$

Therefore

$$H_\rho(z) = \frac{w(z)}{\rho\varphi_1(z) + \varphi_2(z)} > \frac{zw(1)}{\rho z} = H_\rho(1),$$

so 1 is not a maximizer. Thus $\arg \max_{z \in [0,1]} H_\rho(z) = \arg \max_{z \in (0,1)} H_\rho(z)$ for every $\rho > 0$.

Now define the contact set

$$E := \{z \in [0,1] : w(z) = c_1^* z \mathbb{P}(\Theta_2 \leq z) + c_2^* (1-z) \mathbb{P}(\Theta_2 \geq z)\}.$$

By the preceding paragraph, $E = \arg \max_{z \in (0,1)} H_{\rho^*} = \arg \max_{z \in [0,1]} H_{\rho^*}$. Let ν^* be a primal optimizer. By complementary slackness and the proof of Theorem 1, both resource constraints bind and ν^* is supported on E . Hence

$$s \in \text{cone}\{\varphi(z) : z \in E\} \subset \mathbb{R}^2. \quad (22)$$

By Carathéodory's theorem for cones in $\mathbb{R}^2$, there exist $a, b \in E$, with $a \leq b$, and $m_a, m_b \geq 0$ such that $m_a \varphi(a) + m_b \varphi(b) = s$. Now, note that $\frac{\varphi_2(z)}{\varphi_1(z)}$ is strictly decreasing on $(0,1)$. Therefore, for $a \leq b$,

$$s \in \text{cone}\{\varphi(a), \varphi(b)\} \iff \frac{\varphi_2(a)}{\varphi_1(a)} \geq \frac{s_2}{s_1} \geq \frac{\varphi_2(b)}{\varphi_1(b)}.$$

It follows from (22) that the set $\mathcal{P}$ defined in the statement is nonempty. Now fix any $(a, b) \in \mathcal{P}$. Since the ratio condition implies $s \in \text{cone}\{\varphi(a), \varphi(b)\}$, there exist $m_a, m_b \geq 0$ satisfying $m_a \varphi(a) + m_b \varphi(b) = s$. Because $a, b \in E$, the measure $\nu := m_a \delta_a + m_b \delta_b$ is primal feasible and satisfies complementary slackness:

$$\int_{[0,1]} w(z) \nu(dz) = \int_{[0,1]} (c_1^* \varphi_1(z) + c_2^* \varphi_2(z)) \nu(dz) = c_1^* s_1 + c_2^* s_2 = W^*.$$

Thus, the allocation induced by ν is optimal.

Finally, the allocation induced by $\nu = m_a \delta_a + m_b \delta_b$ is $x(1-t, t) = m_a a^a(t) + m_b a^b(t)$. For $a \leq b$, this equals

$$x(1-t, t) = \begin{cases} (am_a + bm_b, 0), & t < a, \\ (bm_b, (1-a)m_a), & a \leq t < b, \\ (0, (1-a)m_a + (1-b)m_b), & t \geq b. \end{cases}$$

Hence the corresponding menu quantities are

$$q_1 = a m_a + b m_b, \quad q_2 = (1-a) m_a + (1-b) m_b, \quad q_1^m = b m_b, \quad q_2^m = (1-a) m_a.$$

If $a = b$ or one of the weights is zero, the mechanism reduces to two pure options.

B.6 Proof of Corollary 2

Under exchangeability, if (c_1, c_2) is dual feasible in the proof of Corollary 1, then so is (c_2, c_1) . Since $s_1 = s_2$, averaging preserves feasibility and leaves the objective unchanged, so there exists an optimal pair with $c_1^* = c_2^*$. Hence $\rho^* = 1$. The symmetric two-option mechanism corresponds to the single threshold $z = 1/2$. Since exchangeability implies $\mathbb{P}(\Theta_2 \leq 1/2) = \mathbb{P}(\Theta_2 \geq 1/2) = \frac{1}{2}$, the supply-balance condition (22) is satisfied by $a = b = 1/2$ and $m = 4s_1$, which yields the two-option menu:

$$\{2s_1 \text{ of good 1}\}, \quad \{2s_1 \text{ of good 2}\}.$$

By Corollary 1, this mechanism is optimal if and only if $H_1(z) \leq H_1(1/2)$ for all $z \in [0, 1]$. By exchangeability, $H_1(z) = H_1(1 - z)$, so it is enough to consider $z \in [1/2, 1]$. Fix $z \in [1/2, 1]$, and let

$$A_z := \{\max_i \Theta_i \leq z\}.$$

Exchangeability of g and λ then gives:

$$\mathbb{E}[\lambda(\Theta) \max\{z\Theta_1, (1 - z)\Theta_2\}] = \frac{1}{2} \mathbb{E}[\lambda(\Theta)(z\mathbf{1}_{A_z} + \max_i \Theta_i \mathbf{1}_{A_z^c})].$$

Indeed, on A_z , the average utility across the two symmetric orientations is $z/2$, while on A_z^c it is $\max_i \Theta_i/2$. Similarly,

$$z\mathbb{P}(\Theta_2 \leq z) + (1 - z)\mathbb{P}(\Theta_2 \geq z) = z\mathbb{P}(A_z) + \frac{1}{2}\mathbb{P}(A_z^c) = \frac{1}{2}(1 + (2z - 1)\mathbb{P}(A_z)).$$

Therefore,

$$H_1(z) = \frac{\mathbb{E}[\lambda(\Theta)(z\mathbf{1}_{A_z} + \max_i \Theta_i \mathbf{1}_{A_z^c})]}{1 + (2z - 1)\mathbb{P}(A_z)}, \quad H_1(1/2) = \mathbb{E}\left[\lambda(\Theta) \max_i \Theta_i\right].$$

Hence, $H_1(z) \leq H_1(1/2)$ is equivalent to

$$\mathbb{E}\left[\lambda(\Theta)(z - \max_i \Theta_i)\mathbf{1}_{A_z}\right] \leq (2z - 1)\mathbb{E}\left[\lambda(\Theta) \max_i \Theta_i\right]\mathbb{P}(A_z).$$

If $\mathbb{P}(A_z) > 0$, dividing by $\mathbb{P}(A_z)$ gives exactly (6). For the sufficient condition, suppose $t\lambda(1 - t, t)$ is non-decreasing in t for $t \in [1/2, 1]$. On the event A_z , we have:

$$\lambda(\Theta)(z - \max_i \Theta_i) \leq (2z - 1)\lambda(\Theta) \max_i \Theta_i.$$

Taking conditional expectations yields

$$\mathbb{E}\left[\lambda(\Theta)(z - \max_i \Theta_i) \mid A_z\right] \leq (2z - 1)\mathbb{E}\left[\lambda(\Theta) \max_i \Theta_i \mid A_z\right]. \quad (23)$$

Now, by exchangeability, $\lambda(\Theta) \max_i \Theta_i = \max_i \Theta_i \lambda(1 - \max_i \Theta_i, \max_i \Theta_i)$, which is nondecreasing in $\max_i \Theta_i$ by assumption. Since conditioning on A_z restricts $\max_i \Theta_i$ to lower values, we obtain

$$\mathbb{E} \left[\lambda(\Theta) \max_i \Theta_i \mid A_z \right] \leq \mathbb{E} \left[\lambda(\Theta) \max_i \Theta_i \right]. \quad (24)$$

Combining (23) and (24) gives (6).

B.7 Proof of Fact 1

For any $q \in \mathbb{R}_{++}^N$, define:

$$x^q(\theta) \in \arg \max_{z \in \{q_1 e_1, \dots, q_N e_N\}} \theta \cdot z.$$

I now show that there exists a unique q for which $\int_{\Gamma} x_i(\theta) g(\theta) d\theta = s_i$ for all i , and thus that there exists a unique pure option mechanism.

Let us write $y_i := \log(1/q_i)$; then choosing an option i to maximize $\theta_i q_i$ is equivalent to choosing it to maximize $\log(\theta_i q_i) = \log \theta_i - y_i$. Thus, the sets of agents choosing each option are given by:

$$\Gamma_i(y) := \left\{ \theta \in \Gamma : \log \theta_i - y_i \geq \log \theta_j - y_j \text{ for all } j \right\}.$$

For any y , the induced aggregate demand for good i then equals $e^{-y_i} m_i(y)$, where $m_i(y) := \int_{\Gamma_i(y)} dG$. Thus, clearing is equivalent to $e^{-y_i} m_i(y) = s_i$ for all i .

Let us now define the potential $\Psi : \mathbb{R}^N \rightarrow \mathbb{R} \cup \{-\infty\}$, with the convention that $\log(0) = -\infty$:

$$\Psi(y) := \int_{\Gamma} \max_{j \in \{1, \dots, N\}} \{\log \theta_j - y_j\} g(\theta) d\theta + \sum_{j=1}^N s_j e^{y_j}. \quad (25)$$

We now show that $\Psi(y)$ is differentiable, and that the FOC $\nabla \Psi(y) = 0$ is equivalent to market clearing. For any y and $i \neq j$, the indifference set between i and j is

$$\{\theta : \log \theta_i - y_i = \log \theta_j - y_j\} = \{\theta : \theta_i = e^{y_i - y_j} \theta_j\},$$

which has measure zero. Hence, the maximizer is unique for a.e. agent. Thus, the map $y \mapsto \max_j \{\log \theta_j - y_j\}$ is differentiable a.e., so by Danskin's theorem and dominated convergence:

$$\frac{\partial \Psi}{\partial y_i} \int_{\Gamma} \max_j \{\log \theta_j - y_j\} g(\theta) d\theta = -m_i(y).$$

This gives $\frac{\partial \Psi}{\partial y_i}(y) = -m_i(y) + s_i e^{y_i}$ and so $\nabla \Psi(y) = 0$ is equivalent to $s_i e^{y_i} = m_i(y)$ for all i , i.e.

$$e^{-y_i} m_i(y) = s_i \quad \text{for all } i.$$

We now show Ψ is strictly convex. For each fixed θ , the map $y \mapsto \max_j \{\log \theta_j - y_j\}$ is the maximum of affine functions of y , hence convex. Since all $s_i > 0$, the second term $\sum_j s_j e^{y_j}$ is strictly

convex in y . Then, since Ψ is strictly convex and differentiable, it has at most one minimizer and the FOC holds there. It therefore remains to show that a minimizer indeed exists.

To that end, we show $\Psi(y) \rightarrow +\infty$ along any sequence with $\|y\| \rightarrow \infty$. If $y_i^n \rightarrow +\infty$, then $s_i e^{y_i^n} \rightarrow +\infty$, hence $\Psi(y^n) \rightarrow +\infty$. If $y_i^n \rightarrow -\infty$, fix $\delta \in (0, 1)$ and set $U_{i,\delta} := \{\theta \in \Gamma : \theta_i > \delta\}$. By full support, $G(U_{i,\delta}) > 0$. On $U_{i,\delta}$, $\max_j \{\log \theta_j - y_j^n\} \geq \log \theta_i - y_i^n \geq \log \delta - y_i^n$, so

$$\Psi(y^n) \geq \int_{\Gamma} \max_j \{\log \theta_j - y_j^n\} g(\theta) d\theta \geq G(U_{i,\delta})(\log \delta - y_i^n) \rightarrow +\infty.$$

Since Ψ is continuous and coercive on $\mathbb{R}^N$, it attains a minimum.

B.8 Proof of Fact 2

Let (p, x) be a CEEI. First, note $p_i > 0$ for every i : if $p_i = 0$, then every type with $\theta_i > 0$ would demand arbitrarily large amounts of good i , contradicting feasibility. Now define $q_i := \frac{1}{p_i}$. For any bundle $z \geq 0$ satisfying $p \cdot z \leq 1$,

$$\theta \cdot z = \sum_j \theta_j z_j \leq \left(\max_j \frac{\theta_j}{p_j} \right) \sum_{j=1}^N p_j z_j \leq \max_j \frac{\theta_j}{p_j} = \max_j \theta_j q_j.$$

This upper bound is attained by any pure option $q_i e_i$ with $i \in \arg \max_j \theta_j q_j$. Thus, for every type, at least one pure option is utility-maximizing in the CEEI demand problem, and for all but a null set of types this pure option is unique. Since the CEEI allocation clears the market, the vector $q = (q_1, \dots, q_N)$ is a market-clearing vector of pure options. By Fact 1, this vector is uniquely pinned down, and the induced allocation coincides with the pure option mechanism up to tie-breaking among a null set of types. The argument for the representative endowment economy is analogous.

Now, consider the choice-based lottery. In any pure-strategy Nash equilibrium, the mass m_i of agents choosing each good i must satisfy $m_i > 0$; otherwise any agent with $\theta_i > 0$ would deviate to good i and receive an infinite allocation. Each agent who chose good i receives $q_i := s_i/m_i$ of it. In equilibrium, every type θ must be choosing a good i that maximizes $\theta_i q_i$; otherwise, switching to a good j with $\theta_j q_j > \theta_i q_i$ would be profitable, since the mass of a single agent is zero and such a deviation does not change m_j . Hence the equilibrium allocation coincides with the pure option mechanism with quantities $q = (q_1, \dots, q_N)$: by Fact 1, this vector is uniquely pinned down, and the induced allocation assigns $q_i e_i$ to almost every type $\theta \in \Gamma_i$.

Conversely, let $q = (q_1, \dots, q_N)$ be the market-clearing vector of pure options. Setting $p_i = 1/q_i$ implements the pure option mechanism as a CEEI. Since market clearing gives $\sum_i s_i/q_i = 1$, the same prices also implement it as a Walrasian equilibrium of the representative endowment economy. Finally, in the choice-based lottery, if each type chooses a good maximizing $\theta_i q_i$, then $m_i = s_i/q_i$ and hence $s_i/m_i = q_i$. No type can profitably deviate, so this action profile is a pure-strategy Nash equilibrium.

B.9 Proof of Fact 4

It is enough to show that J is invertible and that $J^{-1}A$ is strictly positive. Define $k_i := c_i q_i$. Then $Jc = A$ is equivalent to $Hk = A$, where

$$H_{ii} = \frac{M_i}{q_i} + \sum_{j \neq i} T_{ij}, \quad H_{ij} = -T_{ij} \leq 0 \quad (i \neq j).$$

For each row i ,

$$H_{ii} - \sum_{j \neq i} |H_{ij}| = \frac{M_i}{q_i} > 0.$$

Thus H is a strictly diagonally dominant Z -matrix. Hence H is nonsingular and is a nonsingular M -matrix, so $H^{-1} \geq 0$ entrywise. Since $A > 0$, this gives

$$k = H^{-1}A \geq 0.$$

In fact $k > 0$. Indeed, $H^{-1} \geq 0$ and H^{-1} is invertible, so each row of H^{-1} contains at least one strictly positive entry. Since $A > 0$, every component of $H^{-1}A$ is strictly positive. Therefore $k_i > 0$ for every i . Finally, since $q_i > 0$,

$$c_i = \frac{k_i}{q_i} > 0.$$

B.10 Proof of Proposition 2

We verify the condition of Theorem 2 with $\tilde{\mu} = \mu$. The preprocessing condition then holds with equality. Moreover, by the balance condition (13), $\mu_i(\Gamma_i) = 0$, so $\mu_i^+(\Gamma_i) = \mu_i^-(\Gamma_i)$. Since the order relation $\succ_i$ is closed on Γ_i , Theorem 3 implies that it is enough to show $\mu_i(C) \geq 0$ for every closed $\succ_i$ -upper set $C \subseteq \Gamma_i$.

Under exchangeability and equal supplies, the pure option mechanism is symmetric, so $q_1 = \dots = q_N$ and hence $\theta^0 = \frac{1}{N} \mathbf{1}$. Also all A_i are equal; denote the common value by $\bar{A}$. Therefore the shadow costs satisfy $c = N\bar{A} \mathbf{1}$ and $\sum_j c_j = N^2 \bar{A}$. Plugging this into (7), for any Borel set $\Omega \subseteq \Gamma_i$,

$$\mu_i(\Omega) = \int_{\Omega} \lambda \theta_i g d\theta - N^2 \bar{A} \int_{\Omega} \theta_i [\operatorname{div}((\theta - \theta^0)g) + g] d\theta + N^2 \bar{A} \int_{\Omega \cap \partial \Gamma} \theta_i g (\theta - \theta^0) \cdot \nu d\sigma. \quad (26)$$

We first claim that for every $\succ_i$ -upper set $C \subseteq \Gamma_i$,

$$\int_C \lambda \theta_i g d\theta \geq N\bar{A} \int_C g d\theta. \quad (27)$$

Indeed, since $q_1 = \dots = q_N$, the set Γ_i coincides up to a null set with $\{\Theta_i \geq \Theta_j \forall j \neq i\}$. Also, by definition of $\succ_i$, every $\succ_i$ -upper set $C \subseteq \Gamma_i$ can be written as $C = \{\theta \in \Gamma_i : R_i(\theta) \in B\}$ for some $\geq$ -lower set $B \subseteq \mathbb{R}_+^N$. Since B^c is then $\geq$ -upper, the assumed stochastic monotonicity implies that for every $t \geq 0$,

$$\mathbb{P}[\Theta \in C \mid \lambda(\Theta)\Theta_i \geq t, \Theta \in \Gamma_i] \geq \mathbb{P}[\Theta \in C \mid \Theta \in \Gamma_i].$$

Integrating over t yields

$$\mathbb{E} \left[\lambda(\Theta) \Theta_i \mathbf{1}_{\{\Theta \in C\}} \mid \Theta \in \Gamma_i \right] \geq \mathbb{P}[\Theta \in C \mid \Theta \in \Gamma_i] \mathbb{E} \left[\lambda(\Theta) \Theta_i \mid \Theta \in \Gamma_i \right].$$

Equivalently,

$$\int_C \lambda \theta_i g d\theta \geq \frac{\int_{\Gamma_i} \lambda \theta_i g d\theta}{\int_{\Gamma_i} g d\theta} \int_C g d\theta.$$

Under exchangeability, $\int_{\Gamma_i} g d\theta = \frac{1}{N}$ and $\int_{\Gamma_i} \lambda \theta_i g d\theta = \bar{A}$, giving (27).

Next, let $C \subseteq \Gamma_i$ be a $>_i$ -upper set with Lipschitz boundary. Applying Theorem 4 to the tangent field $(\theta - \theta^0)g$ on C , and using $\nabla_H \theta_i = e_i - \frac{1}{N} \mathbf{1}$, we obtain

$$\int_C \theta_i \left[\operatorname{div}((\theta - \theta^0)g) + g \right] d\theta = \int_{\partial C} \theta_i (\theta - \theta^0) \cdot \nu_C g d\sigma + \frac{1}{N} \int_C g d\theta,$$

where ν_C is the outward unit conormal to ∂C . Substituting this into (26) gives

$$\mu_i(C) = \int_C \lambda \theta_i g d\theta - N \bar{A} \int_C g d\theta - N^2 \bar{A} \int_{\partial C \cap \Gamma^\circ} \theta_i (\theta - \theta^0) \cdot \nu_C g d\sigma.$$

By (27), the first two terms are weakly positive. It therefore suffices to show that $(\theta - \theta^0) \cdot \nu_C \leq 0$ for a.e. $\theta \in \partial C \cap \Gamma^\circ$. But if $\theta \in \Gamma_i \cap \Gamma^\circ$, then for every small $t > 0$ we have $\theta + t(\theta - \theta^0) >_i \theta$: indeed, for every $k \neq i$,

$$\frac{\theta_k + t(\theta_k - \theta_k^0)}{\theta_i + t(\theta_i - \theta_i^0)} \leq \frac{\theta_k}{\theta_i} \iff \frac{\theta_k^0}{\theta_i^0} \geq \frac{\theta_k}{\theta_i},$$

and this holds because $\theta \in \Gamma_i$ and $\theta^0 = \frac{1}{N} \mathbf{1}$. Since C is $>_i$ -upper, the ray from θ^0 through any boundary point of C stays in C locally beyond that point, so $\theta - \theta^0$ cannot point outward. Hence $(\theta - \theta^0) \cdot \nu_C \leq 0$ a.e. on $\partial C \cap \Gamma^\circ$, and therefore $\mu_i(C) \geq 0$.

We now extend this logic to all closed $>_i$ -upper sets using the following lemma:

Lemma 4. Fix i . Let $C \subseteq \Gamma_i$ be a closed $>_i$ -upper set. Then there exists a decreasing sequence $(K_m)_{m \geq 1}$ of closed $>_i$ -upper sets such that $K_{m+1} \subseteq K_m$, $\bigcap_{m \geq 1} K_m = C$, and each K_m is a finite union of polytopes in H defined by finitely many inequalities of the form $\theta_k \leq a \theta_i$ for $k \neq i$.

Proof. Define

$$Q_i : \Gamma_i \rightarrow \mathbb{R}_+^{N-1}, \quad Q_i(\theta) := \left(\frac{\theta_k}{\theta_i} \right)_{k \neq i}.$$

This map is injective on Γ_i , and $\theta' >_i \theta$ holds if and only if $Q_i(\theta') \leq Q_i(\theta)$ coordinatewise. Therefore $C \subseteq \Gamma_i$ is $>_i$ -upper if and only if $Q_i(C)$ is a $\geq$ -lower subset of $\mathbb{R}_+^{N-1}$. Since C is closed and Q_i is continuous, $Q_i(C)$ is compact.

Fix $m \geq 1$. Let

$$D_m := \bigcup \left\{ [0, b + \frac{1}{m} \mathbf{1}] : b \in \frac{1}{m} \mathbb{Z}^{N-1}, [0, b] \subseteq Q_i(C) \right\},$$

where $[0, b] := \{r \in \mathbb{R}_+^{N-1} : 0 \leq r \leq b\}$. Then D_m is a closed $\geq$ -lower set and a finite union of

rectangles. Also $Q_i(C) \subseteq D_m$: if $r \in Q_i(C)$, then because $Q_i(C)$ is lower, $[0, r] \subseteq Q_i(C)$. Choose $b \in \frac{1}{m}\mathbb{Z}^{N-1}$ with $b \leq r \leq b + \frac{1}{m}\mathbf{1}$; then $[0, b] \subseteq Q_i(C)$, so $r \in [0, b + \frac{1}{m}\mathbf{1}] \subseteq D_m$.

We claim that $Q_i(C) = \bigcap_{m \geq 1} D_m$. Since $Q_i(C) \subseteq D_m$ for all m , only the reverse inclusion needs proof. Take any $r \in \bigcap_m D_m$. For each m , choose $b_m \in \frac{1}{m}\mathbb{Z}^{N-1} \cap Q_i(C)$ such that $r \leq b_m + \frac{1}{m}\mathbf{1}$. By compactness of $Q_i(C)$, some subsequence of (b_m) converges to $\tilde{b} \in Q_i(C)$, and then $r \leq \tilde{b}$. Since $Q_i(C)$ is lower, this implies $r \in Q_i(C)$.

Now define $C_m := Q_i^{-1}(D_m)$ and set $K_m := \bigcap_{n=1}^m C_n$. Then $K_{m+1} \subseteq K_m$, each K_m is closed and $>_i$ -upper, and

$$\bigcap_{m \geq 1} K_m = \bigcap_{m \geq 1} C_m = Q_i^{-1}\left(\bigcap_{m \geq 1} D_m\right) = Q_i^{-1}(Q_i(C)) = C.$$

Finally, each C_m is a finite union of sets of the form

$$\left\{ \theta \in \Gamma_i : \frac{\theta_k}{\theta_i} \leq b_k \text{ for all } k \neq i \right\} = \left\{ \theta \in \Gamma_i : \theta_k \leq b_k \theta_i \text{ for all } k \neq i \right\},$$

hence a finite union of polytopes in H . The same is therefore true of each K_m . $\square$

Finally, let $C \subseteq \Gamma_i$ be any closed $>_i$ -upper set. By Lemma 4, there exists a decreasing sequence of closed $>_i$ -upper sets $K_m \downarrow C$, each a finite union of polytopes in H , hence each with Lipschitz boundary. By the previous paragraph, $\mu_i(K_m) \geq 0$ for every m . Since μ_i is finite, continuity from above gives $\mu_i(C) = \lim_{m \rightarrow \infty} \mu_i(K_m) \geq 0$. Thus condition 1. of Theorem 3 holds, so μ_i^+ $>_i$ -stochastically dominates μ_i^- . Since this is true for every i , Theorem 2 implies that the pure option mechanism is optimal.

B.11 Proof of Corollary 3

The i.i.d. assumption implies that the induced g and λ are exchangeable. Since supplies are symmetric, Proposition 2 applies once we verify its stochastic monotonicity condition.

Fix i , and write $E_i := \{V_j \geq V_i \text{ for all } j \neq i\}$. Conditional on $V_i = k$ and E_i , the coordinates $\{V_j : j \neq i\}$ are independent and each has distribution F_M truncated to $[0, k]$. Hence, for $j \neq i$ and $t \in (0, 1)$,

$$\mathbb{P}\left[\frac{V_j}{V_i} \geq t \mid V_i = k, E_i\right] = 1 - \frac{F_M(tk)}{F_M(k)}.$$

It is therefore enough to show that $F_M(tk)/F_M(k)$ is non-decreasing in k . Differentiating gives

$$\frac{\partial}{\partial k} \frac{F_M(tk)}{F_M(k)} = \frac{F_M(tk)}{F_M(k)} \left[t \frac{f_M(tk)}{F_M(tk)} - \frac{f_M(k)}{F_M(k)} \right].$$

The bracketed term can be written as

$$\frac{1}{k} \left[tk \frac{f_M(tk)}{F_M(tk)} - k \frac{f_M(k)}{F_M(k)} \right],$$

which is nonnegative by (8), since $tk \leq k$. Thus V_j/V_i is $\geq$ -stochastically decreasing in V_i conditional on E_i . Since the ratios are conditionally independent across $j \neq i$, the whole vector

$$\left(\frac{V_1}{V_i}, \dots, \frac{V_N}{V_i} \right)$$

is $\geq$ -stochastically decreasing in V_i conditional on E_i ; see Theorem 3.3.10 in Müller and Stoyan (2002).

Now, let $C \subseteq \Gamma_i$ be any $>_i$ -upper set. Since $\Theta_j/\Theta_i = V_j/V_i$, membership in C is determined by a coordinatewise lower set of the ratio vector $(V_1/V_i, \dots, V_N/V_i)$. The stochastic monotonicity just proved therefore implies that, for every $t \geq 0$,

$$\mathbb{P}[\Theta \in C \mid V_i \geq t, E_i] \geq \mathbb{P}[\Theta \in C \mid E_i].$$

Integrating over t gives

$$\mathbb{E}[V_i \mathbf{1}_{\{\Theta \in C\}} \mid E_i] \geq \mathbb{P}[\Theta \in C \mid E_i] \mathbb{E}[V_i \mid E_i].$$

Since E_i and C are determined by Θ and $\mathbb{E}[V_i \mid \Theta] = \Theta_i \mathbb{E}[\sum V_j \mid \Theta] = \lambda(\Theta)\Theta_i$, this gives

$$\int_C \lambda(\theta) \theta_i g(\theta) d\theta \geq \frac{\int_{\Gamma_i} \lambda(\theta) \theta_i g(\theta) d\theta}{\int_{\Gamma_i} g(\theta) d\theta} \int_C g(\theta) d\theta.$$

By exchangeability, $\int_{\Gamma_i} g(\theta) d\theta = 1/N$ and $\int_{\Gamma_i} \lambda(\theta) \theta_i g(\theta) d\theta = \bar{A}$, so

$$\int_C \lambda(\theta) \theta_i g(\theta) d\theta \geq N \bar{A} \int_C g(\theta) d\theta.$$

This is exactly the inequality needed in the proof of Proposition 2.

B.12 Proof of Proposition 3

I use a superscript $*$ for objects associated with the baseline primitives p^* . For objects associated with other primitive vectors, I write, for example, $\Gamma_i(p)$, $q(p)$, $c(p)$, and μ^p .

Let $\mathcal{S}_i(p)$ denote the collection of closed $>_i$ -upper sets $C \subseteq \Gamma_i(p)$. For p for which the vector of pure options $q(p)$ is well defined, define the region-identification map $\Phi_i^p : \Gamma_i^* \rightarrow \Gamma_i(p)$ by

$$\Phi_i^p(\theta) = \theta^p,$$

where

$$r_{ik}^p(\theta) := \frac{q_i(p)/q_k(p)}{q_i^*/q_k^*} \frac{\theta_k}{\theta_i}, \quad k \neq i, \quad \theta_k^p = \begin{cases} \frac{1}{1 + \sum_{l \neq i} r_{il}^p(\theta)}, & k = i, \\ r_{ik}^p(\theta) \frac{1}{1 + \sum_{l \neq i} r_{il}^p(\theta)}, & k \neq i. \end{cases}$$

Thus $\Phi_i^{p^*}$ is the identity on Γ_i^* , and Φ_i^p is a homeomorphism from Γ_i^* onto $\Gamma_i(p)$. The following fact comes directly from the definition:

Fact 5. *Let p be any primitive vector for which $q(p) \in \mathbb{R}_{++}^N$ is well defined. Then $\Phi_i^p : \Gamma_i^* \rightarrow \Gamma_i(p)$ is a homeomorphism and an order isomorphism:*

$$\theta \succ_i \theta' \iff \Phi_i^p(\theta) \succ_i \Phi_i^p(\theta'), \quad \text{and thus} \quad C \in \mathcal{S}_i^* \iff \Phi_i^p(C) \in \mathcal{S}_i(p).$$

Moreover, if $q(p_n) \rightarrow q^*$, then $\Phi_i^{p_n} \rightarrow \text{id}$ uniformly on Γ_i^* , and the $(N-1)$ -dimensional interior Jacobians and the $(N-2)$ -dimensional boundary Jacobians of $\Phi_i^{p_n}$ converge uniformly to 1.

Given a signed measure ν^p on Γ , define its pasted pullback $\hat{\nu}^p$ region by region, by requiring that for each i ,

$$\hat{\nu}^p \upharpoonright_{\Gamma_i^*}(C) := \nu^p(\Phi_i^p(C)), \quad C \subseteq \Gamma_i^* \text{ Borel.}$$

Equivalently, whenever $C \subseteq \Gamma_i^*$, the notation $\hat{\nu}^p(C)$ means this i -th regionwise pullback.

Lemma 5. *Assume that μ^* has upper-set slack with constant $\eta > 0$. Let ν^p be a regionwise balanced signed measure on Γ , and suppose that $\mu^* \ll \alpha$ and $\hat{\nu}^p \ll \alpha$ region by region. Define*

$$d_{\text{US}}^p(\nu^p, \mu^*) := \max_i \sup \left\{ \frac{|(\hat{\nu}^p - \mu^*)(C)|}{\min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\}} : C \in \mathcal{S}_i^*, \min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\} > 0 \right\}.$$

If $d_{\text{US}}^p(\nu^p, \mu^*) < \eta$, then, for every i and every $D \in \mathcal{S}_i(p)$, $\nu^p(D) \geq 0$.

Proof. Fix i and $D \in \mathcal{S}_i(p)$. By Fact 5, $C := (\Phi_i^p)^{-1}(D)$ belongs to $\mathcal{S}_i^*$, and $\nu^p(D) = \hat{\nu}^p(C)$.

If $\min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\} = 0$, then either $\alpha(C) = 0$ or $\alpha(\Gamma_i^* \setminus C) = 0$. In the first case, $\hat{\nu}^p \ll \alpha$ gives $\hat{\nu}^p(C) = 0$. In the second case, regionwise balance gives $\hat{\nu}^p(C) = -\hat{\nu}^p(\Gamma_i^* \setminus C) = 0$ again because $\hat{\nu}^p \ll \alpha$. Hence $\nu^p(D) \geq 0$.

Now suppose $\min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\} > 0$. By the definition of d_{US}^p ,

$$\hat{\nu}^p(C) \geq \mu^*(C) - d_{\text{US}}^p(\nu^p, \mu^*) \min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\}.$$

Using (BUS), we obtain

$$\hat{\nu}^p(C) \geq (\eta - d_{\text{US}}^p(\nu^p, \mu^*)) \min\{\alpha(C), \alpha(\Gamma_i^* \setminus C)\} \geq 0.$$

Since $\nu^p(D) = \hat{\nu}^p(C)$, the claim follows. $\square$

The next lemma gives a convenient sufficient condition for being close in the pulled-back upper-set topology.

Lemma 6. *Let ν^p be a signed measure on Γ that is balanced on each $\Gamma_i(p)$, and suppose that μ^* is balanced on each Γ_i^* . For each i , write*

$$m_i^* := m \upharpoonright_{\Gamma_i^*}, \quad \sigma_i^* := \sigma \upharpoonright_{\Gamma_i^* \cap \partial\Gamma}, \quad \alpha_i^* := m_i^* + \sigma_i^*.$$

Suppose that, region by region,

$$\hat{\nu}^p - \mu^* = a_i m_i^* + b_i \sigma_i^* \text{ on } \Gamma_i^*, \quad \text{with} \quad \max_i \left\{ \|a_i\|_{L^\infty(m_i^*)} + \|b_i\|_{L^\infty(\sigma_i^*)} \right\} \leq r.$$

Then $d_{\text{US}}^p(\nu^p, \mu^*) \leq r$.

Proof. Fix i and $C \in \mathcal{S}_i^*$. On Γ_i^* , define $\Delta := \hat{\nu}^p - \mu^*$. By the assumed density representation,

$$|\Delta(C)| \leq \|a_i\|_{L^\infty(m_i^*)} m_i^*(C) + \|b_i\|_{L^\infty(\sigma_i^*)} \sigma_i^*(C) \leq r \alpha_i^*(C).$$

Since ν^p and μ^* are regionwise balanced, $\hat{\nu}^p(\Gamma_i^*) = 0$ and $\mu^*(\Gamma_i^*) = 0$. Hence $\Delta(\Gamma_i^*) = 0$, so $\Delta(C) = -\Delta(\Gamma_i^* \setminus C)$. Applying the same density bound to $\Gamma_i^* \setminus C$ gives

$$|\Delta(C)| = |\Delta(\Gamma_i^* \setminus C)| \leq r \alpha_i^*(\Gamma_i^* \setminus C).$$

Combining the two bounds,

$$|\Delta(C)| \leq r \min\{\alpha_i^*(C), \alpha_i^*(\Gamma_i^* \setminus C)\}.$$

Since α_i^* is the restriction of α to Γ_i^* , this is exactly the bound appearing in d_{US}^p . Taking the supremum over $C \in \mathcal{S}_i^*$, and then the maximum over i , gives $d_{\text{US}}^p(\nu^p, \mu^*) \leq r$. $\square$

We now connect this measure-side topology to primitive perturbations.

Lemma 7. Consider primitives $p = (\lambda, g, s)$ written as

$$\lambda = e^l \lambda^*, \quad g = \frac{e^h g^*}{\int_{\Gamma} e^h g^* dm'},$$

and define

$$d_{\mathcal{P}}(p, p^*) := \|l\|_{C^1(\Gamma)} + \|h\|_{C^2(\Gamma)} + \|s - s^*\|.$$

Then for every $r > 0$, there exists $\varepsilon > 0$ such that, whenever $d_{\mathcal{P}}(p, p^*) < \varepsilon$, the objects

$$q(p), \quad c(p), \quad \Gamma_i(p), \quad \Phi_i^p, \quad \mu^p$$

are all well defined, $q(p) \in \mathbb{R}_{++}^N$, and $d_{\text{US}}^p(\mu^p, \mu^*) < r$.

Proof. By the implicit function theorem applied to the market-clearing equations, there is a $d_{\mathcal{P}}$ -neighborhood of p^* on which the pure-option vector $q(p)$ is uniquely and continuously defined. Shrinking the neighborhood if necessary, $q(p) \in \mathbb{R}_{++}^N$. Hence the regions $\Gamma_i(p)$ and the identification maps $\Phi_i^p : \Gamma_i^* \rightarrow \Gamma_i(p)$ are well defined.

Now let $p_n = (\lambda^n, g^n, s^n)$ be any sequence such that $d_{\mathcal{P}}(p_n, p^*) \rightarrow 0$. Then

$$\lambda^n \rightarrow \lambda^* \quad \text{in } C^1(\Gamma), \quad g^n \rightarrow g^* \quad \text{in } C^2(\Gamma), \quad s^n \rightarrow s^*. \quad (28)$$

By the local continuity of the market-clearing solution, $q(p_n) \rightarrow q^*$. Therefore the maps $\Phi_i^{p_n}$ converge uniformly to the identity on Γ_i^* , and their interior and boundary Jacobians converge uniformly to 1.

The quantities defining the shadow-cost system are continuous in the primitives. Indeed, after pulling all moving regions and faces back to the fixed baseline regions and faces by $\Phi_i^{p_n}$, the domains are fixed, the Jacobians converge uniformly to 1, and the integrands converge uniformly by (28). Thus, the shadow-cost matrices and right-hand sides converge: $J(p_n) \rightarrow J^*$ and $A(p_n) \rightarrow A^*$. Since J^* is nonsingular, $J(p_n)$ is nonsingular for all sufficiently large n , and

$$c(p_n) = J(p_n)^{-1} A(p_n) \rightarrow (J^*)^{-1} A^* = c^*.$$

For each i , write the restriction of the rent measure μ^p to $\Gamma_i(p)$ in density form as

$$d\mu^p = \rho_i(p) dm + \beta_i(p) d\sigma \quad \text{on } \Gamma_i(p),$$

where

$$\rho_i(p)(\theta) = \theta_i \left[\lambda(\theta)g(\theta) + \operatorname{div}_\Gamma \left((c(p) - (\sum_j c_j(p))\theta)g(\theta) \right) - (\sum_j c_j(p))g(\theta) \right],$$

and, on the boundary faces,

$$\beta_i(p)(\theta) = -\theta_i (c(p) - (\sum_j c_j(p))\theta)g(\theta) \cdot \nu(\theta).$$

Write $\hat{\mu}^{p_n}$ for the pasted pullback of μ^{p_n} . That is, on each baseline region Γ_i^* ,

$$\hat{\mu}^{p_n}(C) := \mu^{p_n}(\Phi_i^{p_n}(C)), \quad C \subseteq \Gamma_i^* \text{ Borel.}$$

By the area formula, on Γ_i^* ,

$$d\hat{\mu}^{p_n} = \hat{\rho}_i^n dm_i^* + \hat{\beta}_i^n d\sigma_i^*,$$

where

$$\hat{\rho}_i^n(\theta) = \rho_i(p_n)(\Phi_i^{p_n}(\theta))J_{i,\text{int}}^{p_n}(\theta), \quad \text{and, } \sigma_i^* \text{-a.e.,} \quad \hat{\beta}_i^n(\theta) = \beta_i(p_n)(\Phi_i^{p_n}(\theta))J_{i,\partial}^{p_n}(\theta).$$

Using the convergence of λ^n , g^n , $q(p_n)$, $c(p_n)$, $\Phi_i^{p_n}$, and the corresponding Jacobians, we obtain

$$\|\hat{\rho}_i^n - \rho_i(p^*)\|_{L^\infty(m_i^*)} \rightarrow 0, \quad \|\hat{\beta}_i^n - \beta_i(p^*)\|_{L^\infty(\sigma_i^*)} \rightarrow 0.$$

Hence

$$\hat{\mu}^{p_n} - \mu^* = a_i^n m_i^* + b_i^n \sigma_i^* \quad \text{on } \Gamma_i^*, \quad \text{with} \quad \max_i \left\{ \|a_i^n\|_{L^\infty(m_i^*)} + \|b_i^n\|_{L^\infty(\sigma_i^*)} \right\} \rightarrow 0.$$

Since rent measures are regionwise balanced, Lemma 6 implies $d_{\text{US}}^{p_n}(\mu^{p_n}, \mu^*) \rightarrow 0$. $\square$

We now complete the proof. Apply Lemma 7 with $r = \eta/2$. After shrinking $\varepsilon > 0$ if necessary,

every primitive vector p satisfying the displayed perturbation bound has

$$q(p), \quad c(p), \quad \Gamma_i(p), \quad \Phi_i^p, \quad \mu^p$$

well defined, with $q(p) \in \mathbb{R}_{++}^N$, and satisfies $d_{\text{US}}^p(\mu^p, \mu^*) < \eta/2$. By Lemma 5, it follows that, for every i and every closed $>_i$ -upper set $D \subseteq \Gamma_i(p)$, $\mu^p(D) \geq 0$. Now fix i , and write

$$\mu_i^p := \mu^p \upharpoonright_{\Gamma_i(p)}.$$

Since the rent measure is regionwise balanced, $\mu_i^p(\Gamma_i(p)) = 0$. Let $(\mu_i^p)^+$ and $(\mu_i^p)^-$ denote the positive and negative parts of μ_i^p . Then

$$(\mu_i^p)^+(\Gamma_i(p)) = (\mu_i^p)^-(\Gamma_i(p)).$$

Moreover, for every closed $>_i$ -upper set $D \subseteq \Gamma_i(p)$,

$$(\mu_i^p)^+(D) - (\mu_i^p)^-(D) = \mu_i^p(D) = \mu^p(D) \geq 0; \quad \text{equivalently,} \quad (\mu_i^p)^-(D) \leq (\mu_i^p)^+(D).$$

By Theorem 3, there exists a $>_i$ -monotone transport plan from $(\mu_i^p)^-$ to $(\mu_i^p)^+$. Since i was arbitrary, this verifies the regionwise transport condition in Theorem 2, with the auxiliary measure chosen to be $\tilde{\mu} = \mu^p$. Therefore, Theorem 2 applies to the primitives p , and the pure option mechanism associated with p is optimal.

B.13 Proof of Proposition 4

Fix such an allocation rule x ; note that Pareto-efficiency implies that supply constraints (S') bind for it. Indeed, if it were slack, we could give every agent a representative share of the remaining supply. Since almost all agents have strictly positive values for every good, this would produce a strict Pareto improvement while preserving (IC').

Next, we show that almost every type receives at most one kind of good. Suppose, towards a contradiction, that a positive mass of types receive a bundle. Then there exist $i \neq j$ such that

$$M := \{\theta \in \Gamma : x_i(\theta) > 0, x_j(\theta) > 0\}$$

has positive measure. Since G admits a density on Γ , the faces $\{\theta_i = 0\}$ and $\{\theta_j = 0\}$, as well as every level set $\{\theta_i/\theta_j = t\}$, have G -measure zero. Thus, after discarding a null subset of M , the ratio θ_i/θ_j is finite and is not concentrated at a single value, so there exists $t > 0$ such that both

$$M^- := M \cap \{\theta_i/\theta_j < t\}, \quad M^+ := M \cap \{\theta_i/\theta_j > t\}$$

have positive G -measure. Define

$$m^- := \int_{M^-} x_i(\theta) dG(\theta) > 0, \quad m^+ := \int_{M^+} x_j(\theta) dG(\theta) > 0.$$

Choose $\delta \in (0, 1]$ small enough that $t\delta \frac{m^-}{m^+} \leq 1$ and define $\tilde{x} : \Gamma \rightarrow \mathbb{R}_+^N$ by setting $\tilde{x}_k(\theta) = x_k(\theta)$ for

all $k \notin \{i, j\}$, and

$$(\tilde{x}_i(\theta), \tilde{x}_j(\theta)) = \begin{cases} ((1-\delta)x_i(\theta), x_j(\theta) + t\delta x_i(\theta)), & \theta \in M^-, \\ (x_i(\theta) + \delta \frac{m^-}{m^+} x_j(\theta), (1 - t\delta \frac{m^-}{m^+})x_j(\theta)), & \theta \in M^+, \\ (x_i(\theta), x_j(\theta)), & \theta \notin M^- \cup M^+. \end{cases}$$

This allocation rule is nonnegative by the choice of δ .

The aggregate use of goods i and j is unchanged because

$$\int_{\Gamma} (\tilde{x}_i - x_i) dG = -\delta m^- + \delta \frac{m^-}{m^+} m^+ = 0, \quad \int_{\Gamma} (\tilde{x}_j - x_j) dG = t\delta m^- - t\delta \frac{m^-}{m^+} m^+ = 0.$$

All other goods are unchanged, so $\tilde{x}$ satisfies the same supply constraints as x .

Finally, $\tilde{x}$ Pareto-dominates x . For $\theta \notin M^- \cup M^+$, utility is unchanged. For $\theta \in M^-$,

$$\theta \cdot (\tilde{x}(\theta) - x(\theta)) = -\delta \theta_i x_i(\theta) + t\delta \theta_j x_i(\theta) = \delta x_i(\theta)(t\theta_j - \theta_i) > 0,$$

because $\theta_i/\theta_j < t$ on M^- . Similarly, for $\theta \in M^+$,

$$\theta \cdot (\tilde{x}(\theta) - x(\theta)) = \delta \frac{m^-}{m^+} \theta_i x_j(\theta) - t\delta \frac{m^-}{m^+} \theta_j x_j(\theta) = \delta \frac{m^-}{m^+} x_j(\theta)(\theta_i - t\theta_j) > 0,$$

because $\theta_i/\theta_j > t$ on M^+ . This contradicts Pareto efficiency. Therefore, for every $i \neq j$,

$$G(\{\theta : x_i(\theta) > 0, x_j(\theta) > 0\}) = 0.$$

It follows that x is pure almost everywhere: for almost every θ , either $x(\theta) = 0$ or $x(\theta) = x_i(\theta)e_i$ for some i . For each i , define

$$S_i := \{\theta \in \Gamma : x(\theta) = x_i(\theta)e_i, x_i(\theta) > 0\}.$$

Since the i -th supply constraint binds and $s_i > 0$, each S_i has positive G -measure.

Since G assigns zero mass to the boundary faces, we may ignore types with $\theta_i = 0$ when comparing allocations within S_i . Now take any $\theta, \theta' \in S_i$. If $x_i(\theta) > x_i(\theta')$, then type θ' would strictly prefer to report θ , since

$$\theta' \cdot x(\theta) = \theta'_i x_i(\theta) > \theta'_i x_i(\theta') = \theta' \cdot x(\theta').$$

The reverse inequality is ruled out symmetrically.

Hence there exists $q_i^* > 0$ such that $x(\theta) = q_i^* e_i$ for a.e. $\theta \in S_i$. Define the induced cells $C_i := \{\theta \in \Gamma : \theta_i q_i^* \geq \theta_k q_k^* \text{ for all } k\}$. If $\theta \in S_i$, then incentive compatibility against reports in S_k implies

$$\theta_i q_i^* = \theta \cdot x(\theta) \geq \theta \cdot (q_k^* e_k) = \theta_k q_k^* \quad \text{for every } k.$$

Thus $S_i \subseteq C_i$.

Conversely, consider the strict part of C_i ,

$$C_i^\circ := \{\theta \in \Gamma : \theta_i q_i^* > \theta_k q_k^* \text{ for all } k \neq i\}.$$

For almost every $\theta \in C_i^\circ$, the allocation $x(\theta)$ is pure or zero. Such a type must receive $q_i^* e_i$. Indeed, if it received zero, or if it received $q_k^* e_k$ for some $k \neq i$, then reporting any type in S_i would give utility $\theta_i q_i^*$, which is strictly larger than its utility from the allocation assigned to it. This violates incentive compatibility. Therefore, $x(\theta) = q_i^* e_i$ for almost every $\theta \in C_i^\circ$. The boundaries between the cells C_i have G -measure zero, since they are contained in finitely many hyperplanes of the form $\{\theta : \theta_i q_i^* = \theta_j q_j^*\}$. Hence x coincides almost everywhere with the pure-option allocation that gives $q_i^* e_i$ on C_i° . Since all supply constraints bind, this pure-option allocation clears markets:

$$s_i = \int_{\Gamma} x_i(\theta) dG(\theta) = q_i^* G(C_i) \quad \text{for every } i.$$

Thus $q^* = (q_1^*, \dots, q_N^*)$ is a market-clearing vector of pure options. By Fact 1, the market-clearing vector of pure options is unique. Hence $q^* = q$, where q is the pure-option vector. Moreover, the cells C_i° coincide with the corresponding pure-option regions Γ_i° .

B.14 Proof of Corollary 4

Let μ and μ' denote the rent measures associated with (s, g, λ) and $(s, g, \tilde{\lambda})$, respectively. Since g and s are unchanged, the pure option mechanism and hence θ^0 and the regions Γ_i are unchanged. Moreover, the existence of a $\succ_i$ -monotone transport plan from v_i to $\tilde{v}_i$ implies

$$\int_{\Gamma_i} \theta_i \tilde{\lambda} g d\theta = \int_{\Gamma_i} \theta_i \lambda g d\theta \quad \text{for every } i,$$

so the vector A , and hence the shadow costs $c = J^{-1}A$, are also unchanged.

Define the signed measure $\Delta := \mu' - \mu$. Inspecting the definition of the rent measure, the only term that changes when λ is replaced by $\tilde{\lambda}$ is the first one. Hence, for every measurable $\Omega \subseteq \Gamma_i$,

$$\Delta_i(\Omega) = \mu'_i(\Omega) - \mu_i(\Omega) = \int_{\Omega} (\tilde{\lambda} - \lambda) \theta_i g d\theta = \tilde{v}_i(\Omega) - v_i(\Omega).$$

Thus $\Delta_i(\Gamma_i) = 0$. Also, if $C \subseteq \Gamma_i$ is a closed $\succ_i$ -upper set, then any $\succ_i$ -monotone transport plan from v_i to $\tilde{v}_i$ can only move mass into C , not out of it. Therefore $v_i(C) \leq \tilde{v}_i(C)$, and hence $\Delta_i(C) \geq 0$.

Now let $\tilde{\mu}$ be a certificate for (s, g, λ) in Theorem 2. We claim that $\tilde{\mu} + \Delta$ is a certificate for $(s, g, \tilde{\lambda})$. First, for every continuous, convex $U : \Gamma \rightarrow \mathbb{R}_+$,

$$\int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu'(\theta) = \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\mu(\theta) + \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\Delta(\theta) \leq \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\tilde{\mu}(\theta) + \int_{\Gamma} \frac{U(\theta)}{\theta_*(\theta)} d\Delta(\theta),$$

where the inequality uses the preprocessing condition for $\tilde{\mu}$. Thus the preprocessing condition holds for $\tilde{\mu} + \Delta$.

It remains to verify regionwise transport. Fix i . Since $\tilde{\mu}$ is a certificate, the second condition of Theorem 2 gives a $>_i$ -monotone transport plan from $\tilde{\mu}_i^-$ to $\tilde{\mu}_i^+$. Equivalently, by Theorem 3, this implies $\tilde{\mu}_i(C) \geq 0$ for every closed $>_i$ -upper set $C \subseteq \Gamma_i$. Combining this with $\Delta_i(C) \geq 0$, we get

$$(\tilde{\mu}_i + \Delta_i)(C) \geq 0$$

for every closed $>_i$ -upper set $C \subseteq \Gamma_i$. Also, $\tilde{\mu}_i(\Gamma_i) = 0$ because $\tilde{\mu}_i^-$ can be transported to $\tilde{\mu}_i^+$, and $\Delta_i(\Gamma_i) = 0$ by normalization. Hence $(\tilde{\mu}_i + \Delta_i)(\Gamma_i) = 0$. Applying Theorem 3 again, there exists a $>_i$ -monotone transport plan from $(\tilde{\mu}_i + \Delta_i)^-$ to $(\tilde{\mu}_i + \Delta_i)^+$. Thus $\tilde{\mu} + \Delta$ satisfies the regionwise transport condition. Therefore, by Theorem 2, the pure option mechanism is optimal for $(s, g, \tilde{\lambda})$.

B.15 Proof of Proposition 5

Necessity has been shown in the main body. Let us then show sufficiency. We first show the following lemma.

Lemma 8. *Let $U : \Gamma \rightarrow \mathbb{R}_+$ be convex and satisfy (R). At every $\theta \in \Gamma^\circ$ where $\nabla_H U(\theta)$ exists, define:*

$$x_U(\theta) := \nabla_H U(\theta) - \mathbf{1}(\nabla_H U(\theta) \cdot \theta - U(\theta)).$$

Then:

1. $x_U(\theta) \in \mathbb{R}_+^N$, x_U is uniformly bounded, and $U(\theta) = \theta \cdot x_U(\theta)$ almost everywhere.
2. U is Lipschitz on Γ .

Proof. Since U is finite and convex on Γ , it is locally Lipschitz on Γ° and differentiable almost everywhere on Γ° . Fix a differentiability point $\theta \in \Gamma^\circ$. First note that

$$\theta \cdot x_U(\theta) = \theta \cdot \nabla_H U(\theta) - (\theta \cdot \mathbf{1})(\nabla_H U(\theta) \cdot \theta - U(\theta)) = U(\theta),$$

because $\theta \cdot \mathbf{1} = 1$. We now show that $x_U(\theta) \geq 0$. Fix k . Choose some $i \neq k$, and for $\varepsilon \in (0, 1)$, define

$$\theta^\varepsilon := (1 - \varepsilon)\theta + \varepsilon e_k.$$

Then $\theta >_i \theta^\varepsilon$. Indeed, for $l \neq i, k$,

$$\frac{\theta_l^\varepsilon}{\theta_i^\varepsilon} = \frac{\theta_l}{\theta_i}, \quad \text{while} \quad \frac{\theta_k^\varepsilon}{\theta_i^\varepsilon} = \frac{(1 - \varepsilon)\theta_k + \varepsilon}{(1 - \varepsilon)\theta_i} \geq \frac{\theta_k}{\theta_i}.$$

Therefore (R) implies $\frac{U(\theta^\varepsilon)}{\theta_i^\varepsilon} \geq \frac{U(\theta)}{\theta_i}$. Since $\theta_i^\varepsilon = (1 - \varepsilon)\theta_i$, this gives $U(\theta^\varepsilon) \geq (1 - \varepsilon)U(\theta)$, and hence

$$\frac{U(\theta^\varepsilon) - U(\theta)}{\varepsilon} \geq -U(\theta).$$

Letting $\varepsilon \downarrow 0$, and using differentiability of U at θ , gives $D_{e_k-\theta}U(\theta) + U(\theta) \geq 0$, but

$$D_{e_k-\theta}U(\theta) + U(\theta) = \nabla_H U(\theta) \cdot (e_k - \theta) + U(\theta) = (x_U)_k(\theta). \quad (29)$$

Therefore $(x_U)_k(\theta) \geq 0$.

We next prove Lipschitzness. Let $M_U := \max_k U(e_k) < \infty$. At every differentiability point $\theta \in \Gamma^\circ$, convexity along the segment from θ to e_k gives $D_{e_k-\theta}U(\theta) \leq U(e_k) - U(\theta)$. Using (29) gives:

$$(x_U)_k(\theta) = D_{e_k-\theta}U(\theta) + U(\theta) \leq U(e_k) \leq M_U.$$

Together with nonnegativity, this gives $0 \leq (x_U)_k(\theta) \leq M_U$ for every k at every differentiability point $\theta \in \Gamma^\circ$. Now, take any tangent vector $v \in TH$. Since $v \cdot \mathbf{1} = 0$,

$$D_v U(\theta) = \nabla_H U(\theta) \cdot v = x_U(\theta) \cdot v.$$

Therefore, at every differentiability point, $|D_v U(\theta)| \leq M_U \|v\|_1$. It follows that U is M_U -Lipschitz on Γ° with respect to the ℓ^1 -norm: for any $\theta, \theta' \in \Gamma^\circ$, integrating the a.e. derivative of U along the segment from θ' to θ gives

$$|U(\theta) - U(\theta')| \leq M_U \|\theta - \theta'\|_1.$$

It remains to extend this Lipschitz bound to the boundary. Fix $\theta \in \partial\Gamma$, and choose i such that $\theta_i > 0$. Define

$$\zeta_i = 0, \quad \zeta_j = \frac{1}{N-1} \quad \text{for } j \neq i, \quad \text{and} \quad \theta^\varepsilon := (1-\varepsilon)\theta + \varepsilon\zeta.$$

Then $\theta^\varepsilon \in \Gamma^\circ$ for every $\varepsilon \in (0, 1)$, and $\theta \succ_i \theta^\varepsilon$. Hence (R) gives $\frac{U(\theta^\varepsilon)}{\theta_i^\varepsilon} \geq \frac{U(\theta)}{\theta_i}$. Since $\theta_i^\varepsilon = (1-\varepsilon)\theta_i$, we get $U(\theta^\varepsilon) \geq (1-\varepsilon)U(\theta)$. On the other hand, convexity gives

$$U(\theta^\varepsilon) \leq (1-\varepsilon)U(\theta) + \varepsilon U(\zeta).$$

Therefore, $U(\theta^\varepsilon) \rightarrow U(\theta)$. Now, let $\eta^m \in \Gamma^\circ$ be any sequence with $\eta^m \rightarrow \theta$. Choose any sequence $\varepsilon_m \downarrow 0$. Since both η^m and θ^{ε_m} lie in Γ° , the interior Lipschitz bound gives

$$|U(\eta^m) - U(\theta^{\varepsilon_m})| \leq M_U \|\eta^m - \theta^{\varepsilon_m}\|_1 \rightarrow 0.$$

Since $U(\theta^{\varepsilon_m}) \rightarrow U(\theta)$, it follows that $U(\eta^m) \rightarrow U(\theta)$. Thus U is continuous on Γ . Finally, take arbitrary $\theta, \theta' \in \Gamma$. Choose sequences $\theta^m, \theta'^m \in \Gamma^\circ$ such that $\theta^m \rightarrow \theta$ and $\theta'^m \rightarrow \theta'$. By the interior Lipschitz bound,

$$|U(\theta^m) - U(\theta'^m)| \leq M_U \|\theta^m - \theta'^m\|_1.$$

Passing to the limit using continuity gives

$$|U(\theta) - U(\theta')| \leq M_U \|\theta - \theta'\|_1.$$

Hence U is Lipschitz on all of Γ . □

Assume that $U : \Gamma \rightarrow \mathbb{R}_+$ is convex and satisfies (R). Let $\mathcal{D} \subseteq \Gamma^\circ$ denote the set of points at which

$\nabla_H U$ exists. For $\theta \in \mathcal{D}$, define $x_U(\theta)$ by (9). By Lemma 8, there is some $M_U > 0$ such that

$$0 \leq (x_U)_k(\theta) \leq M_U \quad \text{for every } k, \theta \in \mathcal{D}$$

and U is Lipschitz on Γ . We now extend x_U to all of Γ . For each $\theta \in \Gamma$, let

$$X(\theta) := \left\{ z \in \mathbb{R}_+^N : \begin{array}{l} \text{there exists a sequence } \theta^m \in \mathcal{D} \text{ such that } \theta^m \rightarrow \theta \\ \text{and } x_U(\theta^m) \rightarrow z \end{array} \right\}.$$

This set is nonempty because $\mathcal{D}$ is dense in Γ , and it is compact because $x_U(\mathcal{D}) \subseteq [0, M_U]^N$. The graph of the correspondence X is closed, and its values are nonempty compact subsets of $[0, M_U]^N$. Hence, by the measurable selection theorem, there exists a Borel measurable selection $x(\theta) \in X(\theta)$. Then $x(\theta) \in \mathbb{R}_+^N$ for every $\theta \in \Gamma$.

We claim that this allocation rule implements U . Fix $\theta \in \Gamma$, and take a sequence $\theta^m \in \mathcal{D}$ such that $\theta^m \rightarrow \theta$ and $x_U(\theta^m) \rightarrow x(\theta)$. Since $U(\theta^m) = \theta^m \cdot x_U(\theta^m)$ for every m , and since U is continuous, passing to the limit gives

$$U(\theta) = \theta \cdot x(\theta). \quad (30)$$

We now verify incentive compatibility. To that end, define $p(\theta) := x(\theta) - \mathbf{1} \frac{1}{N} \sum x_j(\theta)$. We show:

$$U(\theta') \geq U(\theta) + p(\theta) \cdot (\theta' - \theta) \quad \text{for every } \theta' \in \Gamma. \quad (31)$$

Indeed, for each m , set

$$p^m := x_U(\theta^m) - \mathbf{1} \frac{1}{N} \sum (x_U)_j(\theta^m).$$

Because $\nabla_H U(\theta^m) \in TH$, the definition of x_U implies $p^m = \nabla_H U(\theta^m)$. Hence, by convexity,

$$U(\theta') \geq U(\theta^m) + p^m \cdot (\theta' - \theta^m) \quad \text{for every } \theta' \in \Gamma.$$

Since $x_U(\theta^m) \rightarrow x(\theta)$, we have $p^m \rightarrow p(\theta)$. Passing to the limit gives (31).

Now, fix $\theta, \theta' \in \Gamma$. Using (30) at θ' , we have

$$\theta \cdot x(\theta') = U(\theta') + x(\theta') \cdot (\theta - \theta').$$

Since $\theta - \theta' \in TH$, the component of $x(\theta')$ parallel to $\mathbf{1}$ drops out, so

$$x(\theta') \cdot (\theta - \theta') = p(\theta') \cdot (\theta - \theta').$$

By (31), applied at θ' ,

$$U(\theta) \geq U(\theta') + p(\theta') \cdot (\theta - \theta').$$

Combining the last three statements gives $U(\theta) \geq \theta \cdot x(\theta')$. Using again (30), this is exactly $\theta \cdot x(\theta) \geq \theta \cdot x(\theta')$ for all $\theta, \theta' \in \Gamma$.

B.16 Proof of Proposition 6

For $U \in \mathcal{U}$, write

$$\Phi(U) := \int_{\Gamma} \lambda U g d\theta, \quad \Pi(U) := \int_{\Gamma} x_U g d\theta.$$

Recall that by definition, $\Pi(U_{\text{pure}}) = s$. By Fact 4, $c = J^{-1}A \in \mathbb{R}_{++}^N$. First suppose U_{pure} solves the shadow-cost problem (12). Let $U \in \mathcal{U}$ be feasible for (10). Then $\Pi(U) \leq s$. Since $c \geq 0$,

$$\Phi(U) \leq \Phi(U) - c \cdot (\Pi(U) - s).$$

By optimality of U_{pure} in (12),

$$\Phi(U) - c \cdot (\Pi(U) - s) \leq \Phi(U_{\text{pure}}) - c \cdot (\Pi(U_{\text{pure}}) - s) = \Phi(U_{\text{pure}}).$$

Therefore $\Phi(U) \leq \Phi(U_{\text{pure}})$ for every feasible U , so U_{pure} solves (10).

Conversely, suppose U_{pure} solves (10). Define

$$P(r) := \sup_{U \in \mathcal{U}} \{\Phi(U) : \Pi(U) \leq r\}, \quad r \in \mathbb{R}_+^N.$$

The function P is finite, increasing, and concave. Indeed, if $\Pi(U) \leq r$, then

$$0 \leq U(\theta) = \theta \cdot x_U(\theta) \leq \sum_i (x_U)_i(\theta) \quad \text{a.e.,}$$

so, writing $\bar{\lambda}$ for an upper bound on $\lambda(\theta)$,

$$\Phi(U) \leq \bar{\lambda} \int_{\Gamma} U dG \leq \bar{\lambda} \sum_i \Pi_i(U) \leq \bar{\lambda} \sum_i r_i.$$

Thus $P(r) < \infty$. Monotonicity is immediate. Concavity follows from the convexity of $\mathcal{U}$ and the identity

$$x_{tU+(1-t)V} = tx_U + (1-t)x_V \quad \text{a.e.}$$

Since $s \in \mathbb{R}_{++}^N$, s is an interior point of $\mathbb{R}_+^N$. Hence the finite concave function P has a supergradient at s . Thus there exists $c^* \in \mathbb{R}^N$ such that

$$P(r) \leq P(s) + c^* \cdot (r - s) \quad \text{for all } r \in \mathbb{R}_+^N.$$

Because P is increasing, necessarily $c^* \geq 0$.

Since U_{pure} solves (10) and $\Pi(U_{\text{pure}}) = s$, we have $P(s) = \Phi(U_{\text{pure}})$. Now fix any $U \in \mathcal{U}$, and set $r := \Pi(U)$. Then

$$\Phi(U) \leq P(r) \leq P(s) + c^* \cdot (r - s).$$

Hence

$$\Phi(U) - c^* \cdot (\Pi(U) - s) \leq \Phi(U_{\text{pure}}).$$

Using $\Pi(U_{\text{pure}}) = s$, this becomes

$$\Phi(U) - c^* \cdot (\Pi(U) - s) \leq \Phi(U_{\text{pure}}) - c^* \cdot (\Pi(U_{\text{pure}}) - s).$$

Thus U_{pure} maximizes the Lagrangian with multiplier c^* .

It remains to identify this multiplier. To that end, we show the following lemma.

Lemma 9. For $q' \in \mathbb{R}_{++}^N$, define $U_{q'}(\theta) := \max_i \theta_i q'_i$. Then, at q , for every $h \in \mathbb{R}^N$,

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\Gamma} \lambda U_{q+th} g d\theta = A \cdot h \quad \text{and} \quad \left. \frac{d}{dt} \right|_{t=0} \int_{\Gamma} (x_{U_{q+th}})_j g d\theta = \sum_i J_{ij} h_i \quad \text{for every } j.$$

Proof. Because indifference boundaries have G -measure zero, Danskin's theorem gives

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\Gamma} \lambda U_{q+th} g d\theta = \sum_i h_i \int_{\Gamma_i} \theta_i \lambda g d\theta = A \cdot h.$$

For the resource derivative, note that

$$\int_{\Gamma} (x_{U_{q'}})_j g d\theta = q'_j G(\Gamma_j(q')).$$

The direct effect of changing q_j is M_j . The remaining effect comes from boundary movements. Along $\Gamma_i \cap \Gamma_j$, the separating surface is $q_i \theta_i = q_j \theta_j$. Its intrinsic normal has norm

$$\|\nabla_H(q_i \theta_i - q_j \theta_j)\| = \sqrt{q_i^2 + q_j^2 - \frac{1}{N}(q_i - q_j)^2}.$$

Therefore,

$$\frac{\partial}{\partial q_j} \int_{\Gamma} (x_{U_{q'}})_j g d\theta = M_j + q_j \sum_{i \neq j} T_{ji} = J_{jj} \quad \text{and, for } i \neq j, \quad \frac{\partial}{\partial q_i} \int_{\Gamma} (x_{U_{q'}})_j g d\theta = -q_j T_{ij} = J_{ij}.$$

Thus

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\Gamma} (x_{U_{q+th}})_j g d\theta = \sum_i J_{ij} h_i.$$

□

Let $h \in \mathbb{R}^N$. For all sufficiently small t , the perturbation

$$U_{q+th}(\theta) := \max_i \theta_i (q_i + th_i)$$

is admissible in $\mathcal{U}$. Therefore the function

$$t \mapsto \Phi(U_{q+th}) - c^* \cdot (\Pi(U_{q+th}) - s)$$

has a local maximum at $t = 0$. Hence

$$0 = \frac{d}{dt} \Big|_{t=0} \left[\Phi(U_{q+th}) - c^* \cdot (\Pi(U_{q+th}) - s) \right].$$

By Lemma 9,

$$0 = A \cdot h - \sum_j c_j^* \sum_i J_{ij} h_i = \sum_i \left(A_i - \sum_j J_{ij} c_j^* \right) h_i.$$

Since $h \in \mathbb{R}^N$ was arbitrary, $Jc^* = A$. By Fact 4, $c^* = J^{-1}A = c$, so U_{pure} solves (12).

B.17 Proof of Lemma 2

Step 1: Measure formulation. Dropping the constant $c \cdot s$, the objective is

$$\int_{\Gamma} \lambda U g d\theta - \int_{\Gamma} c \cdot x_U g d\theta. \quad (32)$$

Using the definition of x_U ,

$$\begin{aligned} \int_{\Gamma} x_U \cdot c g d\theta &= \int_{\Gamma} (\nabla_H U - \mathbf{1}(\nabla_H U \cdot \theta - U)) \cdot c g d\theta \\ &= \int_{\Gamma} (\nabla_H U - (\nabla_H U \cdot \theta) \mathbf{1}) \cdot c g d\theta + (\sum_j c_j) \int_{\Gamma} U g d\theta \\ &= \int_{\Gamma} (c - (\sum_j c_j) \theta) g \cdot \nabla_H U d\theta + (\sum_j c_j) \int_{\Gamma} U g d\theta. \end{aligned}$$

Now, $c - (\sum_j c_j) \theta \in TH$ because $(c - (\sum_j c_j) \theta) \cdot \mathbf{1} = 0$; also, U is Lipschitz by Corollary 5. We can therefore apply Theorem 4 to the former integral on the RHS above. This gives:

$$\int_{\Gamma} (c - (\sum_j c_j) \theta) g \cdot \nabla_H U d\theta = - \int_{\Gamma} U \operatorname{div} [(c - (\sum_j c_j) \theta) g] d\theta + \int_{\partial\Gamma} U (c - (\sum_j c_j) \theta) g \cdot \nu d\sigma.$$

We therefore get:

$$\int_{\Gamma} x_U \cdot c g d\theta = \int_{\Gamma} U \left((\sum_j c_j) g - \operatorname{div} [(c - (\sum_j c_j) \theta) g] \right) d\theta + \int_{\partial\Gamma} U (c - (\sum_j c_j) \theta) g \cdot \nu d\sigma.$$

Plugging back into (32) and collecting terms gives:

$$\int_{\Gamma} U \left[\lambda g + \operatorname{div} [(c - (\sum_j c_j) \theta) g] - (\sum_j c_j) g \right] d\theta - \int_{\partial\Gamma} U (c - (\sum_j c_j) \theta) g \cdot \nu d\sigma.$$

Step 2: Balance on each region Γ_i . We now prove that μ is balanced on each region Γ_i . Applying Theorem 4 with $\Omega = \Gamma_i^\circ$, $\eta(\theta) = \theta_i$ and $X(\theta) = (c - (\sum_j c_j) \theta) g(\theta)$ we get:

$$\int_{\Gamma_i} \theta_i \operatorname{div} [(c - (\sum_j c_j) \theta) g] d\theta + \int_{\Gamma_i} \nabla_H \theta_i \cdot (c - (\sum_j c_j) \theta) g d\theta = \int_{\partial\Gamma_i} \theta_i (c - (\sum_j c_j) \theta) g \cdot \nu d\sigma.$$

Substitute this into the definition of $\mu(\Gamma_i)$ to obtain:

$$\begin{aligned}\mu(\Gamma_i) &= A_i - \int_{\Gamma_i} \nabla_H \theta_i \cdot (c - (\sum c_j) \theta) g d\theta - (\sum c_j) \int_{\Gamma_i} \theta_i g d\theta \\ &\quad + \int_{\partial \Gamma_i} \theta_i (c - (\sum c_j) \theta) g \cdot \nu d\sigma - \int_{\partial \Gamma \cap \Gamma_i} \theta_i (c - (\sum c_j) \theta) g \cdot \nu d\sigma.\end{aligned}\quad (33)$$

Note that $\nabla_H \theta_i = e_i - \frac{1}{N} \mathbf{1}$, and hence:

$$- \int_{\Gamma_i} \nabla_H \theta_i \cdot (c - (\sum c_j) \theta) g d\theta = - \int_{\Gamma_i} (c_i - (\sum c_j) \theta_i) g d\theta = -c_i M_i + (\sum c_j) \int_{\Gamma_i} \theta_i g d\theta.$$

Substituting into (33) and simplifying gives:

$$\mu(\Gamma_i) = A_i - c_i M_i + \int_{\partial \Gamma_i} \theta_i (c - (\sum c_j) \theta) g \cdot \nu d\sigma - \int_{\partial \Gamma \cap \Gamma_i} \theta_i (c - (\sum c_j) \theta) g \cdot \nu d\sigma.$$

Combining the boundary terms gives

$$\mu(\Gamma_i) = A_i - c_i M_i + \int_{\partial \Gamma_i \setminus (\partial \Gamma \cap \Gamma_i)} \theta_i (c - (\sum c_j) \theta) g \cdot \nu d\sigma.$$

Note that, up to lower-dimensional edges, we have $\partial \Gamma_i \setminus (\partial \Gamma \cap \Gamma_i) = \bigcup_{k \neq i} \Gamma_i \cap \Gamma_k$, giving:

$$\mu(\Gamma_i) = A_i - c_i M_i + \sum_{k \neq i} \int_{\Gamma_i \cap \Gamma_k} \theta_i (c - (\sum c_j) \theta) g \cdot \nu_{ik}^{(i)} d\sigma, \quad (34)$$

where $\nu_{ik}^{(i)}$ is the outward unit conormal from Γ_i into Γ_k along $\Gamma_i \cap \Gamma_k$. Now, fix $k \neq i$ and note $\Gamma_i \cap \Gamma_k$ is the level set of $q_i \theta_i - q_k \theta_k$, with $q_i \theta_i - q_k \theta_k = 0$ on $\Gamma_i \cap \Gamma_k$, and:

$$\nu_{ik}^{(i)} = - \frac{\nabla_H (q_i \theta_i - q_k \theta_k)}{\|\nabla_H (q_i \theta_i - q_k \theta_k)\|}.$$

Thus, the integrand in the last term of (34) becomes:

$$\begin{aligned}\theta_i (c - (\sum c_j) \theta) g \cdot \nu_{ik}^{(i)} &= -(c - (\sum c_j) \theta) \cdot \nabla_H (q_i \theta_i - q_k \theta_k) g \frac{\theta_i}{\|\nabla_H (q_i \theta_i - q_k \theta_k)\|} \\ &= -(c - (\sum c_j) \theta) \cdot (q_i (e_i - \frac{1}{N} \mathbf{1}) - q_k (e_k - \frac{1}{N} \mathbf{1})) g \frac{\theta_i}{\|\nabla_H (q_i \theta_i - q_k \theta_k)\|} \\ &= -[q_i (c_i - (\sum c_j) \theta_i) - q_k (c_k - (\sum c_j) \theta_k)] g \frac{\theta_i}{\|\nabla_H (q_i \theta_i - q_k \theta_k)\|} \\ &= (q_k c_k - q_i c_i) g \frac{\theta_i}{\|\nabla_H (q_i \theta_i - q_k \theta_k)\|},\end{aligned}$$

where the last line follows because on $\Gamma_i \cap \Gamma_k$ we have $q_i \theta_i = q_k \theta_k$, so the $(\sum c_j)$ -terms cancel.

Since $q_k c_k - q_i c_i$ is constant along $\Gamma_i \cap \Gamma_k$, substituting into (34) gives:

$$\mu(\Gamma_i) = A_i - c_i M_i + \sum_{k \neq i} (q_k c_k - q_i c_i) \int_{\Gamma_i \cap \Gamma_k} g \frac{\theta_i}{\|\nabla_H(q_i \theta_i - q_k \theta_k)\|} d\sigma. \quad (35)$$

Moreover:

$$\nabla_H(q_i \theta_i - q_k \theta_k) = q_i(e_i - \frac{1}{N}\mathbf{1}) - q_k(e_k - \frac{1}{N}\mathbf{1}) \implies \|\nabla_H(q_i \theta_i - q_k \theta_k)\|^2 = (q_i^2 + q_k^2) - \frac{(q_i - q_k)^2}{N},$$

and therefore:

$$\mu(\Gamma_i) = A_i - c_i M_i + \sum_{k \neq i} (q_k c_k - q_i c_i) T_{ik}.$$

Finally, the i th row of the system $Jc = A$ gives exactly:

$$A_i - c_i M_i + \sum_{k \neq i} (q_k c_k - q_i c_i) T_{ik} = 0,$$

so $\mu(\Gamma_i) = 0$ by the construction of the cost vector c .

B.18 Proof of Proposition 7

($\Leftarrow$). Suppose there exists a dual certificate $(\gamma_1, \dots, \gamma_N)$. Then, for every $U \in K$,

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{i=1}^N \int_{R_i} B_i U d\gamma_i.$$

If U is feasible for (15), then $B_i U \leq 0$ on R_i for every i . Since each γ_i is nonnegative,

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{i=1}^N \int_{R_i} B_i U d\gamma_i \leq 0.$$

By (14), we have that $\int_{\Gamma} \frac{U_{\text{pure}}}{\theta_*} d\mu = 0$, which gives

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \int_{\Gamma} \frac{U_{\text{pure}}}{\theta_*} d\mu.$$

Thus U_{pure} is primal optimal.

($\Rightarrow$). We introduce the ordered Banach space

$$Y := \prod_{i=1}^N C(R_i),$$

with the product sup norm and the pointwise order. The primal constraints are $B_i U \leq 0$ on R_i ,

for every i . Note also that $\mathcal{U}$, defined in (11), can be written as

$$\mathcal{U} = \{U \in K : B_i U \leq 0 \text{ on } R_i \text{ for every } i\}.$$

We first show the following lemma:

Lemma 10. *There exists $C < \infty$ such that the following holds. Let $\bar{U} \in K$ and let $\delta \geq 0$. Suppose*

$$B_i \bar{U} \leq \delta \quad \text{on } R_i \text{ for every } i.$$

Then there exists $U \in \mathcal{U}$ such that $\|\bar{U} - U\|_\infty \leq C\delta$.

Proof. Extend $\bar{U}$ homogeneously to the positive cone by

$$W(z) := \begin{cases} (\sum z_l) \bar{U}(z / \sum z_l), & z \neq 0, \\ 0, & z = 0. \end{cases}$$

Since $\bar{U}$ is convex and nonnegative on Γ , the function W is nonnegative, convex, and positively homogeneous. Hence W is sublinear on $\mathbb{R}_+^N$.¹⁷

Fix a coordinate j , take $r \in \mathbb{R}_+^N$ with $r_j = 0$, and let $0 \leq s \leq t$ with $\sum r_l + t \leq 1$. We first prove the following bound:

Fact 6. $W(r + se_j) \leq W(r + te_j) + (N - 1)\delta$.

Proof. If $r = 0$, then by homogeneity and nonnegativity,

$$W(se_j) = s\bar{U}(e_j) \leq t\bar{U}(e_j) = W(te_j),$$

so the statement holds. Now suppose $r \neq 0$. Choose $i \neq j$ such that $r_i \geq \frac{\sum r_l}{N-1}$ and define

$$\alpha := \frac{r}{\sum r_l}, \quad \beta := \frac{r + te_j}{\sum r_l + t}.$$

Then $\alpha, \beta \in \Gamma$. Moreover, $(\beta, \alpha) \in R_i$. Indeed, for $k \neq i, j$,

$$\frac{\beta_k}{\beta_i} = \frac{r_k}{r_i} = \frac{\alpha_k}{\alpha_i}, \quad \text{and} \quad \frac{\beta_j}{\beta_i} = \frac{t}{r_i} \geq 0 = \frac{\alpha_j}{\alpha_i}.$$

Equivalently, $\beta_k \alpha_i \geq \alpha_k \beta_i$ for every $k \neq i$.

Since $B_i \bar{U} \leq \delta$ on R_i , we get

$$\beta_i \bar{U}(\alpha) - \alpha_i \bar{U}(\beta) \leq \delta.$$

¹⁷A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *sublinear* if it satisfies $f(x + y) \leq f(x) + f(y)$ and $f(\lambda x) = \lambda f(x)$ for all $x, y \in \mathbb{R}^n$ and all $\lambda \geq 0$. Equivalently, f is sublinear if and only if it is convex and positively homogeneous.

Using $\beta_i = \frac{r_i}{\sum r_l + t}$, $\alpha_i = \frac{r_i}{\sum r_l}$ and the definition of W , this becomes

$$\frac{r_i}{(\sum r_l)(\sum r_l + t)} [W(r) - W(r + te_j)] \leq \delta.$$

Therefore

$$W(r) - W(r + te_j) \leq \delta \frac{(\sum r_l)(\sum r_l + t)}{r_i} \leq (N-1)\delta(\sum r_l + t) \leq (N-1)\delta.$$

Now, the function $a \mapsto W(r + ae_j)$ is convex on $[0, t]$. Hence, for $0 \leq s \leq t$,

$$W(r + se_j) \leq \left(1 - \frac{s}{t}\right) W(r) + \frac{s}{t} W(r + te_j).$$

Thus

$$W(r + se_j) - W(r + te_j) \leq \left(1 - \frac{s}{t}\right) (W(r) - W(r + te_j)) \leq (N-1)\delta.$$

□

Now, take $x, y \in \mathbb{R}_+^N$ such that $0 \leq x \leq y$ coordinatewise, and $\sum y_l \leq 1$. By applying Fact 6 at most N times, we then get that:

$$W(x) \leq W(y) + N(N-1)\delta. \quad (36)$$

Now, define:

$$W^\uparrow(y) := \sup\{W(x) : 0 \leq x \leq y\}.$$

In particular, this implies $W(y) \leq W^\uparrow(y)$. On the other hand, if $y \in \Gamma$, then $\sum y_l = 1$, so (36) gives

$$W(x) \leq W(y) + N(N-1)\delta \quad \text{for every } 0 \leq x \leq y.$$

Taking the supremum over $x \leq y$, we obtain

$$0 \leq W^\uparrow(y) - W(y) \leq N(N-1)\delta \quad \text{for every } y \in \Gamma. \quad (37)$$

Fact 7. $W^\uparrow$ is coordinatewise increasing and sublinear.

Proof. Coordinatewise monotonicity and positive homogeneity are immediate from the definition. It therefore suffices to show subadditivity, i.e. that for any $y, y' \in \mathbb{R}_+^N$ we have

$$W^\uparrow(y + y') \leq W^\uparrow(y) + W^\uparrow(y'). \quad (38)$$

Fix such y, y' and take any z such that $0 \leq z \leq y + y'$. Define $z_k^1 := \min\{z_k, y_k\}$ and $z^2 := z - z^1$. Then

$$0 \leq z^1 \leq y, \quad 0 \leq z^2 \leq y', \quad z = z^1 + z^2.$$

Since W is sublinear,

$$W(z) \leq W(z^1) + W(z^2) \leq W^\uparrow(y) + W^\uparrow(y').$$

Taking the supremum over all $z \leq y + y'$ yields (38). □

Now, set

$$U(\theta) := W^\uparrow(\theta), \quad \theta \in \Gamma.$$

Note (37) gives

$$\|\bar{U} - U\|_\infty \leq N(N-1)\delta.$$

It therefore suffices to show that $U \in \mathcal{U}$. Since $W^\uparrow$ is sublinear, its restriction to Γ is convex. Also $U \geq 0$. To see that U is continuous, let $M := \max_i W^\uparrow(e_i) < \infty$. For $\theta, \theta' \in \Gamma$, sublinearity and coordinatewise monotonicity give

$$W^\uparrow(\theta) \leq W^\uparrow(\theta' + (\theta - \theta')_+) \leq W^\uparrow(\theta') + W^\uparrow((\theta - \theta')_+) \leq W^\uparrow(\theta') + M\|\theta - \theta'\|_1.$$

Interchanging θ and θ' gives $|U(\theta) - U(\theta')| \leq M\|\theta - \theta'\|_1$, giving $U \in C(\Gamma)$ and thus $U \in K$.

U satisfies (R). Fix i and $(\theta, \theta') \in R_i$. If $\theta_i = 0$, then $B_i U(\theta, \theta') = -\theta'_i U(\theta) \leq 0$. Now suppose $\theta_i > 0$. Define $z := \frac{\theta'_i}{\theta_i} \theta$. Then $z_i = \theta'_i$, and the inequalities defining R_i imply

$$z_k = \frac{\theta'_i}{\theta_i} \theta_k \geq \theta'_k \quad \text{for every } k \neq i.$$

Thus $z \geq \theta'$ coordinatewise. Since $W^\uparrow$ is coordinatewise increasing and homogeneous,

$$U(\theta') = W^\uparrow(\theta') \leq W^\uparrow(z) = \frac{\theta'_i}{\theta_i} W^\uparrow(\theta) = \frac{\theta'_i}{\theta_i} U(\theta); \quad \text{equivalently, } \theta_i U(\theta') - \theta'_i U(\theta) \leq 0.$$

□

We now use Lemma 10 to prove existence of a certificate. Recall that by (14),

$$\int_\Gamma \frac{U_{\text{pure}}}{\theta_*} d\mu = 0.$$

Since U_{pure} is optimal for (15), we have

$$\int_\Gamma \frac{U}{\theta_*} d\mu \leq 0 \quad \text{for all } U \in \mathcal{U}. \quad (39)$$

Let

$$\mathcal{C} := \left\{ \left((-B_i U - w_i)_{i=1}^N, \int_\Gamma \frac{U}{\theta_*} d\mu - r \right) : U \in K, (w_i)_{i=1}^N \in Y_+, r \geq 0 \right\} \subseteq Y \times \mathbb{R},$$

where Y_+ denotes the positive cone of Y . We now show the following fact:

Fact 8. $(0, 1) \notin \bar{\mathcal{C}}$.

Proof. Suppose not. Then there are $U_n \in K$, $w_n = (w_{n,i})_{i=1}^N \in Y_+$, and $r_n \geq 0$ such that

$$(-B_i U_n - w_{n,i})_{i=1}^N \rightarrow 0 \quad \text{in } Y, \quad \int_\Gamma \frac{U_n}{\theta_*} d\mu - r_n \rightarrow 1.$$

Set

$$\eta_n := \|(-B_i U_n - w_{n,i})_{i=1}^N\|_Y.$$

Then $\eta_n \rightarrow 0$. Since $w_{n,i} \geq 0$ for every i , we have

$$B_i U_n \leq \eta_n \quad \text{on } R_i \text{ for every } i.$$

By Lemma 10, there exists $\hat{U}_n \in \mathcal{U}$ such that $\|U_n - \hat{U}_n\|_\infty \leq C\eta_n$. Using (39),

$$\int_\Gamma \frac{\hat{U}_n}{\theta_*} d\mu \leq 0.$$

Therefore

$$\int_\Gamma \frac{U_n}{\theta_*} d\mu \leq \int_\Gamma \frac{\hat{U}_n}{\theta_*} d\mu + \left\| \frac{1}{\theta_*} \mu \right\|_{TV} \|U_n - \hat{U}_n\|_\infty \leq C \left\| \frac{1}{\theta_*} \mu \right\|_{TV} \eta_n \rightarrow 0.$$

But since $r_n \geq 0$,

$$\int_\Gamma \frac{U_n}{\theta_*} d\mu \geq \int_\Gamma \frac{U_n}{\theta_*} d\mu - r_n \rightarrow 1,$$

a contradiction. □

Since $\bar{\mathcal{C}}$ is a closed convex cone and $(0, 1) \notin \bar{\mathcal{C}}$, Hahn–Banach separation gives a nonzero continuous linear functional

$$(\Lambda, \lambda) \in Y^* \times \mathbb{R} \quad \text{such that} \quad \Lambda(z) + \lambda q \leq 0 \quad \text{for all } (z, q) \in \bar{\mathcal{C}}.$$

Since $0 \in \bar{\mathcal{C}}$, we can orient the separating functional so that it strictly separates $(0, 1)$ from $\bar{\mathcal{C}}$: $\Lambda(z) + \lambda q \leq 0 < \Lambda(0) + \lambda = \lambda$ for all $(z, q) \in \bar{\mathcal{C}}$. Thus $\lambda > 0$. We normalize $\lambda = 1$.

Now take any tuple $w = (w_1, \dots, w_N)$ with $w_i \in C(R_i)$ and $w_i \geq 0$ on R_i for every i . Then $(-w, 0) \in \mathcal{C}$, so the separating inequality gives $-\Lambda(w) \leq 0$. Thus, $\Lambda(w) \geq 0$ for every componentwise nonnegative tuple w . For every $U \in K$, taking $w = 0$ and $r = 0$ gives

$$\left((-B_i U)_{i=1}^N, \int_\Gamma \frac{U}{\theta_*} d\mu \right) \in \mathcal{C}.$$

Therefore

$$-\Lambda((B_i U)_{i=1}^N) + \int_\Gamma \frac{U}{\theta_*} d\mu \leq 0, \quad \text{or equivalently,} \quad \int_\Gamma \frac{U}{\theta_*} d\mu \leq \Lambda((B_i U)_{i=1}^N) \quad \text{for all } U \in K. \quad (40)$$

Since Λ is a positive continuous linear functional on $\prod_{i=1}^N C(R_i)$, the Riesz representation theorem gives finite positive Borel measures $\gamma_i \in \mathcal{M}_+(R_i)$, $i = 1, \dots, N$, such that

$$\Lambda((B_i U)_{i=1}^N) = \sum_{i=1}^N \int_{R_i} (B_i U)(\theta, \theta') d\gamma_i(\theta, \theta') \quad \text{for all } U \in K.$$

Substituting this representation into (40) gives

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{i=1}^N \int_{R_i} (B_i U)(\theta, \theta') d\gamma_i(\theta, \theta') \quad \text{for all } U \in K.$$

B.19 Proof of Lemma 3

Let $(\gamma_1, \dots, \gamma_N)$ be a dual certificate. Applying (16) to U_{pure} , using (14) and $B_i U_{\text{pure}} \leq 0$ gives

$$0 \leq \sum_{i=1}^N \int_{R_i} B_i U_{\text{pure}} d\gamma_i \leq 0.$$

Thus each integral is zero. Since $B_i U_{\text{pure}} \leq 0$ and $\gamma_i \geq 0$, we may restrict attention to

$$E_i := \{(\theta, \theta') \in R_i : B_i U_{\text{pure}}(\theta, \theta') = 0\}.$$

We first remove the part of the measure supported where both coefficients in B_i vanish. Let

$$Z_i := \{(\theta, \theta') \in R_i : \theta_i = \theta'_i = 0\}.$$

For every $U \in K$, $B_i U(\theta, \theta') = 0$ on Z_i . Thus replacing γ_i by $\gamma_i|_{R_i \setminus Z_i}$ does not change the functional $U \mapsto \int_{R_i} B_i U d\gamma_i$. Hence, without loss, assume $\gamma_i(Z_i) = 0$.

Fact 9. *Every point in $E_i \setminus Z_i$ belongs to a common pure-option region, that is,*

$$E_i \setminus Z_i \subseteq \bigcup_l (\Gamma_l \times \Gamma_l).$$

Proof. Fix $(\theta, \theta') \in E_i \setminus Z_i$. Then $\theta_i > 0$ and $\theta'_i > 0$.¹⁸

Since $(\theta, \theta') \in R_i$, for every $k \neq i$ and $\theta_i > 0, \theta'_i > 0$, we get

$$\theta_k \theta'_i \geq \theta'_k \theta_i, \quad \text{and thus} \quad \frac{q_k \theta_k}{\theta_i} \geq \frac{q_k \theta'_k}{\theta'_i} \quad \text{for } k \neq i.$$

For $k = i$, the two sides are both equal to q_i and so

$$\frac{q_k \theta_k}{\theta_i} \geq \frac{q_k \theta'_k}{\theta'_i} \quad \text{for every } k.$$

Because $(\theta, \theta') \in E_i$, we have

$$\theta_i U_{\text{pure}}(\theta') = \theta'_i U_{\text{pure}}(\theta),$$

¹⁸Indeed, if one of them were zero and the other were positive, the equality $B_i U_{\text{pure}}(\theta, \theta') = 0$ would be impossible because $U_{\text{pure}} > 0$ on Γ ; and the case $\theta_i = \theta'_i = 0$ has been removed.

and therefore

$$\max_k \frac{q_k \theta_k}{\theta_i} = \frac{U_{\text{pure}}(\theta)}{\theta_i} = \frac{U_{\text{pure}}(\theta')}{\theta'_i} = \max_k \frac{q_k \theta'_k}{\theta'_i}.$$

Choose j such that

$$\frac{q_j \theta'_j}{\theta'_i} = \max_k \frac{q_k \theta'_k}{\theta'_i}.$$

Then

$$\frac{q_j \theta'_j}{\theta'_i} \leq \frac{q_j \theta_j}{\theta_i} \leq \max_k \frac{q_k \theta_k}{\theta_i} = \max_k \frac{q_k \theta'_k}{\theta'_i} = \frac{q_j \theta'_j}{\theta'_i}.$$

Since the first and last terms in this chain are equal, all inequalities in the chain must bind. Multiplying by θ'_i and θ_i , respectively, gives

$$q_j \theta'_j = U_{\text{pure}}(\theta'), \quad q_j \theta_j = U_{\text{pure}}(\theta).$$

Therefore, $\theta, \theta' \in \Gamma_j$. □

Now partition each $E_i \setminus Z_i$ into such common regions. Set

$$S_{i1} := (E_i \setminus Z_i) \cap (\Gamma_1 \times \Gamma_1),$$

and, for $j \geq 2$,

$$S_{ij} := (E_i \setminus Z_i) \cap (\Gamma_j \times \Gamma_j) \setminus \bigcup_{l < j} (\Gamma_l \times \Gamma_l).$$

Then

$$E_i \setminus Z_i = \bigcup_{j=1}^N S_{ij}, \quad \eta_{ij} := \gamma_i|_{S_{ij}}, \quad \gamma_i = \sum_{j=1}^N \eta_{ij}.$$

We now reassign η_{ij} from constraint i to constraint j . Fix i, j . On S_{ij} , we have

$$U_{\text{pure}}(\theta) = q_j \theta_j, \quad U_{\text{pure}}(\theta') = q_j \theta'_j.$$

Since $B_i U_{\text{pure}} = 0$ on S_{ij} , we have $\theta'_i q_j \theta_j = \theta_i q_j \theta'_j$ and therefore

$$\frac{\theta_i}{\theta_j} = \frac{\theta'_i}{\theta'_j}. \tag{41}$$

Next, we show that $S_{ij} \subseteq R_j$. Take $(\theta, \theta') \in S_{ij}$ and fix $k \neq j$. If $k = i$, the required inequality follows from (41). If $k \neq i, j$, then since $(\theta, \theta') \in R_i$,

$$\frac{\theta_k}{\theta_i} \geq \frac{\theta'_k}{\theta'_i}.$$

Using (41), we get

$$\frac{\theta_k}{\theta_j} = \frac{\theta_k/\theta_i}{\theta_j/\theta_i} \geq \frac{\theta'_k/\theta'_i}{\theta'_j/\theta'_i} = \frac{\theta'_k}{\theta'_j}.$$

Therefore $(\theta, \theta') \in R_j$.

We now define a positive Borel measure $\tilde{\eta}_{ij} \in \mathcal{M}_+(R_j)$ by

$$d\tilde{\eta}_{ij}(\theta, \theta') := \frac{\theta_i}{\theta_j} d\eta_{ij}(\theta, \theta').$$

This measure is finite because θ_j is bounded away from zero on Γ_j .¹⁹ Moreover,

$$\int_{R_j} B_j U d\tilde{\eta}_{ij} = \int_{S_{ij}} B_i U d\eta_{ij} \quad \text{for all } U \in K. \quad (42)$$

Indeed, on S_{ij} , we have

$$\frac{\theta_i}{\theta_j} \theta_j = \theta_i \quad \text{and} \quad \frac{\theta_i}{\theta_j} \theta'_j = \frac{\theta'_i}{\theta'_j} \theta'_j = \theta'_i, \quad \text{so} \quad \frac{\theta_i}{\theta_j} (\theta_j U(\theta') - \theta'_j U(\theta)) = \theta_i U(\theta') - \theta'_i U(\theta).$$

Finally, for each j , define $\bar{\gamma}_j := \sum_{i=1}^N \tilde{\eta}_{ij}$. Then

$$\bar{\gamma}_j \in \mathcal{M}_+(R_j), \quad \text{supp}(\bar{\gamma}_j) \subseteq R_j \cap (\Gamma_j \times \Gamma_j).$$

Summing over (42) gives

$$\sum_{j=1}^N \int_{R_j} B_j U d\bar{\gamma}_j = \sum_{i=1}^N \int_{R_i} B_i U d\gamma_i \quad \text{for all } U \in K.$$

Since the original certificate gave

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{i=1}^N \int_{R_i} B_i U d\gamma_i \quad \text{for all } U \in K,$$

we obtain

$$\int_{\Gamma} \frac{U}{\theta_*} d\mu \leq \sum_{j=1}^N \int_{R_j} B_j U d\bar{\gamma}_j \quad \text{for all } U \in K.$$

¹⁹ θ_i/θ_j is bounded on S_{ij} since $\theta \in \Gamma_j$ gives $q_j \theta_j = U_{\text{pure}}(\theta) \geq \frac{\min_k q_k}{N}$, and thus θ_j is bounded away from zero on Γ_j .

C Deriving examples

C.1 Example 1

Let us compute $H_\rho(z)$ for this value distribution. For $z \in [0, \frac{1}{2}]$,

$$\mathbb{P}(\Theta_2 \leq z) = \frac{z}{2(1-z)}, \quad \mathbb{P}(\Theta_2 \geq z) = \frac{2-3z}{2(1-z)}, \quad \mathbb{E}[\max\{zV_1, (1-z)V_2\}] = \frac{4z^2 - 6z + 3}{6(1-z)}.$$

For $z \in [\frac{1}{2}, 1]$,

$$\mathbb{P}(\Theta_2 \leq z) = \frac{3z-1}{2z}, \quad \mathbb{P}(\Theta_2 \geq z) = \frac{1-z}{2z}, \quad \mathbb{E}[\max\{zV_1, (1-z)V_2\}] = \frac{4z^2 - 2z + 1}{6z}.$$

Since $\mathbb{E}[\lambda(\Theta) \max\{z\Theta_1, (1-z)\Theta_2\}] = \mathbb{E}[\max\{zV_1, (1-z)V_2\}]$, we get:

$$H_\rho(z) = \begin{cases} \frac{1}{3} \frac{4z^2 - 6z + 3}{(\rho + 3)z^2 - 5z + 2}, & z \in [0, \frac{1}{2}], \\ \frac{1}{3} \frac{4z^2 - 2z + 1}{(3\rho + 1)z^2 - (\rho + 2)z + 1}, & z \in [\frac{1}{2}, 1]. \end{cases}$$

Note also that $H_\rho(z) = \rho^{-1} H_{1/\rho}(1-z)$, so it is enough to consider $\rho \leq 1$. On $[\frac{1}{2}, 1]$,

$$H'_\rho(z) = \frac{2(2\rho z^2 - 6\rho z + \rho - 6z^2 + 6z)}{3((3\rho + 1)z^2 - (\rho + 2)z + 1)^2}.$$

Its numerator is strictly decreasing in z , positive at $z = \frac{1}{2}$, and negative at $z = 1$. Thus H_ρ has a unique maximizer on $[\frac{1}{2}, 1]$, and therefore also on all of $[0, 1]$.

Let z^* be the unique maximizer of H_{ρ^*} , where ρ^* is as in Theorem 1. Since $\arg \max H_{\rho^*} = \{z^*\}$, the theorem implies $a = b = z^*$, so the bundle is redundant. Hence the optimal mechanism offers only two pure options. By Theorem 1, the quantities offered by the pure options are chosen so that supply constraints hold with equality if all agents choose their favorite options.

C.2 Example 3

For the power family,

$$x \frac{f_M(x)}{F_M(x)} = \alpha,$$

which is constant, hence non-increasing. The uniform distribution is the special case $\alpha = 1$. For the truncated Weibull family,

$$x \frac{f_M(x)}{F_M(x)} = \frac{\alpha \beta x^\alpha}{e^{\beta x^\alpha} - 1}.$$

Writing $z = \beta x^\alpha$, this becomes $\alpha \frac{z}{e^z - 1}$. The function $z \mapsto z/(e^z - 1)$ is decreasing on $(0, \infty)$, since

$$\frac{d}{dz} \frac{z}{e^z - 1} = \frac{e^z(1 - z) - 1}{(e^z - 1)^2} \leq 0.$$

Therefore $xf_M(x)/F_M(x)$ is non-increasing in x . The truncated exponential distribution is the special case $\alpha = 1$. The result follows from Corollary 3.

C.3 Examples 2 and 4

Let us first consider Example 4. By exchangeability and equal supplies, the pure option mechanism is symmetric. Thus, each pure option gives Ns units of the corresponding good. Its welfare is $W_{\text{pure}} = Ns \mathbb{E}[\max_j V_j]$. We now construct a feasible incentive-compatible menu that yields strictly higher welfare. Fix some $\tau \in (1/N, 1)$, and define $p_\tau := \mathbb{P}(\max_j V_j \leq \tau \sum_j V_j)$. Consider the menu consisting of N pure options and one balanced mixed option:

$$\{q_\tau e_1, \dots, q_\tau e_N, \tau q_\tau \mathbf{1}\}, \quad \text{where} \quad q_\tau := \frac{Ns}{1 + p_\tau(N\tau - 1)}.$$

A type V gets utility $q_\tau \max_j V_j$ from her favorite pure option and utility $\tau q_\tau \sum_j V_j$ from the mixed option. Hence she chooses the mixed option exactly when $\max_j V_j \leq \tau \sum_j V_j$, and otherwise chooses her favorite pure option. The menu is feasible. Indeed, by exchangeability, conditional on choosing a pure option, each good is chosen with equal probability. Therefore, the amount of any good i used by this menu is

$$q_\tau \frac{1 - p_\tau}{N} + \tau q_\tau p_\tau = q_\tau \left(\frac{1 - p_\tau}{N} + \tau p_\tau \right) = q_\tau \frac{1 + p_\tau(N\tau - 1)}{N} = s.$$

Thus every supply constraint binds.

The welfare from this menu is $W_\tau = q_\tau \mathbb{E}[\max\{\max_j V_j, \tau \sum_j V_j\}]$. Since

$$\max\{\max_j V_j, \tau \sum_j V_j\} = \max_j V_j + (\tau \sum_j V_j - \max_j V_j)^+,$$

we have

$$W_\tau > W_{\text{pure}} \quad \text{if and only if} \quad \frac{Ns}{1 + p_\tau(N\tau - 1)} \mathbb{E}[\max_j V_j + (\tau \sum_j V_j - \max_j V_j)^+] > Ns \mathbb{E}[\max_j V_j].$$

Equivalently,

$$\mathbb{E}[(\tau \sum_j V_j - \max_j V_j)^+] > p_\tau(N\tau - 1) \mathbb{E}[\max_j V_j]. \quad (43)$$

It remains to verify (43). Consider first the limiting case $\varepsilon = 0$. Common-need agents have $\sum_j V_j = N$ and $\max_j V_j = 1$. Since $\tau > 1/N$, they satisfy

$$\max_j V_j = 1 < \tau N = \tau \sum_j V_j.$$

Specialized-need agents have exactly one nonzero coordinate. If that coordinate is j , then, except at the null type with $V_j = 0$, $\sum_k V_k = V_j$ and $\max_k V_k = V_j$. Since $\tau < 1$, they satisfy

$$\max_k V_k = V_j > \tau V_j = \tau \sum_k V_k.$$

Therefore, in the limit,

$$p_\tau = p, \quad \mathbb{E}[(\tau \sum_j V_j - \max_j V_j)^+] = p(N\tau - 1), \quad \text{and} \quad \mathbb{E}[\max_j V_j] = p + (1 - p)\frac{1}{2}.$$

Hence

$$\begin{aligned} \mathbb{E}[(\tau \sum_j V_j - \max_j V_j)^+] - p_\tau(N\tau - 1)\mathbb{E}[\max_j V_j] &= p(N\tau - 1) - p(N\tau - 1)(p + (1 - p)\frac{1}{2}) \\ &= p(1 - p)(N\tau - 1)(1 - \frac{1}{2}) > 0. \end{aligned}$$

Finally, the objects appearing in (43) vary continuously as $\varepsilon \downarrow 0$. Indeed, the values are uniformly bounded, and the event $\{\max_j V_j = \tau \sum_j V_j\}$ has probability zero in the limiting distribution. Therefore, the strict inequality continues to hold for all sufficiently small $\varepsilon > 0$. Hence, $W_\tau > W_{\text{pure}}$, so the pure option mechanism is not optimal. This verifies Example 4.

For Example 2, the argument showing that the pure option mechanism is not optimal is analogous. Recall also that by Theorem 1 an optimal mechanism has either two pure options or two pure options and one bundle. Since the pure option mechanism is not optimal, the optimal mechanism must take the latter form. Since the bundle is not redundant, it must deliver strictly more total quantity than either pure option.

C.4 Example 5

By symmetry and equal supplies, the market-clearing pure-option vector is symmetric. Hence, up to null indifference sets,

$$\Gamma_i = \{\theta \in \Gamma : \theta_i \geq \theta_k \text{ for all } k\}.$$

Fix i . Identify Γ_i with the cube $[0, 1]^{N-1}$ by the ratio-coordinate map

$$R_i(\theta) := \left(\frac{\theta_k}{\theta_i} \right)_{k \neq i}.$$

For $r = (r_k)_{k \neq i} \in [0, 1]^{N-1}$, the inverse map is

$$\theta_i(r) = \frac{1}{1 + \sum_{k \neq i} r_k}, \quad \theta_k(r) = \frac{r_k}{1 + \sum_{l \neq i} r_l} \quad (k \neq i). \quad (44)$$

Moreover, if $C \subseteq \Gamma_i$ is a closed $\succ_i$ -upper set, then $D := R_i(C)$ is a closed downset in $[0, 1]^{N-1}$: whenever $r \in D$ and $0 \leq r' \leq r$ coordinatewise, we have $r' \in D$.

Since V is uniformly distributed on $[0, 1]^N$, the ray $\{t\theta : t \geq 0\}$ remains in $[0, 1]^N$ exactly for

$0 \leq t \leq 1/\max_j \theta_j$. Therefore, for a normalizing constant $K_N > 0$,

$$g(\theta) = K_N \max_j \theta_j^{-N}, \quad \lambda(\theta) = \frac{\int_0^{1/\max_j \theta_j} t \cdot t^{N-1} dt}{\int_0^{1/\max_j \theta_j} t^{N-1} dt} = \frac{N}{N+1} \frac{1}{\max_j \theta_j}.$$

On Γ_i , $\max_j \theta_j = \theta_i$. Hence

$$g(\theta) = K_N \theta_i^{-N}, \quad \lambda(\theta) \theta_i = \frac{N}{N+1}.$$

The Jacobian of R_i^{-1} is proportional to $(1 + \sum_{k \neq i} r_k)^{-N} = \theta_i^N$. Thus the pushforward of $g dm$ under R_i is constant. Since, by symmetry, $G(\Gamma_i) = 1/N$, we have

$$(R_i)_\#(g dm) = \frac{1}{N} dr,$$

where dr denotes Lebesgue measure on $[0, 1]^{N-1}$. Therefore

$$M_i = \int_{\Gamma_i} g(\theta) dm(\theta) = \frac{1}{N}, \quad A_i = \int_{\Gamma_i} \lambda(\theta) \theta_i g(\theta) dm(\theta) = \frac{1}{N+1}.$$

By symmetry, the shadow costs are all equal. Since the pure-option quantities are equal, each row of the shadow-cost matrix sums to $M_i = 1/N$. Therefore the equation $Jc = A$ gives, for all i , $c_i = c_0 := \frac{N}{N+1}$.

We now compute the rent measure on Γ_i . The change of variables above gives

$$(R_i)_\#(g dm) = \frac{1}{N} dr.$$

In ratio coordinates, the vector field $(c - (\sum_j c_j) \theta) g = c_0(1 - N\theta) g$ has r_k -component density

$$b_k(r) = \frac{c_0}{N} (1 - r_k) \left(1 + \sum_{l \neq i} r_l \right). \quad \text{hence} \quad \theta_i(r) \sum_{k \neq i} \partial_{r_k} b_k(r) = -\frac{c_0 \sum_{k \neq i} r_k}{1 + \sum_{k \neq i} r_k}.$$

The remaining interior terms $\theta_i(\lambda g - N c_0 g) dm$ contribute $\frac{c_0}{N} (1 - N/1 + \sum_{k \neq i} r_k) dr$. Adding the two interior contributions gives the constant density $-\frac{N-1}{N+1} dr$. Finally, on the boundary face $\{r_k = 0\}$, the conormal boundary term in the definition of μ contributes

$$\frac{1}{N+1} \mathcal{H}^{N-2} \upharpoonright_{\{r_k=0\}}.$$

Therefore

$$(R_i)_\# \mu_i = \frac{1}{N+1} \left[\sum_{k \neq i} \mathcal{H}^{N-2} \upharpoonright_{\{r_k=0\}} - (N-1) \mathcal{L}^{N-1} \upharpoonright_Q \right].$$

Equivalently, for every closed $\succ_i$ -upper set $C \subseteq \Gamma_i$, with $D := R_i(C)$,

$$\mu_i(C) = \frac{1}{N+1} \left[\sum_{k \neq i} \mathcal{H}^{N-2}(D \cap \{r_k = 0\}) - (N-1) \mathcal{L}^{N-1}(D) \right]. \quad (45)$$

We next prove the geometric estimate. For a Borel set $B \subseteq Q$, define

$$\bar{\alpha}(B) := \mathcal{L}^{N-1}(B) + \sum_{k \neq i} \mathcal{H}^{N-2}(B \cap \{r_k = 0\}).$$

We claim that there exists $\kappa > 0$ such that for every closed downset $D \subseteq [0, 1]^{N-1}$,

$$\sum_{k \neq i} \mathcal{H}^{N-2}(D \cap \{r_k = 0\}) - (N-1) \mathcal{L}^{N-1}(D) \geq \kappa \min\{\bar{\alpha}(D), \bar{\alpha}(Q \setminus D)\}. \quad (46)$$

Let $V := \mathcal{L}^{N-1}(D)$ and $P_k := \mathcal{H}^{N-2}(D \cap \{r_k = 0\})$. If $D = \emptyset$ or $D = [0, 1]^{N-1}$, the claim is immediate. If $\mathcal{L}^{N-1}(D) = 0$, then $\sum_{k \neq i} P_k - (N-1)V = \sum_{k \neq i} P_k = \bar{\alpha}(D)$, so (46) holds, for instance with any $\kappa \leq 1$. Hence assume from now on that $0 < V < 1$.

First consider $N = 2$. Then the cube becomes $[0, 1]$, and a nonempty proper closed downset is of the form $D = [0, a]$ with $0 \leq a < 1$. Thus $V = a$, $\sum_{k \neq i} P_k = 1$, and $\sum_{k \neq i} P_k - (N-1)V = 1 - a = \bar{\alpha}([0, 1]^{N-1} \setminus D)$. Hence (46) holds for $N = 2$, with $\kappa = 1$.

Now suppose $N \geq 3$. Since D is a downset, $D \cap \{r_k = 0\}$ is the coordinate projection of D onto the face $\{r_k = 0\}$. By the Loomis–Whitney inequality,

$$V^{N-2} \leq \prod_{k \neq i} P_k \leq \left(\frac{\sum_{k \neq i} P_k}{N-1} \right)^{N-1}, \quad \text{and therefore} \quad \sum_{k \neq i} P_k \geq (N-1) V^{(N-2)/(N-1)}.$$

Hence

$$\sum_{k \neq i} P_k - (N-1)V \geq (N-1)(V^{(N-2)/(N-1)} - V) = (N-1)V^{(N-2)/(N-1)}(1 - V^{1/(N-1)}).$$

If $V \leq 1/2$, then

$$\frac{\sum_{k \neq i} P_k - (N-1)V}{\sum_{k \neq i} P_k + V} \geq \frac{(N-1)(1 - V^{1/(N-1)})}{N-1 + V^{1/(N-1)}} \geq \frac{(N-1)(1 - 2^{-1/(N-1)})}{N-1 + 2^{-1/(N-1)}} =: \kappa_1 > 0.$$

Since $\sum_{k \neq i} P_k + V = \bar{\alpha}(D)$, this gives $\sum_{k \neq i} P_k - (N-1)V \geq \kappa_1 \bar{\alpha}(D)$. If $V \geq 1/2$, write $t := V^{1/(N-1)}$. Then

$$1 - V = 1 - t^{N-1} = (1-t)(1+t+\dots+t^{N-2}),$$

and so

$$\frac{\sum_{k \neq i} P_k - (N-1)V}{1 - V} \geq \frac{(N-1)t^{N-2}}{1+t+\dots+t^{N-2}} \geq 2^{-\frac{(N-2)}{(N-1)}} =: \kappa_2 > 0.$$

Moreover,

$$\bar{\alpha}([0, 1]^{N-1} \setminus D) = (1 - V) + \sum_{k \neq i} (1 - P_k) = N(1 - V) - \left(\sum_{k \neq i} P_k - (N - 1)V \right) \leq N(1 - V).$$

Therefore

$$\sum_{k \neq i} P_k - (N - 1)V \geq \frac{\kappa_2}{N} \bar{\alpha}([0, 1]^{N-1} \setminus D).$$

Combining the cases proves (46), with $\kappa := \min\{1, \kappa_1, \kappa_2/N\} > 0$.

Finally, recall the inverse ratio map $R_i^{-1} : [0, 1]^{N-1} \rightarrow \Gamma_i$ is given by (44). Since $r \in [0, 1]^{N-1}$, the denominator is bounded above and below away from zero. Hence the Jacobian of R_i^{-1} , as well as the Jacobians of its restrictions to the lower faces $\{r_k = 0\}$, are bounded by a constant depending only on N . Therefore, by the area formula, there exists $B_N < \infty$ such that, for every Borel $B \subseteq [0, 1]^{N-1}$,

$$\alpha(R_i^{-1}(B)) \leq B_N \left[\mathcal{L}^{N-1}(B) + \sum_{k \neq i} \mathcal{H}^{N-2}(B \cap \{r_k = 0\}) \right] = B_N \bar{\alpha}(B).$$

Applying this to $B = D$ and $B = [0, 1]^{N-1} \setminus D$, we get

$$\min\{\bar{\alpha}(D), \bar{\alpha}(Q \setminus D)\} \geq \frac{1}{B_N} \min\{\alpha(C), \alpha(\Gamma_i \setminus C)\}.$$

Combining this with (45) and (46) yields

$$\mu_i(C) \geq \frac{\kappa}{(N+1)B_N} \min\{\alpha(C), \alpha(\Gamma_i \setminus C)\}.$$

Thus (BUS) holds with $\eta := \frac{\kappa}{(N+1)B_N} > 0$.